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Life Cycle Inventories of Waste Treatment Services

Data v2.1 (2009)



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Part I

General introduction
Waste material compositions
Municipal waste collection

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Summary

This report is a general part on waste management and waste compositions. A general introduction to the waste disposal inventories of ecoinvent 2000 is given. General statistical data on waste in Switzerland and Europe is presented. The larger part of the report details the waste compositions used to calculate the disposal inventories in the ecoinvent database for waste incineration, landfilling, wastewater treatment, and building material disposal.

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1 Introduction

Life cycle assessments are also called 'cradle-to-grave' analysis, because they assess the process chain of a product from production, use to disposal. Often only cradle-to-gate or gate-to-gate assessments are performed in LCA. This report addresses the 'graves'.

Waste disposal is a somewhat neglected part in life cycle inventories. While for the *production and use phase* elaborate data gathering and computation of emissions and burdens is performed, the disposal phase receives often comparatively inferior treatment. For example, mere waste masses are summed up into a single figure, or masses to few classes of wastes are inventoried (see e.g. Boustead 1999). Sometimes only land occupation for landfills is inventoried. Such inventoried parameters pose an unnecessary complication in the valuation step of LCA. Seeing that waste disposal processes are indeed man-made technosphere processes that emit pollutants like any other technosphere process, there seems no need to assign special indirect inventory parameters like 'waste mass' to disposal processes. In the case of inventorying only land occupation for landfills, it is not clear, why for landfills only land occupation should be relevant and not direct emissions.

An improvement is the use of *average* disposal emissions. For example the air emissions, water emissions and auxiliary material inputs during *average operation* of a waste incinerator can be inventoried. However, seeing that very dissimilar waste materials are depicted with this procedure (paper, plastic, kitchen waste glass, cans) this is by comparison as accurate as if the emissions for *any* industrial material (metals, plastic, paper etc.) were inventoried using some *average industry mix* emissions.

Calculating disposal emissions in a way which is as detailed as for the production and use phase means to heed the *waste-specific* burdens generated by or attributable to a *specific waste composition*. With data from substance flow analysis in disposal processes, it has become feasible to model the fate of single chemical elements in disposal processes. With this information it is also possible to calculate waste-specific disposal inventories. Such inventories were first performed in (Zimmermann et al. 1996, Sundqvist et al. 1997).

The aim of the reports in the disposal part of ecoinvent 2000 is to create waste-specific life cycle inventories for different waste compositions, based on the Swiss disposal processes. The report is structured into several parts¹:

- **Part I** General statistical information, inventory of waste collection and documentation of waste compositions (This part)
- **Part II** Inventory model of municipal waste incineration, hazardous waste incineration
- **Part III** Inventory model of sanitary landfills, slag compartments, residual material landfills, inert material landfills, subsurface deposits (salt mines) and surface spreading
- **Part IV** Inventory model of municipal wastewater treatment
- **Part V** Inventory of building material disposal

All the induced further downstream processes are included in the inventories, e.g. for wastewater treatment, the disposal of resulting wastewater treatment sludge is included. Or for incineration processes, the landfilling of solid incineration residues and resulting burdens are included in the inventories as well.

Recycling processes per se are not considered in these inventories. Although for building material disposal, a sorting plant is inventoried which can be seen as a prerequisite for recycling (see part V).

¹ Treatment and disposal of *radioactive wastes* is inventoried in (Dones 2003).

Some limited information on electronic dismantling prior to recycling is given in chapter 4.2.8 'Plastics from electronics products' on page 32.

The aim of these reports is to document *calculation tools* that allow the consistent calculation of life cycle inventory (LCI) data from *any* waste composition, i.e. not only the disposal of *average* waste is inventoried by these tools, but also the inventories attributable to the disposal of *a specified waste fraction*. The Excel calculation tools are part of this report and the user shall be enabled to compute inventories of the disposal of user-specified waste materials, that can virtually have any arbitrary combination of characteristics. Some waste material disposals that are common and/or occurring within other parts of the ecoinvent 2000 are already calculated and available in the database. A restriction is however, that only chemical elements are modelled. The fate of individual chemical substances (e.g. PCB, hexachlorobenzene) is not modelled². The conception and application of the calculation models for different disposal processes is presented in the respective parts of this report. The specific waste compositions for different inventoried wastes are presented for all wastes in chapter 4 'Compositions of waste fractions' on page 28.

These reports differ from other reports in the ecoinvent 2000 project, as not only static data tables for some pre-defined processes are provided and documented, but actually *software tools*. The inventory process therefore aims to anticipate characteristics in any imaginable waste composition and make sure that the software delivers reasonable data, not only for average waste compositions, but also for particular waste fractions.

The reporting of this kind of LCI user tools tends to be more complex than for static LCI data tables encountered in other LCI reports. The reason for this is that some calculation steps, that would seem natural or obvious, if performed by a human for a specific waste, need to be formalised, automated and implemented in the software in order to deliver reasonable data for any waste composition. Also a lot of information is cross-connected, and all these co-dependencies need to be formalised³. Almost any section of the report is therefore cross-referenced to another part and the whole calculation procedure is rather elaborate and intricate. I apologise to the reader, if this makes for difficult reading. The reviewers and me tried our best to structure the report in a clear way, without too much redundancy. The advantage is of course that in the end the calculation tool behaves 'intelligently' (reacts correctly for any waste) and besides the waste composition data only little control information is needed to calculate the waste disposal LCI data.

Please consult the list of abbreviations and the glossary given prior to the appendices of each report.

1.1.1 Internal references and navigation through the report

Due to the complex nature of the topic, the reports contain a lot of internal references. Internal references to distant tables, figures and chapters within a part are given with the page number for convenience.

In the electronic versions of the reports (Adobe Acrobat PDFs) the page references are invisible or boxed 'hotlinks'. They will take you directly to the indicated table, figure or chapter. Move the cursor over the indicated page number. The cursor will change to a pointer ()^h). Click and Acrobat takes you to this page. *To return to the page you came from use the arrow buttons in the menu bar of Acrobat Reader* (cf. Fig. 1.1).

² For wastewaters however generic fate of COD, BOD, DOC, TOC, NH₄, NO₃, NO₂, N_{org}, SKN, TKN, PO₄, P_{part.}, P_{tot.}, SO₄, S_{part.}, S_{tot.} is heeded as well as for other chemical elements.

³ These dependencies not only affect the calculation of mean values, but also the calculation of uncertainty ranges.



Fig. 1.1 Use the arrow buttons in the Acrobat Reader menu bar to return from hotlinks.

Literature references for internet sources are given as [underlined weblinks](#). Move the cursor over the link. The cursor will change to a web pointer in Acrobat Reader (). Click and your web browser should start with the indicated URL address.

1.2 Other general remarks

This report deviates in some other respects from other reports in the ecoinvent 2000 project. These differences are pointed out below to avoid misunderstandings.

1.2.1 Displayed vs. significant digits

Figures in the inventory tables of this report often feature several digits. This is not to imply that all the digits are really significant or that the data displayed is very precise (it is mostly not). Showing several digits helps to minimise the avoidable accumulation of rounding mistakes along the chain of calculations performed here, and in possible future studies referring to this data.

The level of significance or the uncertainty of the inventoried values is specifically expressed with *uncertainty values* in ecoinvent 2000. Those uncertainty values shall be used to judge the significance of data values, and not the number formatting in tables. Unnecessary rounding can even introduce new rounding errors which are *not* accounted in ecoinvent 2000 and which are entirely avoidable⁴.

1.2.2 Literature references

Literature references are given in round brackets, e.g. (BUWAL 1999). Different references from the same source and year are discerned by lower case letters, e.g. (BUWAL 2000a, BUWAL 2000b etc.). Page numbers for a particular information reference are separated by a colon, e.g. (Leuenberger & Spittel 2001:20). If page numbers are used discontinuously throughout a report, the part of the report is included in the page number reference, e.g. (Zimmermann et al. 1996:B.17, Frischknecht et al. 1996:V.55).

1.2.3 Uncertainty values GSD and GSD²

Uncertainty ranges can be expressed for lognormal distributions by the geometric standard deviation GSD. The GSD is an 'uncertainty factor' for the confidence range around a mean value (cf. Tab. 1.1). The uncertainty information in the ecoinvent 2000 database relates to a 95%-confidence range.

⁴ For example, if a parameter has a mean value of 0.123 and a GSD² of 123%, its lower to upper boundary values are 0.1 – 0.151 (mean*GSD^{±2}). The practice of keeping only to the significant digits would suggest to use a 'significant' mean value of 0.1. The rounding mistake is 0.023. *The introduced rounding error is comparable to the original parameter uncertainty.* Using significant digits only, the original *lower boundary value* (0.1) is used as the new *mean value* in this example. The data quality has deteriorated by using significant digits only. In the opinion of the author this is unnecessary and avoidable. The new 'significant' mean value of 0.1 carries less – or less appropriate – information than the original figure 0.123. A rounded mean value of 0.1 can potentially originate from any figure between 0.050 and 0.149. The uncertainty of this latter range has GSD² values of approximately 150% to 200%, i.e. is *larger than the original parameter uncertainty*. Hence, rounding to only *one* significant digit can lead to intolerable rounding mistakes, *which are larger than the original parameter uncertainty*. This effect is increasingly alleviated, the larger the one retained significant digit is: for 0.9 the original figure can be between 0.85 and 0.949, which corresponds to a GSD² of approx. 106%. Rounding to significant digits makes sense, *only if* the original parameter uncertainty so low that the introduced rounding mistake is *much smaller* than the original parameter uncertainty. In LCA this is often *not* the case.

Tab. 1.1 Calculation of confidence ranges expressed with the GSD value.

	<i>lower confidence limit</i>	<i>upper confidence limit</i>
68% confidence range:	mean/GSD	mean*GSD
95% confidence range:	mean/GSD ²	mean*GSD ²

In ecoinvent 2000 uncertainty values are often estimated using the Pedigree approach introduced in (Frischknecht et al. 2003a). In this report GSD values derived from the Pedigree approach are usually given as **GSD²** i.e. *GSD squared*. Uncertainty values derived from other sources are sometimes also given as *non-squared* GSD values **GSD**.

In ECOSPOLD percent values are entered *without multiplication with a factor of 100*⁵. I.e. a value of 175% is entered as '175%' or the mathematical equivalent '1.75', *but not as '175'*.

Please note the difference between GSD and GSD²,
the latter also denoted as GSD2 or GSD^{^2}.

In any case a value of e.g. 175% equals 1.75 (percent is not a unit)

1.2.4 Uncertainty of transport services

In almost every unit process inventory of the ecoinvent 2000 project transport services for transport of materials are inventoried. Transport services can be fulfilled by lorries, rail, ship and air cargo. The functional unit of transport services for goods is ton-kilometer, tkm, i.e. one ton of goods transported for the distance of one kilometer. Average load factors are included in the standard transport modules in the ecoinvent database.

During the inventory process the situation often arises that inventoried *masses* are known, but the *transport distances* are not. To fill the data gap of transport distances, standard distances are devised for several materials in (Frischknecht et al. 2003a).

Within the ecoinvent 2000 project, the uncertainty of transport services when using standard distances, is generally described by a GSD² value of 209%, using the Pedigree approach (Frischknecht et al. 2003a). Tab. 1.2 shows this generic Pedigree uncertainty. This uncertainty is understood to cover the uncertainty in the transport distance *and* the uncertainty in the transported mass, i.e. the mathematical product expressed in ton-kilometers. Most reports within the ecoinvent project use this generic uncertainty for transport services when applying the ecoinvent standard distances.

Tab. 1.2 Generic Pedigree uncertainty of transport services (not used in this report)

Exchange	GSD ² value	Pedigree codes	Comment
Uncertainty of transport services	209%	(4,5,n.a.,n.a.,n.a.,n.a.)	Basic uncertainty of 200%; uncertainty of transport services (tkm) using ecoinvent standard distances (km)

In this report however, input masses are often *calculated* depending on waste compositions. The masses vary and so do the associated uncertainties. For the *transport distances* usually the ecoinvent standard distances are used. Since the transport services are the product of 'mass' times 'distance', the uncertainties of the transport services are *not constant*, because the uncertainty in the mass changes.

⁵ This is a deviation in ECOSPOLD from the earlier SPOLD format. In the SPOLD format, percent values were entered with a factor 100, i.e. 175% was entered as 175 (Weidema 1999:16).

This contrasts with the notion expressed in Tab. 1.2, that devises one constant GSD^2 value for the uncertainty of a transport service.

Uncertainties of transport services are calculated in this report by first calculating the uncertainty of the product of 'mass' times 'distance' for each individual material. The uncertainty of this product depends on the uncertainties of the factors 'mass of material' and 'transport distance', i.e. standard distance. Next, the uncertainty of the sum of the transports for all the individual materials is calculated using the Wilkinson-Fenton approximation for sums of lognormal distributed summands.

To minimise the discrepancy between the procedure in this report and the other ecoinvent reports, the uncertainty of the ecoinvent standard *distances* is altered to a lower, constant value so that the calculated uncertainties of transport *services* are in the approximate vicinity of the value given in Tab. 1.2. The uncertainties in the transported *masses* is usually already quite high (GSD^2 typically ranges between 200% and 300%). The uncertainty of the final sum decreases somewhat by use of the Wilkinson-Fenton approximation. By setting the uncertainty of the ecoinvent standard distances to the minimum of 100%, final GSD^2 values in the approximate range of 180% to 260% are obtained (compared to a constant value of 209% using the Pedigree approach above).

Tab. 1.3 Calculated uncertainties of transport services in this report

Exchange	GSD^2 ranges	Comment
Uncertainty of masses	approx. 200% – 300%	Variable, dependent on waste composition.
Uncertainty of ecoinvent standard transport distances	100%	Constant value
Uncertainty of transport services	approx. 180% – 260%	Calculated depending on the figures above. Variable, dependent on waste composition.

This deviation from other project members – although considered an improvement by the author – has earned the respective data modules a review comment: *"Passed. Calculation for standard deviation 95 of transport services can differ from standard Pedigree approach used in this project"*.

1.2.5 Uncertainty of infrastructure exchanges

The Pedigree approach suggest a basic uncertainty of 300% for infrastructure exchanges. Some administrators in the ecoinvent 2000 project have used this basic uncertainty only for infrastructure *materials* (bricks, concrete, steel etc.), but for infrastructure construction *energy* used the basic uncertainty for generic energy exchanges (i.e. 105%).

In this report the suggested basic uncertainty of 300% for infrastructure exchanges has been applied for both infrastructure materials *and* infrastructure construction energy. The reason for this is that it seems not feasible that – e.g. in the maximal case – three times as much materials are processed with only 5% more energy⁶. This deviation – although considered entirely appropriate by the author – has earned the respective data modules a review comment: *"Passed. Calculation for standard deviation 95 of infrastructure can differ from standard Pedigree approach used in this project"*.

⁶ This would mean that the *specific energy consumption* per kg built material would *decrease* by a factor 2.86. Some decrease in specific energy consumption can be expected for larger building volumes, but not to the extent that the *absolute energy consumption* is almost identical for any size of building. I.e. if three times as much materials are processed, then energy consumption is bound to be (almost) threefold as well, and unlikely to increase only 5%.

2 Waste Constituents

The waste compositions use for the life-cycle inventories of disposal processes generated in this study are documented in chapter 4 'Compositions of waste fractions' on page 28. In this chapter a general overview of municipal and hazardous waste is given.

2.1 Municipal waste composition

Municipal waste is household refuse and similar waste from small businesses or industry, excluding hazardous or special wastes. Main constituents are kitchen waste (biomass), paper, plastics, metals, glass and fine material. The composition of the waste being disposed of in municipal facilities depends on recycling rates for individual fractions, which are collected separately (e.g. paper, glass, metals, cf. Fig. 3.1 on page 19 and Fig. 3.6 on page 22).



Fig. 2.1 Municipal solid waste sample, sorted according to defined size fractions (Kost & Rotter 2000)

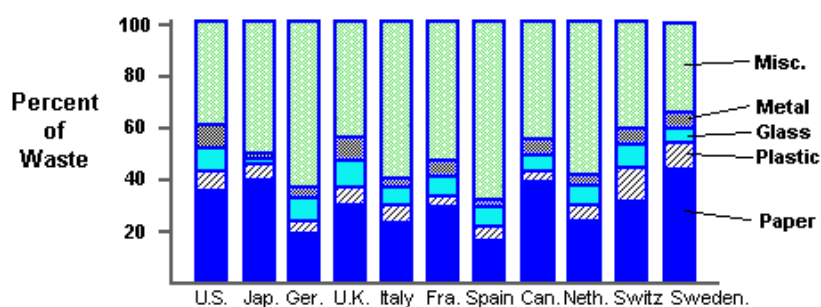


Fig. 2.2 Composition of municipal waste in several countries as an illustration of waste variability. Data is evaluated by EPA from OECD data of 1989 (EPA 1997)

2.2 Hazardous waste composition

Hazardous wastes are a very heterogeneous group of materials. In Switzerland 170 different types of waste are considered as hazardous wastes according to the ordinance on hazardous waste transactions (VVS). The largest type of hazardous waste by weight is 'soil contaminated with mineral oil or other substances' (214'000 tons/a). Some other important hazardous waste types are for example: acids, solvents, emulsions with metals, oil mixtures like oil separator sludge, non-metallic shredder residues,

dusts, electrostatic precipitator ashes from waste incineration, wastes and distillation residues from organic synthesis, lead batteries, or road gutter sludge. Figures for 1999 are given in Tab. 2.1.

The 170 types of hazardous wastes can be grouped into classes with similar characteristics or treatment methods. Fig. 2.3 shows the development of these classes of hazardous waste fractions generated in Switzerland. One of the largest classes – contaminated soils – is very variable in time, and essentially determines the overall trend of the total amount of hazardous wastes generated. Wastes with high heating value are dominantly incinerated (waste solvents, mineral oil mixtures, organic synthesis waste, residues from decanting).

Tab. 2.1 Hazardous waste generated in Switzerland 1999 (BUWAL 2001a). The 21 itemised waste types comprise 85% of the total hazardous waste mass in 1999

English	German	waste generated t/a
soil contaminated with other substances	mit anderen Substanzen verunreinigtes Erdreich	185'436
gasoline and oil separator wastes	Oelabscheider-, Benzinabscheiderabfaelle	69'168
road gutter sludge	Strassensammlerschlaemme	66'074
mineral oil mixtures	Mineraloelgemische	49'814
contaminated materials and appliances	verunreinigte Materialien und Geraete	47'547
non-metallic shredder residues (mostly from cars)	nichtmetallische Schrederabfaelle	44'590
electrostatic precipitator ash from MSWI	Elektrofilterstaub aus KVA	42'475
halogen-free or weakly halogenated solvents	nicht oder schwach halogenierte Loesungsmittel (Chlorgehalt <= 1%)	41'970
waste, malfunctions, and by-products from organic synthesis	Fehlchargen, Ausschussware und Nebenprodukte aus Organischensynthesen	39'335
dusts, fines, fly ashes	Staeube, Feinstteile, Flugaschen	37'825
chlorine-free solvents/mixtures	chlorfreie Loesungsmittelgemische, auch stark verschmutzt	33'926
aqueous wastes not contaminated with halogenated solvents	waessrige, mit nicht halogenierten Loesungsmitteln verunreinigte Abfaelle	30'308
soil contaminated with mineral oil	mit Mineraloelprodukten verunreinigtes Erdreich	29'079
distillation residues from organic synthesis	fluessige Dest.-Rueckstaende aus der Synthese organischer Produkte	22'972
residues from decanting, filtration and centrifugation	Rueckstaende von der Dekantierung, Filtration und Zentrifugierung	21'527
lead batteries	Bleiakkumulatoren	21'362
weakly halogenated solvents	schwach halogenierte Loesungsmittel (Chlorgehalt <= 2%)	18'099
aqueous wastes contaminated with halogenated solvents	waessrige, mit halogenierten Loesungsmitteln Verunreinigteabfaelle	17'447
halogenated solvents	halogenierte Loesungsmittel (Chlorgehalt > 2%)	16'726
acids, free of metals (except iron)	Saeuren, metallfrei oder nur eisenhaltig	12'741
chlorinated solvents	chlorhaltige, leichtentzuendliche Loesungsmittelgemische	12'016
others	weitere	156'262
Total		1'016'699

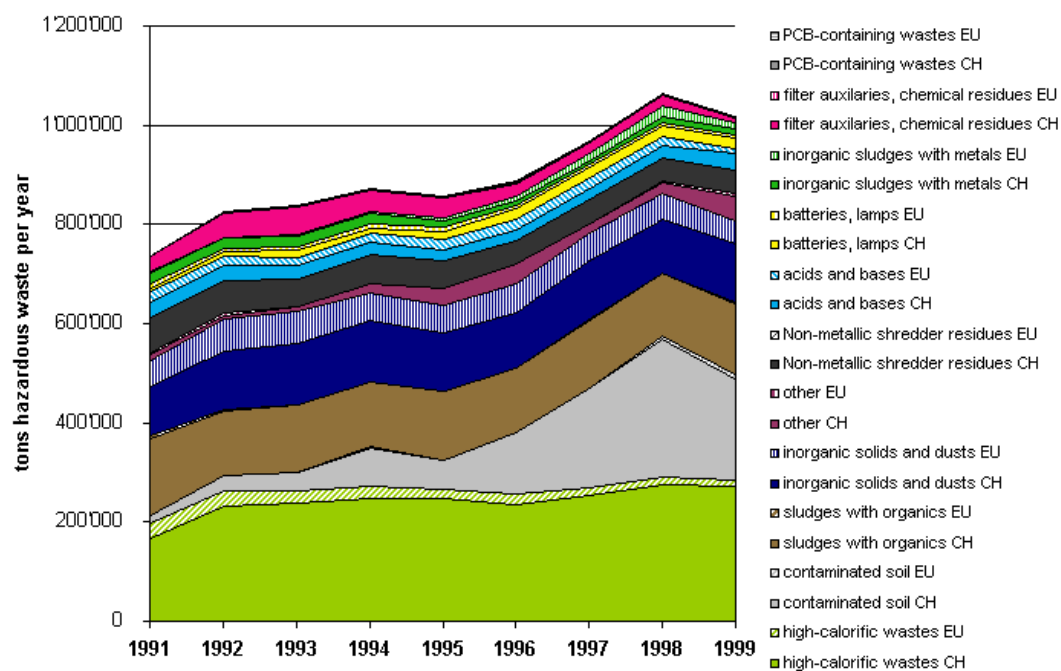


Fig. 2.3 Development of classes of hazardous waste fractions generated in Switzerland from 1991-1999 (BUWAL 2001a). 'CH' denotes masses disposed in Switzerland; 'EU' denotes exported masses

3 Waste Generation and Treatment

3.1 Municipal waste generation and treatment

3.1.1 Swiss municipal waste generation and treatment

In 2000 Switzerland generated 2.68 Mio. tons of MSW⁷. Of that amount 2.4 Mio tons (89%) were incinerated, 291'700 tons (11%) were landfilled. Landfilling is phased out. The waste generation is 368 kg per capita (BUWAL 2001b). Additionally, major waste masses are generated from construction activities: approximately 6 Mio tons from road construction, 5 Mio tons from building construction waste, and an estimated 30 to 40 Mio. cubic meters or roughly 70 Mio tons of excavated earth (BUWAL 2001e). Smaller amounts come from hazardous waste (cf. chapter 3.2 'Swiss hazardous waste generation and treatment' on page 24), WWT sludge and recycling of paper, glass and biomass. Fig. 3.1 shows these major waste flows in Switzerland in 2000.

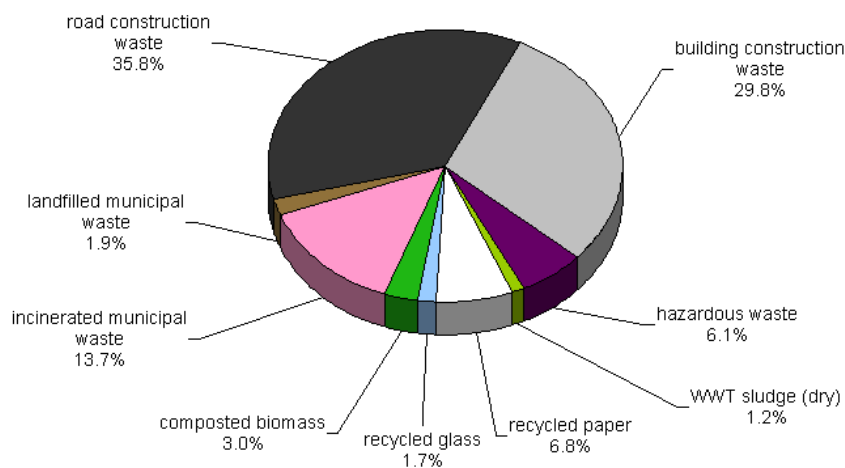


Fig. 3.1 Major waste flows generated in Switzerland in 2000 either to disposal or recycling. 100% = 16.8 Mio tons of material, excluding an estimated 70 Mio tons of excavated earth from construction

3.1.2 Municipal waste generation and treatment in other countries

The quantity of municipal waste generated in the OECD area has risen by 35% between 1980 and 1997 and reached 540 million tonnes in the late 1990s, with however a slight slowdown in recent years. Generation intensity per capita has risen mostly in line with private final consumption expenditure and gross domestic product and is currently at 500 kg per capita.

⁷ Municipal solid waste is household refuse and similar waste from businesses and industries (e.g. packaging waste), but excluding recycled waste, burnable construction waste, hazardous waste and WWT sludge.

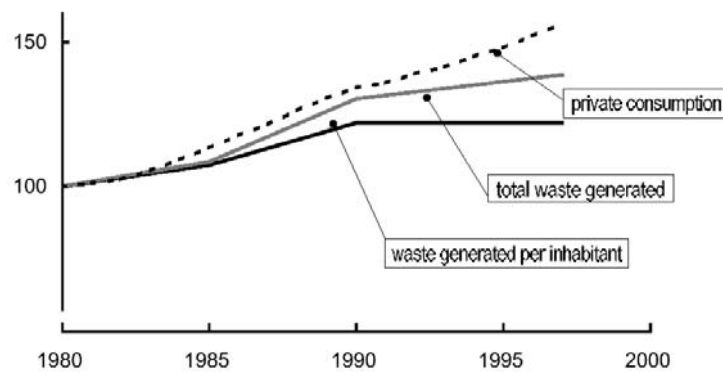


Fig. 3.2 Municipal waste generation and private final consumption expenditure in OECD countries (Index 1980 = 100) (OECD 2001)

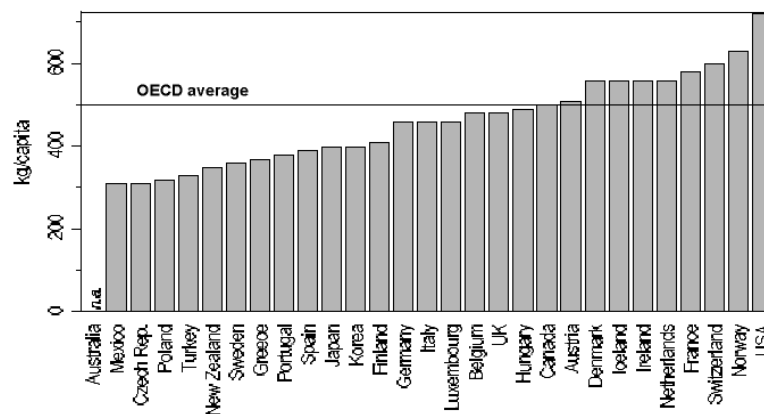


Fig. 3.3 Municipal waste generation per capita in OECD countries (OECD 2001). Includes waste to recycling.

The amount and the composition of municipal waste varies widely among OECD countries, being directly related to levels and patterns of consumption and also depending on national waste management practices. Only a few countries have succeeded in reducing the quantity of solid waste to be disposed of. In most countries for which data are available, increased affluence, associated with economic growth and changes in consumption patterns, tends to generate higher rates of waste per capita (OECD 2001). The plot of per-capita municipal waste production versus per-capita gross domestic product in Fig. 3.4 shows a clear increasing trend, with just a hint of a decrease at the high-GDP end. The industrial waste production per capita shows no clear trend. High or low waste industrial productions are possible at the low-GDP and the high-GDP end, which probably strongly depends on the industries involved.

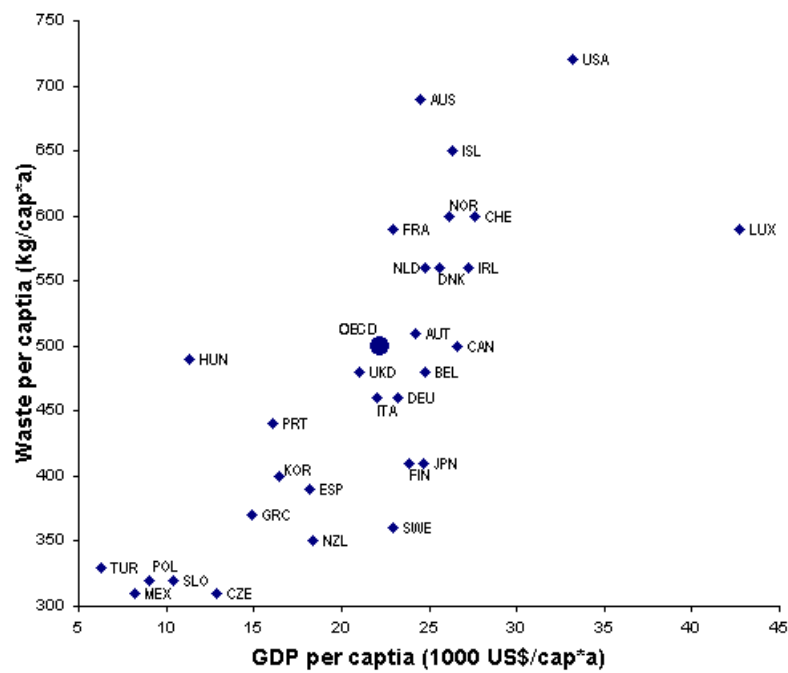


Fig. 3.4 Annual municipal waste generation per capita in OECD countries vs. gross domestic product. Includes recycled waste. Data from (OECD 2002a, OECD 2002b)

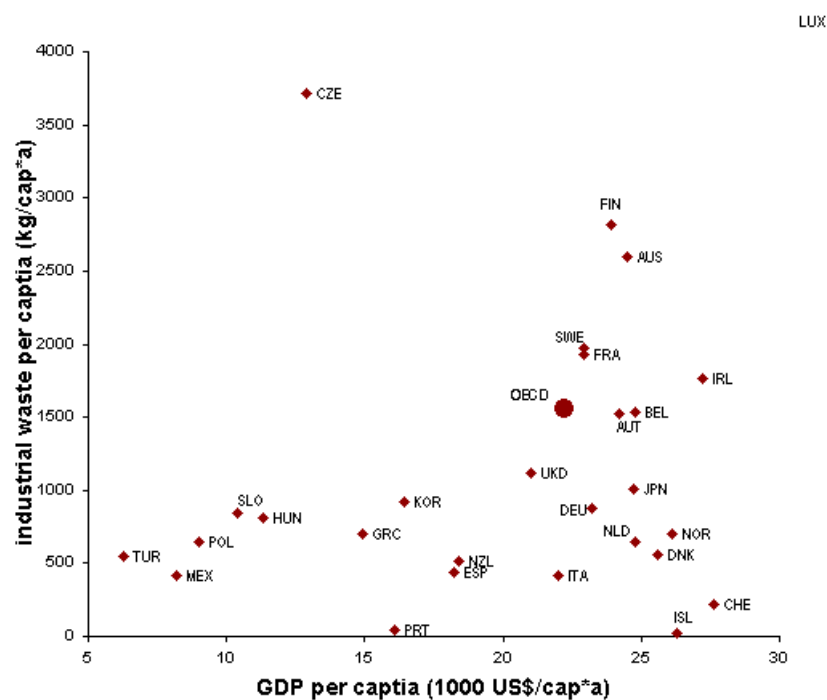


Fig. 3.5 Annual industrial waste generation per capita in OECD countries vs. gross domestic product. Data from (OECD 2002a, OECD 2002b)

Most waste is dealt with by the cheapest available method: in OECD Europe⁸ during 1991-95, 66% of municipal waste was landfilled, 18% was incinerated, 9% was recycled, 6% was composted and 1% was treated in other ways (UNEP 2001). More detailed and recent data is shown in Fig. 3.7. Recycling of waste in most Western European countries is increasing. Recycling rates in Switzerland and other European countries are shown in Fig. 3.6.

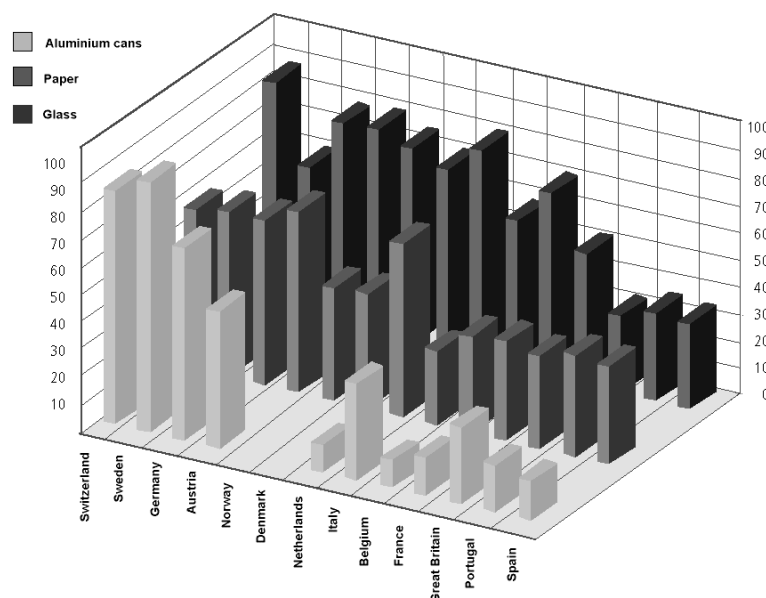


Fig. 3.6 Recycling rates in some European countries. No data on aluminium can recycling for Norway and Denmark (BFS 1999:144)

Large differences between individual European countries exist regarding waste management. Traditionally landfilling is predominant. Enforced creation of incinerator capacities and phase out of landfilling – especially in smaller countries – has lead to a large incineration share e.g. in Denmark, the Netherlands, Switzerland, Luxembourg, Sweden. In the USA and Canada landfilling shares were 80-90% in 1995, while in Japan it was 15% (EPA 1997)

⁸ Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Poland, Portugal, Slovak Republic, Spain, Sweden, Switzerland, Turkey, United Kingdom.

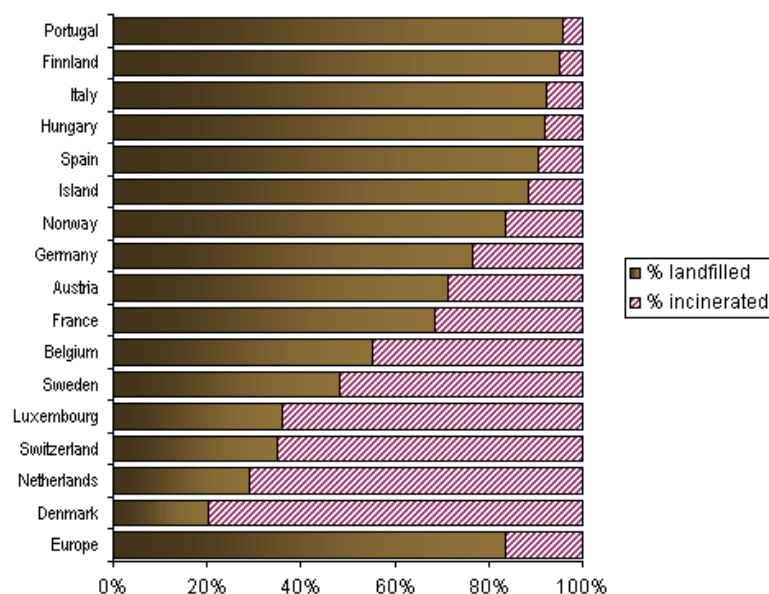


Fig. 3.7 Disposal fates of municipal waste in European countries (without recycling). Average data from reporting countries during 1995-1999 (Germany for 1993) from (ETCW 2002)

3.1.3 Beyond developed waste treatment

Within the ecoinvent 2000 project the Swiss disposal technologies are inventoried, which can be characterised as modern waste disposal processes. In other parts of the world quite different management of waste are commonplace. Some impressions shall be given here, though this information will not be reflected in the inventories of this report.

Patterns of household waste disposal in developing countries vary according to climate, housing conditions, the existing collection service, local regulations, cultural and other factors. Also, the characteristics of the waste itself (smell, attractiveness to flies and rodents) determines disposal habits. Households in hot, humid climates have incentive to dispose of waste immediately. In communities where the primary waste collection does not exist – like in many rural areas – open burning is a common disposal method. Where collection exists, the collected waste is usually hauled to open dumps. These open dumps are often uncontrolled landfills without leachate collection and often pose hazards in the form of vermin, uncontrolled fires and unsanitary conditions. In this environment waste scavengers – often children – pick out reusable material (ISWA 2002). Recycled material from such sources can be essential to local enterprises. Dumps can be rendered unstable by uncontrolled fires eating cavities into the compacted waste. Deaths of scavengers from collapsing waste dumps are common. The recovery of materials from waste represents an important survival strategy for disadvantaged populations throughout the developing world. In many cases, they are subject to exploitation by middlemen and discriminated by local and federal government policies. There are however also scavenger co-operatives in Mexico, Venezuela, Peru, Ecuador, Guatemala, Costa Rica, Colombia, Brazil, Philippines, Indonesia, India and other countries (Medina 1998).

Kraxner et al. (2001:17) point out that in dry or cold climates the physical-chemical development of (inorganic) landfills is dominated by mechanical erosion and emissions of pollutants occurs mainly by wind erosion, while in temperate zones chemical weathering prevails and pollutants are emitted via water.



Fig. 3.8 Waste scavenging children in Bolivia (Yee et al. 2000)

3.2 Swiss hazardous waste generation and treatment

Approximately 1 million ton of hazardous wastes were generated in Switzerland in 1999 (BUWAL 2001a). Most hazardous waste is generated in industries and only a minor fraction in households like pharmaceuticals, fluorescent light tubes, or paint.

Most Swiss hazardous waste is disposed in Switzerland by incineration, landfilling, transformation (precipitation, physical-chemical treatment) or recycling (cf. Fig. 3.9). Only 12% of the hazardous waste is exported – in decreasing order – to Germany, France, Belgium, the Netherlands, Spain and other, mostly European countries (cf. Fig. 3.10). Transfers and trade with hazardous waste is regulated by the ordinance on hazardous waste transfers (VVS) since 1987. Since 1990 Switzerland is a party to the Basel convention on international transfers of hazardous wastes⁹.

⁹ In 1999 the Basel convention had 133 country parties, while USA and Afghanistan are merely signatories.

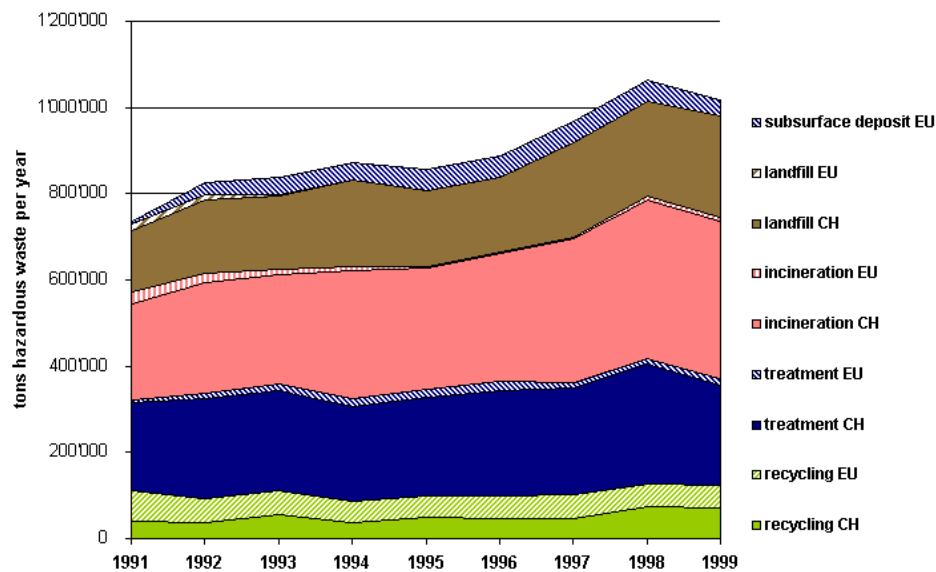


Fig. 3.9 Fate of hazardous waste generated in Switzerland 1991-1999 (BUWAL 2001a). 'CH' denotes masses disposed in Switzerland; 'EU' exported masses

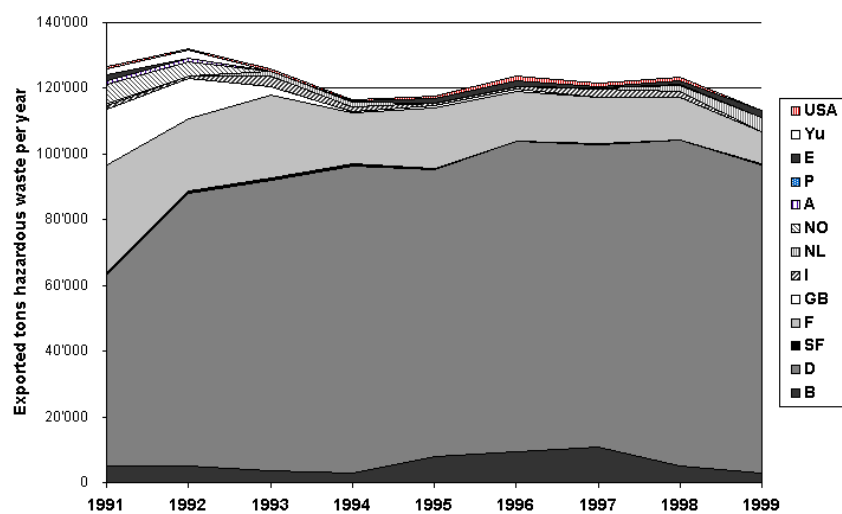


Fig. 3.10 Receiving countries of Swiss hazardous waste exports 1991-1999 (BUWAL 2001a)

3.3 Hazardous waste generation and treatment in other countries

Fig. 3.10 shows data for the total annual hazardous waste generation and hazardous waste generation per capita in EU countries. The figures per capita depend on the industry mixture present in the country.

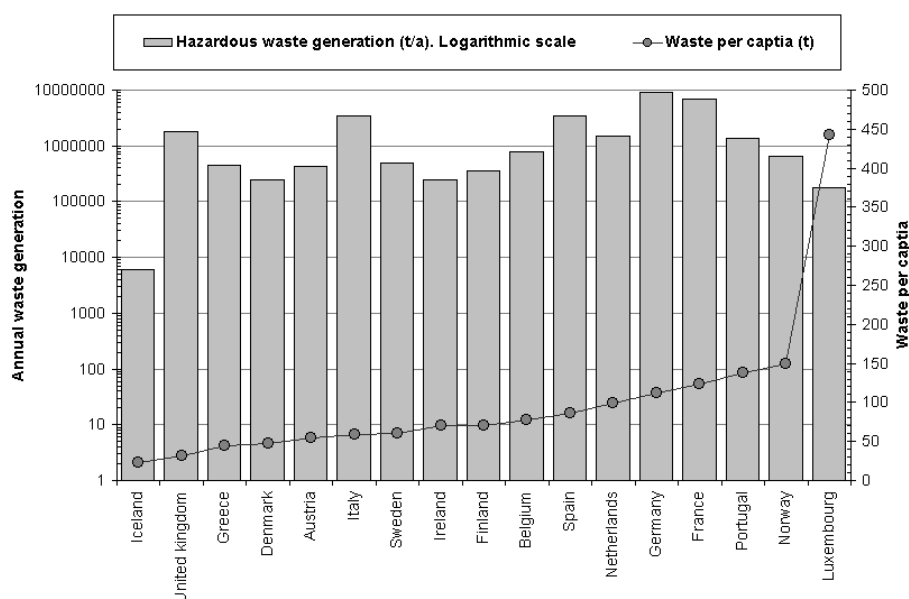


Fig. 3.11 Total hazardous waste generation in EU countries: tonnes per year and tonnes per capita (EEA 2002)

Industrial waste production and sources in selected countries from around 1990 are shown in Tab. 3.1 (WRI 1997). Industrial waste data are collected by various means, and definitions might vary across countries. Comparisons should be made cautiously, because (a) definitions vary from country to country, (b) the mix of hazardous materials in each category also varies, (c) these data do not include all industrial or hazardous waste (some data are based only on surveys of particular segments of an industry), and (d) these data do not measure potential toxicity.

Tab. 3.1 Industrial waste in selected countries. Taken from (WRI 1997:292)

	Year of Estimate	Waste Generated from		Waste (metric tons)			Waste from Production and Use of (metric tons)			
		Surface Treatment of Metals and Plastics (metric tons)	Biocide Production (metric tons)	Oil	Containing PCBs	Clinical and Pharmaceutical	Photographic Materials	Organic Solvents	Paints and Pigments	Resins and Latex
Austria	1990	14,731	450	60,300	81	8,254	1,400	27,253	15,000	X
Canada	1985	186,200	4,500	367,000	120,000	X	X	262,000	72,700	74,000 a
Czechoslovakia (former)	1987	2,561,174	183	565,764	X	X	X	20,723	13,875	131,519
Finland	1987	1,813	361	35,684	1,789	97	547	7,384	5,787	2,123
France	1990	X	X	409,000	17,000 b	X	X	285,000 b	X	X
Germany, Fed Rep	1987	219,527	X	859,456	10,537	X	X	454,489	225,525	867,015
Greece	1990	X	X	25,000	1,800	1,500	X	21,000	6,000	150
Hungary	1989	12,000	10,300	455,000	134	X	X	49,000	11,000	X
Ireland	1991	7,000	5	1,000 c	X	X	X	12,500	X	45,000
Japan	1985	8,877,000 d	X	3,672,000 d	X	X	X	X	X	2,894,000
Luxembourg	1990	22,200	5	3,900	480	356	29	284	540	X
Netherlands	1990	22,000	1,800	279,000	400	1,000	21,000	69,000	25,000	20,000
New Zealand	1990	3,030	1,100	18,151	4	2,770	451	3,690	29,381	12,892
Norway	1988	8,000	400	55,000	2,000 e	X	6,000	9,000	16,000	X
Poland	1990	X	X	41,400	X	X	X	X	175,900	X
Portugal	1989	X	X	16,473	703	X	X	X	X	X
Spain	1990	X	X	320,000	2,200	X	X	5,400	X	X
United States (f)	1990	1,982,379	13,216	4,960,000	5,015,060	2,800,000	X	70,000,000	693,833	41,000,000

Source: Organisation for Economic Co-Operation and Development.

Notes: a. Data for resins and latex are from 1987; PCB waste includes 6,500 metric tons in storage. b. Data for PCBs and organic solvents are from 1989.

c. Waste oil is only lubricating oil; data for organic solvents include miscellaneous chemical wastes.

d. Waste oil includes waste solvents; waste generated from the surface treatment of metals is total waste metals; data for resins and latex refer to plastics and rubber.

e. Data for PCBs refer to 1987. f. Data are from 1989 through 1991; PCB data are from a survey and do not represent total PCB waste; organic solvents include all organic chemicals; resins and latex refer to plastics and rubber. X = not available.

For additional information, see Sources and Technical Notes.

The waste generated from surface treatment of metals and plastics includes acids and alkalis as well as other pollutants. Surface metal treatment is the largest source of acid wastes. Waste generated from

biocide production results from the manufacturing and use of insecticides, herbicides, and fungicides (not including those quantities applied correctly, but including spills, residues, etc.). Waste oil includes used motor oil, contaminated fuel oils, waste from industrial processes, and waste vegetable oils, among others. Waste containing PCBs includes waste from their manufacture, from the scraping of equipment containing PCBs, and from certain hydraulic fluids used in mining equipment and aircrafts. Clinical and pharmaceutical waste includes waste pharmaceuticals, laboratory chemical residues arising from their production and preparation, and clinical (i.e., infectious) waste from hospitals, medical centres, clinics, and research institutions. Waste from the production and use of photographic materials includes waste chemicals from photographic processing. Waste organic solvents arise from dry cleaning and metal cleaning, from chemical processes, as well as from the production of numerous manufactured products such as paints, toiletries, thinners, and degreasing agents. Waste from paints and pigments includes waste from the manufacture and use of inks, dyes, pigments, paints, lacquers, and varnishes. Waste from resins and latex comes from the production, formulation, and use of resins, latex, plasticizers, glues, and other adhesives (WRI 1997).

4 Compositions of waste fractions

The waste compositions are an essential part of calculating the waste-specific disposal inventories. Various waste compositions were established from literature data, estimates and theoretical considerations. The figures are targeted to represent compositions *after use*, i.e. can contain traces from the use phase. In some cases data is complemented with figures from similar materials to avoid data gaps. In the chapters below the sources and assumptions are described. The established waste compositions used for the calculation of the disposal inventories are listed in the appendix of this report (cf. chapter 7.1 'Synopsis of solid waste compositions' on page 73 and chapter 7.2 'Synopsis of wastewater compositions' on page 99).

More information on the necessary waste information (like heating values, biogenic vs. fossil carbon, share of magnetic iron, generic uncertainty estimates for solid waste composition data) is compiled in the chapters 'Necessary waste data' and 'Uncertainty of waste composition data' in part II (incineration).

Guidance to this chapter

The entries to the various waste materials *for general or common waste materials* are roughly ordered according to material type or class:

- **Average waste mixtures** (municipal waste, residual landfill, inert landfill, sewage)
- Waste **polymers** (PE, PP, PS, PVC, PET, PU, mixture, electronics plastic, PVF, rubber)
- Waste **metal materials** (steel, aluminium, tin sheet)
- Waste **biomass materials** (paper, packaging paper, newspaper, packaging cardboard, untreated wood, chrome preserved wood, textiles)
- Waste **inert materials**
- Waste **building materials** (concrete, gypsum boards, mineral wool, expanded polystyrene, wood, paints, bulk plastics, cement-fibre slab, plaster, plastic sheets, electric wiring)

After the unspecific, generic fractions, wastes which originate *from a specific process type* are shown per process (e.g. all specific wastes from aluminium production). Efforts were made to make the disposal module names as distinct as possible. These efforts are sometimes obstructed by the 80 character limit in module names. Therefore, the text in the ECOSPOLD data field 'Synonyms' contains additional clarifications regarding the waste type and origin. Processes are arranged in the order listed below. In case of several wastes per process type a sequence according to life-cycle chain is pursued.

- Wastes from **power plants**
- Wastes from **fossil energy** production and conversion (coal, oil, gas)
- Wastes from **renewable energy** production and conversion (photovoltaic, solar heat, wood)
- Wastes from **materials production** (paper & cardboard, wood building materials and concrete, glass)
- Wastes from **metals production** (aluminium, steel and iron, silicon, other metals, black chrome coating)
- Wastes from **inorganic chemicals production** (iron chloride, titanium dioxide, phosphoric acid, chlorine, potash, soda, water purification)
- Wastes from **organic chemicals production** (formaldehyde, ethylene oxide, ethylene dichloride)
- Wastes from **agricultural products** (starch)
- Wastes from **transport services** (rail, road, ship)

4.1 Unspecific, general waste fractions

4.1.1 Average municipal waste mixture

As average municipal waste the composition derived in (Hellweg 2000) is used. This composition is based on information from (Morf et al. 1997, Belevi 1998, Zimmermann et al. 1996) for average municipal solid waste. The waste is made up of fractions of different burnability and decomposability. The composition of the different fractions is given in the appendix of part II. Necessary additional information is given in Tab. 4.1.

Tab. 4.1 Additional information on the waste fractions in average municipal solid waste

Waste fraction	Share of fraction in waste	Burnable (1) or inert (0)	Share of carbon in waste that is biogenic	Share of iron in waste that is metallic/recyclable	Degradability of waste in a municipal landfill within 100 years
Compostable material	22.0%	1	100.0%	0.0%	27.0%
Paper	21.0%	1	100.0%	0.0%	27.0%
Plastics	15.0%	1	0.0%	0.0%	1.0%
Natural products	9.0%	1	100.0%	0.0%	27.0%
Mixed cardboard	8.0%	1	100.0%	0.0%	32.4%
Minerals	8.0%	0	0.0%	0.0%	0.0%
Laminated materials	3.0%	1	80.0%	0.0%	18.0%
Combined goods: diapers	3.0%	1	50.0%	0.0%	18.0%
Glass	3.0%	0	0.0%	0.0%	0.0%
Inert metals	2.7%	0	0.0%	83.0%	10.0%
Laminated packaging	2.0%	1	80.0%	0.0%	18.0%
Textiles	2.0%	1	80.0%	0.0%	12.0%
Volatile metals	1.0%	1	0.0%	0.0%	0.0%
Electronic goods	0.3%	1	0.0%	60.0%	0.0%
Batteries	0.0%	1	0.0%	100.0%	0.0%

4.1.2 Average residual material waste mixture

Different waste materials are deposited in residual material landfills. Statistical figures on deposited amounts are difficult to obtain. From MSWI alone, 44'100 tons of fly ashes and 3160 tons of scrubber residues were deposited in residual landfills (BUWAL 2001m, BUWAL 2001n).

Also industrial wastes are deposited in residual material landfills. No literature references for this latter waste stream, nor compositions, could be found. The total amount of *landfilled hazardous wastes* was 234'407 tons in 1999 (BUWAL 2001a). Approximately 162'000 tons of hazardous wastes¹⁰ were landfilled in sanitary landfills in 2000 (BUWAL 2001d). The remainder of approximately 72'000 tons per year is landfilled in residual material landfills. A total amount of approx. 120'000 tons per year (40% from MSWI, 60% from industry) is deposited in residual material landfills.

The composition of industrial hazardous wastes landfilled in residual material landfills are not known. Hence, no average residual material waste mixture can be estimated. As a proxy the known fraction – incineration residues from MSWI – is used here. The composition is derived from average municipal waste composition (cf. chapter 4.1.1 'Average municipal waste mixture' on page 29) and the municipal

¹⁰ The categories 'hazardous waste' (Sonderabfall) and 'polluted excavation material' (verschmutzter & tolerierbarer Aushub) from (BUWAL 2001d) were added for this figure. Excluded is an amount of 270'000 tons of unpolluted excavation material.

waste incineration model as detailed in part II. The waste material is the incineration residue (fly ash + scrubber sludge) of average municipal waste. Solidification with cement is assumed. A share of 61% biogenic carbon is assumed, which corresponds to the figure of the burnable fraction in MSW. The calculated composition of the unsolidified residual material has high content of zinc (7.2%), sulfur (2.8%) and lead (2.1%). Various other metal and non-metal traces <1% are present.

This waste can be used as a proxy for inorganic process residues or incineration remains of unknown composition. It is used e.g. to represent unspecified solid wastes of type 'inert chemical' adapted from APME LCI reports on plastics (see Althaus et al. 2003a). Solid wastes of type 'inert chemical' in APME/Boustead sources can be described as a wide range of chemical wastes that can be landfilled directly. The term 'inert' in APME/Boustead sources *does not* match the meaning of 'inert material' as understood by Swiss waste regulations for 'inert material landfills' (TVA 2000). The Swiss regulation means 'rock-like' material by 'inert' e.g. building and excavation material with low pollutant content. Boustead means by 'inert' that these materials are not volatile, corrosive or explosive and *'could, in principle, be sent to landfill sites without further treatment'* (Boustead 1999). This does not exclude possibly serious pollution potential. Alas, since no further characterisation of the wastes is given, the module 'average incineration residues to residual landfill' is used as a proxy for 'inert chemical' from APME/Boustead data.

This module is also used as a proxy for ashes of unknown composition resulting from the incineration of non-radioactive wastes in the production of uranium hexafluoride from uranium oxide (see chapter 4.12.3 'Uranium hexafluoride production' on page 48).

This module is *not* used for incineration residues of waste fractions incinerated in municipal or hazardous waste incinerators. Residues of those processes are calculated *waste-specifically* from the incinerated waste composition (cf. part II).

4.1.3 Average hazardous waste to incineration

This waste is average hazardous waste of unspecified origin, that is incinerated in Swiss hazardous waste incinerators. Waste composition for H₂O, O, H, C, S, N, P, B, Cl, F, Cd, Co, Cr, Cu, Hg, Ni, Pb, and Zn is adopted from (BZL 2000b:129, Suisselectra 1988). The remainder to 100% is taken to be silicon. A lower heating value of 17 MJ/kg is adopted from (Suisselectra 1988). the upper heating value is estimated with (Michel 1938).

4.1.4 Average hazardous waste to underground deposit

No waste-specific emission are currently inventoried for underground deposits. Accordingly, no waste composition must be specified. For generic, average waste to underground deposit deposition in a 200 liter steel drum without cement solidification is assumed.

4.1.5 Average wastewater to municipal treatment

For average wastewater treated in Swiss municipal wastewater treatment plants, input data is presented in part IV of this report for BOD, COD, DOC, TOC, sulfur, nitrogen, and phosphorus species. Data is chiefly based on the large BUWAL database on Swiss WWTPs (DBGS 1994). For these species changes in time are unlikely, as they are dominated by human excretions.

Content of trace elements in sewage was calculated in reverse from sewage sludge composition and transfer coefficients for those elements to sewage sludge. The latter are given in part IV. A conversion factor of 0.0813 kg digested sludge kg per m³ wastewater is applied. Traces of Cd, Co, Cr, Cu, Hg, Mo, Ni, Pb, and Zn in sewage sludge composition are taken from Herter et al. (2001:69) for weighted average 1999 Swiss sewage sludge¹¹. For other elements¹² sludge data from (Stäubli & Keller 1993,

¹¹ Herter et al. (2001:69) also give a time series 1975–1999 of the development of these traces in sludge. Major decreases in pollutant content generally occurred early and mid-80ties. Decreases have been smaller in recent years. For cobalt and →

von Rackeck 1993:20) is used. For direct concentrations of As, Mn, and Sn in sewage unreferenced values given in (Zimmermann et al. 1996) are used.

Uncertainty of average sewage composition

The uncertainty of the sewage composition is estimated with the Pedigree approach and given in Tab. 4.2¹³.

Tab. 4.2 GSD² values for average Swiss sewage composition for 2000

Pollutants in sewage	GSD ²	Pedigree scores	comment
BOD; COD, DOC, TOC	150%	(1,2,2,1,1,1)	basic uncertainty of 1.5; from BUWAL database on Swiss WWTP (DBGS 1994)
S, N, and P species	150%	(1,2,2,1,1,1)	basic uncertainty of 1.5; from BUWAL database on Swiss WWTP (DBGS 1994)
Cd, Co, Cr, Cu, Hg, Mo, Ni, Pb, and Zn	501%	(2,3,1,1,1,1)	basic uncertainty of 5; calculated in reverse from sludge using current sludge data (Herter et al. 2001) and WWTP model transfer coefficients
O, H, C, S, N, P, Cl, F, Si, Fe, Ca, Al, K, Mg, and Na	502%	(2,3,3,2,1,2)	basic uncertainty of 5; calculated in reverse from sludge using older sludge data (Stäubli & Keller 1993, von Rackeck 1993:20) and WWTP model transfer coefficients
As, Mn, and Sn	502%	(1,3,3,2,1,2)	basic uncertainty of 5; unreferenced values in (Zimmermann et al. 1996)

4.1.6 Solidifying cement to residual material landfill

Residual materials from MSWI (boiler ash, precipitator ash, scrubber sludge) are solidified and landfilled in a residual material landfill. For solidification cement and water are added to the raw residual material. This module heeds the hydrated cement in such solidified waste materials (cement + water). This inventory is only applicable to solidifying cement, *not to cement waste from concrete or building waste*, which is usually not disposed in residual landfills (cf. chapter 4.6.1 'Structural inorganic building materials' on page 38).

Though being regarded of minor importance, the trace heavy metals in hydrated cement are heeded. Cement trace metal composition is calculated from data on raw materials and transfer coefficients for cement kilns given in (BZL 2000a) and data for Portland cement in (GTK 1997) and (Renz et al. 1997:21). A mixture of 40% cement and 60% water (as O and H) is assumed for hydrated cement.

For solidified residual material landfill *from average municipal waste* the heavy metal input from cement to the landfilled residue is negligible for most metals. Relevant changes value occur for arsenic, cobalt and vanadium. This observation, however, depends of course on the composition of the incinerated waste.

molybdenum even *increases* of up to 25% have been observed since 1994. However, for lead and cadmium decreases are still large (-20 to -30% since 1994).

¹² Those elements are O, H, C, S, N, P, Cl, F, Si, Fe, Ca, Al, K, Mg, and Na.

¹³ Other wastewater compositions in this report are specific to a energy chain or a production process. Uncertainties of those wastewater compositions are defined in the respective inventory reports for these processes. Uncertainties of wastewater compositions are however reproduced in the appendix (7.2 'Synopsis of wastewater compositions' on page 99ff.).

4.2 Plastic waste material

For plastic materials (PE, PP, PS, PVC, PET, PU, plastics mixture) in municipal solid waste several literature sources were compiled in (Zimmermann et al. 1996)¹⁴. The following materials are based on these sources, except where noted. Missing heating values were estimated from (Michel 1938).

4.2.1 Polyethylene PE

The composition of polyethylene wastes is based on data from literature (cf. footnote 14 on page 32). Data gaps were filled with data from unspecified average plastics (for As, Se, Sr, Be, Tl, Ti, Al, Na) from the same literature sources. The composition is normalised to 100%.

4.2.2 Polypropylene PP

Polystyrene wastes are often not discerned from polyethylene wastes in waste analyses. For this reason a similar composition to polyethylene is used here but with a larger water content.

4.2.3 Polystyrene PS

The composition of polystyrene wastes is based on data from literature (cf. footnote 14 on page 32). Data gaps were filled with data from unspecified average plastics (for F, As, Ba, Co, Cr, Hg, Mn, Ni, Sb, Se, Sn, V, Sr, Be, Tl, Ti, Fe, Al, Na) from the same literature sources. The composition is normalised to 100%.

4.2.4 Polyvinylchloride PVC

The composition of polyvinylchloride wastes is based on data from literature (cf. footnote 14 on page 32). Data gaps were filled with data from unspecified average plastics (for Br, F, As, Ba, Co, Hg, Mn, Sb, Se, V, Sr, Be, Tl, Ti, Fe, Al, Na) from the same literature sources. The composition is normalised to 100%.

4.2.5 Polyethylene terephthalate PET

The composition of polyvinylchloride wastes is based on data from literature (cf. footnote 14 on page 32). Data gaps were filled with data from unspecified average plastics (for N, Br, As, Be, Sr, Ti, Tl, Al, Na) from the same literature sources. The composition is normalised to 100%.

4.2.6 Polyurethane PU

The composition of polyvinylchloride wastes is based on data from literature (cf. footnote 14 on page 32). Data gaps were filled with data from unspecified municipal plastics (for Br, F, As, Ba, Co, Cr, Hg, Mn, Ni, Sb, Se, V, Sr, Be, Tl, Ti, Fe, Al, Na) from the same literature sources. The composition is normalised to 100%.

4.2.7 Unspecified municipal plastics mixture

In waste analyses of municipal wastes, plastic might not be discerned according to polymer type as above. This composition is derived from such information from literature (cf. footnote 14 on page 32) representing an average plastics mixture in municipal waste.

4.2.8 Plastics from electronics products dismantling

Plastics from electronics products (or electronics plastics) is the fraction of electronic products which are not recycled after separation and sorting. Waste plastics are generally incinerated in MSWIs,

¹⁴ Barrage et al. 1995, Baumann et al. 1993, Bildingmaier 1990, Bilitewski et al. 1991, Brahms et al. 1989, BUWAL 1995b, Domalski et al. 1987, Feess-Dörr et al. 1991, Frankenhäuser et al. 1995, Franssen et al. 1990, Grünwald et al. 1988, Haber et al. 1990, Hamm et al. 1986, IFEU 1991, IFEU 1992, Kaiser 1975, Keilen et al. 1977, Knezevic 1988, Mark et al. 1994a, Mark et al. 1994b, Mark et al. 1995, Maystre et al. 1994, Paterna 1995, Rasp 1988, Salami et al. 1992, Stumpf 1994, Thalmann et al. 1982, Tillman 1991, Tuminski 1988, Wünschmann et al. 1995.

cement kilns or HWIs. In Switzerland, only a small amount of expanded polystyrene EPS was transferred to recycling from electronics dismantling (SENS 2001).

Electronic wastes are a very heterogeneous class of wastes. If complete electronic appliances are meant by this waste category, they can contain metal or plastic casings, bulk metal parts from voltage transformers, glass from cathode ray tubes (CRT), printed wiring boards (PWB) with electronic components and tin solder, cables (wire and insulation), batteries and accumulators.

Proper disposal of electronics products is take-back, dismantling and recycling. In Switzerland electronic consumer goods ('brown goods' like audio- & video equipment, TVs, personal computers) can be returned to the dealer without charge since 2002 (Monteil 2000, ERZ 2002). In Switzerland, 38'800 tons of old electronic goods¹⁵ were collected in 2000. Estimated from sales figures, the expected yearly volume of old electrical and electronic goods is 100'000 tons per year. The collection gap is 61%. The Swiss electronic disposal foundation S.EN.S assumes that a large part of the not collected appliances are sold and re-used as second-hand products in Switzerland or abroad, and that there are still considerable quantities disposed of through cheaper and uncontrolled channels (SENS 2001).

For the generic disposal of electronic equipment following assumptions can be made. Large metal and plastic parts can be separated. Metal parts are recycled, plastic parts (casings, wire insulation) are incinerated in MSWIs. Printed wiring boards PWB are separated and recycled in copper smelters¹⁶. Thus only plastic parts are not recycled. Hence only disposal of electronics plastic is inventoried here.

For dismantling an electricity consumption of 20 to 110 kWh per ton of electronic input is given by Behrendt et al. (1998). The company Electrorecycling reports 200 to 220 kWh/t¹⁷. This electricity consumption is *not* inventoried in the inventory for plastic incineration, as the latter relates to the plastic *delivered to the incinerator plant* and not any upstream processes. However, these energy demands can be helpful to inventory complete electronics disposal.

A disposal of electronic control units¹⁸ for chemicals production is inventoried in (Althaus et al. 2003a).

The detailed compositions of two types of electronics plastics (from consumer and industrial electronic waste) is given in Table 2 of (APME 1997)¹⁹. These compositions are complemented with data for Co, Se, Be, Sr, and Tl from average plastics mixture in municipal waste (cf. chapter 4.2.7 'Unspecified municipal plastics mixture' on page 32). Incineration in a MSWI is inventoried for both plastic types.

4.2.9 Polyvinylfluoride to municipal waste incineration

Polyvinylfluoride (PVF, Tedlar) is used for example as laminated film in the production of solar panels (Jungbluth 2003a). No composition data of waste PVF is available from literature. PVF with the chemical formula $(CH_2CHF)_n$ is analogous to the more common polyvinylchloride (PVC,

¹⁵ Electronic appliances collected comprise 51% small electrical and electronic goods, 25% large electrical goods, 18% refrigeration appliances, 6% var. dismantled parts.

¹⁶ Copper smelters produce black copper from copper wastes of varying purity. The PWB plastic part is used as a reducing fuel. A dust rich in zinc, tin and lead is obtained as additional output which is recycled in zinc furnaces (Waelz kiln). Black copper is further purified in electrochemical processing, which yields nickel sulfate and precious metals as by-products. The Waelz kiln produces purified zinc and an a lead oxide dust that is further processed in lead furnaces.

¹⁷ Personal communication by Dr. Georg Fröhlich, Electrorecycling, Goslar, Germany, of August 13, 2002.

¹⁸ With a composition of 46% steel (housing), 32% plastics, 14% printed wiring boards and 8% cables.

¹⁹ For Copper, Antimony and lower heating value the figures from Table 4 of (APME 1997) for electronics plastics mixtures are adopted.

$(\text{CH}_2\text{CHCl})_n$)²⁰. The theoretical composition of pure PVF is 52.2 w% carbon, 6.5 w% hydrogen and 41.3 w% fluorine. In reality, the polymer is mixed with additives, plasticizers etc. and the theoretical composition differs from the one encountered in waste. Based on the differences between *theoretical PVC* composition and *observed PVC* composition, a waste-PVF composition of 57.01 w% carbon, 7.11 w% hydrogen and 32.96 w% fluorine is estimated. The remaining 2.9 w% are complemented with traces and water as observed in PVC waste²¹. Heating values are estimated from (Michel 1938).

4.2.10 Rubber to municipal waste incineration

Rubber wastes are e.g. car tyres and conveyor belts. These wastes also contain embedded steel parts, which behave dissimilar in incineration and must be inventoried separately (i.e. steel to municipal incineration). Rubber wastes can also be used as alternative fuel in cement kilns. Rubber incineration in cement kilns and the issue of allocation is discussed in chapter 4.27.7 'Automobile tyres disposal' on page 61. For rubber incineration in *municipal waste incineration* plants the full incineration burdens are allocated to the waste producer.

Rubber composition data for S, Cl, Cd, Hg, Pb and lower heating value are available from (BZL 2000b:81); a value for Zn for tyre rubber is adapted from (Neumann 1980, cited in Frischknecht et al. 1996:B.10). The remaining composition is complemented with data for carbon and hydrogen from theoretical rubber composition (isoprene monomers made up of C_4H_6 units resulting in 88.9% C and 11.1% H for pure rubber). Water content is assumed to be 0%. Upper heating value is estimated from (Michel 1938).

4.3 Biomass waste

See also disposal of wood ash is chapter 4.16.1 'Ash from natural wood incineration' on page 49.

4.3.1 Average paper

For average paper in MSW a mixture of 10% newspaper and 90% packaging paper is assumed. Compositions see below.

4.3.2 Packaging paper

A composition of average paper and different paper fractions in MSW (without newspapers) is adopted from literature (BUWAL 1998b:426, Wünschmann et al. 1995, Maystre et al. 1994, Baumann et al. 1993, Franke et al. 1991, Franke et al. 1992, BUWAL 1990, Tillman 1991, Domalski et al. 1987, Knezevic 1988, Grünewald et al. 1988, Tuminski 1988, Hamm et al. 1986, Thalmann et al. 1982). The composition is normalised to 100%.

4.3.3 Newspaper

A composition of newspapers in MSW is adopted from literature (BUWAL 1998b:426, Baumann et al. 1993, Grünewald et al. 1988, Hamm 1986, Tillmann 1991, Mayestre et al. 1994, Domalski et al. 1987, Tuminski 1988). The composition is normalised to 100%.

4.3.4 Packaging cardboard

A composition of packaging cardboard in MSW is adopted from literature (BUWAL 1998b:426, Wünschmann et al. 1995, Baumann et al. 1993, Tillmann 1991, Mayestre et al. 1994, Knezevic 1988,

²⁰ Other fluoridated polymers are for example: poly(vinylidene fluoride) or PVDF, which is the *difluoridated* ethylene $(\text{CH}_2\text{CF}_2)_n$. The familiar non-stick coating Teflon® (polytetrafluoroethylene or PTFE) is the *tetra-* or *perfluoridated* ethylene $(\text{CF}_2\text{CF}_2)_n$. The considered PVF, as the *monofluoridated* ethylene, contains the lowest fluorine content in all these.

²¹ This procedure yields no value for chlorine, although it is likely to be present in waste PVF.

Thalmann et al. 1982, Hamm et al. 1986, Domalski et al. 1987). The composition is normalised to 100%.

4.3.5 Untreated wood

A composition of untreated, natural wood is adopted from several literature sources (Dones et al. 2001, EMPA 2000, Nussbaumer 2000, Skreiberg 1997, BUWAL 1996, Frischknecht et al. 1996:IX.12, Mayestre et al. 1994). Values of "<x" were interpreted as "x/2". The composition is normalised to 100%. Upper heating value is estimated with (Michel 1938).

More information on disposal of "waste wood" in general – including treated wood – can be found in part V of this report in chapter 3.6 "Remark on current wood disposal practices".

4.3.6 Chrome preserved wood pole

Electricity and telephone poles are preserved with chrome containing preservatives. The types of preservatives used are chromium-copper-fluoride salts CKF, chromium-copper-boron salts CKB, or chromium-copper-oxides CCO (also contains boron). Minimal initial application doses for electricity poles are 10 – 50 kg preservatives per m³ wood (Frischknecht et al. 1996:C.7). Additional impregnation during use increases the total applied preservative mass.

According to the guidelines of the Swiss Electricity Supply Association (SESA or VSE) the *minimal* applied preservative mass for electricity poles in Switzerland is 12 kg/m³ wood and the allowed preservatives are CKB or CCO (VSE 1997). According to (Künniger & Richter 1995:15) the actually applied doses are 16 kg/m³ for CKB. The mass for additional impregnation is 1.8 kg/m³. A total applied preservative mass of **15.7 kg/m³** during the whole lifecycle of the pole of 30 years is used in this study.

Preservative composition can be derived from composition data of the federally permitted wood preservatives in Switzerland. In (BUWAL 2002) the composition of the currently permitted Swiss CKB or CCO wood preservatives are given and shown in Tab. 4.3.

Tab. 4.3 Currently permitted Swiss CKB or CCO wood preservatives (BUWAL 2002)

Agent	Impralit CCO liquid (F,*)	Pentol CCO liquid (F,*)	Impralit CKB liquid (F,*)	Arbezol CKB liquid (F)	Pentol CKB (F,*)	Pentol CKB-P (F,*)	Adolit CKB-P (F,*)
Boric acid H ₃ BO ₄	3.8%	3.8%	12%	12%	26%	25%	25%
Disodium octaborate Na ₂ B ₈ O ₁₃			10%	10%			
Chromium-VI oxide CrO ₃	30%	30%	25%	25%		5.5%	5.5%
Copper-II oxide CuO	12%	12%	10%	10%		1%	
Sodium dichromate Na ₂ Cr ₂ O ₇					38%	30%	30%
Copper sulfate CuSO ₄					34%	31%	31%
% agent	45.8%	46%	57%	57%	98%	92.5%	91.5%

F Toxic for fish

* Applicable with restrictions to wood for agricultural barns, milk and cattle feed storage.

From Tab. 4.3 a mean composition of the active agents in the available CKB or CCO wood preservatives is calculated (arithmetic mean). The mean elemental preservative composition is given in Tab. 4.4.

Tab. 4.4 Mean elemental composition of Swiss CKB/CCO wood preservatives, leaching rates during use, resulting waste wood trace concentrations and measured concentrations in older poles from Germany .

	Mean preservative composition w-%	Leaching rate during use (Künniger & Richter 1995:18)	Resulting waste wood traces mg/kg	Measured waste wood poles Germany (Künniger & Richter 1995:186)
Cr	24.79%	60%	2290	2747
Cu	13.61%	75%	785	1726
B	3.29%	95%	57	190
S	3.56%	80% ¹	164	–
Na	6.96%	80% ¹	321	–

¹ adopted from fluorine (soluble).

During use the preservatives leach out (to ground). This leaching is inventoried in (Frischknecht & Faist 2003). Leaching rates are given in (Künniger & Richter 1995:18) for different elements and reproduced in Tab. 4.4. From the applied preservative mass per m³ wood, the preservative composition and the leaching rates the *remaining* preservative traces in waste wood poles can be calculated²². The calculated preservative traces are smaller than the traces in measured in older waste wood pole samples from Germany (see Tab. 4.4). This might reflect the decreasing preservative application or higher application rates in Germany.

Other composition data is taken from untreated natural wood (chapter 4.3.5 'Untreated wood' on page 35). The composition is normalised to 100%. Disposal in a municipal waste incineration is assumed.

4.3.7 Chrome preserved building wood

Building wood for outdoor application can be preserved. Inorganic or organic preservatives can be used. Inorganic preservatives can contain chrome, copper, fluoride or boron. Organic preservatives are e.g. creosote (tar oil from hard coal), permethrine, or various halo-organic compounds. Since the waste disposal models in this project heed only *elemental compositions* and not individual substances, the *organic* treatments have little effect on the waste wood composition (C, H, O, Cl etc.).

Inorganic preservatives however can introduce significant levels of heavy metals and non-metals. Remaining preservative traces in building wood are estimated similar to the calculations performed for electricity poles in chapter 4.3.6 'Chrome preserved wood pole'. For weather exposed wood logs 4 to 7.5 kg/m³ of Cr-containing preservatives are used, for timber 6 to 11.25 kg/m³ (data from Werner et al. 2003, based on Künniger & Richter 2000). For this study an average application of **7 kg/m³** is used. The preservative composition given in Tab. 4.4 for CKB/COO preservatives is used. Leaching during the use phase reduces the levels of the initially applied preservative components. Leaching rates are given in Tab. 4.4. A wood density of 650 kg/m³ is used. Other traces are taken from untreated natural wood (chapter 4.3.5 'Untreated wood' on page 35). The composition is normalised to 100%. Disposal in a municipal waste incineration is assumed.

4.3.8 Textiles to municipal waste incineration

See chapter 4.11.2 'Soiled textiles to municipal waste incineration' on page 46.

²² A wood density of 650 kg/m³ is used based on information of Imprägnierwerk AG Willisau, Switzerland, <http://www.impraegnierwerk.ch/produkt1.332.htm>

4.4 Metal and glass waste

Usually metal wastes are recycled. This is especially the case for bulk metal parts. Metals in composite materials and small parts can however enter disposal processes.

See also electrical wiring (building wastes, see chapter 4.6.11 'Electric wiring' on page 41) and metals in car shredder residue (see chapter 4.27.6 'Automotive shredder residue (ASR) to incineration' on page 60).

4.4.1 Steel

Steel in MSW is usually steel from small parts, like screws, paperclips, or not recycled tin sheet packaging. Large bulk metal parts are not suitable for MSWIs due to their lack of heating value and inertia in combustion. Larger metal parts can enter the MSW stream for example in furniture. Steel is partially removed from the incinerator bottom ash for recycling (cf. part II).

A composition of iron scrap from MSWI (S, P, Cr, Cu, Mn, Ni, Sb, Sn, V, Si, Fe, and Al) is adopted from (Doka 2002:19&27). This composition is based on measurements in (Traber 2001:63, Hoeffken et al. 1988, ²³). Traces of Cd, Pb, Zn and Cl as found in MSW iron cans are adopted from (Mayestre et al. 1994). The remainder to 100% is taken to be oxygen.

Heating value of steel in incineration

According to Lichtensteiger (1995) approximately one third of the metallic iron in MSW is oxidised in a grate MSWI to magnetite (Fe_3O_4) and wüstite ($\text{Fe}_{0.947}\text{O}$). Oxidation is an exothermic reaction and heat is liberated from the steel waste. The heat of formation (for $\text{Fe} + x\text{O} = \text{FeO}_x$) gives an approximate figure of the produced heat energy. The heat of formation of magnetite is -6.68 MJ/kg Fe, for wüstite -5.03 MJ/kg Fe (CRC 1985:D-70). Thus, per kilogram iron in MSWI approximately 2 MJ/kg is produced, which could be used as a proxy for the upper heating value of steel in MSWI. This heating value would lead to some waste heat emissions. No net energy is generated as the energy consumption to *process* the waste is larger than the gross energy that is *produced* from 2 MJ/kg waste.

However, in ecoinvent the storage or liberation of chemical energy is not inventoried as such (Frischknecht et al. 2003a). For example, in iron smelters a part of the necessary energy is in reality stored as chemical energy in the metallic iron. In ecoinvent the *full* amount of energy in the utilised smelter fuels is inventoried as waste heat. The inventoried waste heat is therefore larger than the actual waste heat from the smelter²⁴. Accordingly, the chemical energy released during oxidation of steel products in a MSWI (which is chemically the reversal of the smelting reaction), as described above, should *not* be heeded in the inventory to avoid double counting of waste heat. This influences the net energy consumption and the waste heat inventory of the incinerated steel waste. Since in this study no emissions are allocated to the energy production from a waste, only the waste heat balance is affected by this methodological choice. The inventoried waste heat is smaller than the actual waste heat from steel waste. To avoid double counting a upper heating value of zero is inventoried for steel in MSWI.

4.4.2 Aluminium

Aluminium in MSW are usually thin layers in laminated materials or non-recycled beverage and food cans. A partial composition analysis of aluminium in MSW (Cl, Cd, Cu, Pb, Zn) is available from (Mayestre et al. 1994). The remainder is assumed to be Al. Thin layers of aluminium can be completely oxidised, volatilised by entrainment and transferred to the fly ash. This is not heeded here and the material is assumed to be bulky and inert (100% to bottom ash). For a more volatile aluminium material see chapter 4.27.6 'Automotive shredder residue (ASR) to incineration' on page 60.

²³ Personal communication with Christoph Zeltner, Stahlwerk Gerlafingen, Switzerland, November 14, 2001

²⁴ The rationale behind this choice is that this way the waste heat reflects the sum total of energy resources consumed.

Heating value of aluminium in incineration

Aluminium is partly oxidised in a MSWI. Before incineration approximately 40% of the aluminium in waste is metallic (see contributions from inert metals, volatile metals, laminated materials, and electronic goods to average municipal solid waste, appendix of part II). After incineration approximately 34% of the aluminium in waste is metallic (Zeltner & Lichtensteiger 2002:79). So on average 15%²⁵ of the metallic aluminium is oxidised during incineration. The heat of full oxidation is -30.83 MJ per kilogram aluminium (value for α -corundum from CRC 1985:D-51). So per kilogram aluminium in MSWI an upper heating value of 4.62 MJ/kg could be inventoried due to oxidation. As explained in chapter 4.4.1 'Steel' on page 37 the heating value of metals is not heeded in ecoinvent. To avoid a double count with waste heat from aluminium production an upper heating value of zero is inventoried for aluminium in MSWI.

After incineration approximately 85% of aluminium metal waste is still metallic and is transferred to the slag. In the slag compartment the aluminium can be oxidised and can form hydrogen gas (H_2) above a pH of 10 (Lichtensteiger 2002:34, Sabbas 2002:21), i.e. in slag storage and the early years after deposition. This poses some risk of deflagration (German 'Verpuffung'), i.e. vigorous combustion of H_2 with oxygen to form H_2O , which is observed in the field (Söllner 1999). Neither the heat of oxidation in the landfill (double count with waste heat from aluminium production) nor the possible emission of hydrogen gas (rather an occupational hazard to landfill workers than an environmental burden) is inventoried in ecoinvent.

4.4.3 Tin sheet

Tin sheet cans consist of tin coated steel containers. If not recycled these containers for food usually are disposed via municipal solid waste. Tin sheet contains 0.39% tin (BUWAL 1998b:427). There is a certain volatility of tin during incineration. 50% of the tin is recorded as being volatile. The remaining composition is adopted from (Mayestre et al. 1994). The remainder to 100% is assumed to be iron.

4.4.4 Glass packaging

For glass disposed via MSWI is completely transferred to bottom ash and then landfilled in a slag compartment. Emissions from slag compartments are calculated. A glass composition found in MSW is taken from (Mayestre et al. 1994, Domalski et al. 1987, Bilitewski et al. 1991, IFEU 1992). The composition is normalised to 100%.

4.5 Inert materials to inert landfill

Inert materials are materials usually landfilled in an inert material landfill (inorganic building materials, excavation material, similar materials). No landfill emissions are inventoried in the inert material landfill model and accordingly no waste compositions must be known to inventory such waste disposal.

4.6 Building materials

In this chapter the compositions of building material waste fractions are described, which are generated e.g. from *dismantling* of buildings (End-of-Life). These are not the wastes from building material *production* which are described in chapter 4.18 'Building materials production' on page 50.

4.6.1 Structural inorganic building materials

Inert materials like concrete and bricks etc. lead to no emissions in inert material landfills. After sorting in a building waste sorting plant, inert materials like cement from concrete can also be transferred to sanitary landfills as part of a landfilled fine fraction²⁶. For inert material to sanitary

²⁵ $(40\%-34\%)/40\% = 15\%$.

²⁶ Fine fractions are usually not suitable for inert material landfills due to their pollutant content (cf. part V).

landfill a composition from hydrated cement is adopted (cf. chapter 4.1.6 'Solidifying cement to residual material landfill' on page 31).

4.6.2 Gypsum products

Gypsum in building products is assumed to be 100% natural gypsum (calcium sulfate dihydrate $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$). Desulphurisation gypsum which can be used for gypsum products is not considered. Composition of natural gypsum including traces is taken from (Renz et al. 1997:23ff., Lorenz & Gwosdz:61f.). This composition is used for plaster boards, gypsum plaster, plaster-cardboard sandwich boards and reinforced plaster boards.

Gypsum can enter a sanitary landfill as part of a fine fraction from building material sorting plants. This issue of gypsum disposal *in sanitary landfill* has led to some debate after the publication of (Doka 2000). The characteristics of gypsum material in landfills including degradability is discussed in the appendix of part V on building material disposal.

4.6.3 Insulation materials

Mineral wool is treated as inert material with no emission potential.

Polystyrene isolation (EPS) is flame-retarded with hexa-bromo-cyclo-dodecan and thus contains 0.67 w% bromine (Weibel & Stritz 1995:14.2). The EPS matrix composition is derived from polystyrene waste (PS) in municipal waste (see chapter 4.2 'Plastic waste material' on page 32).

4.6.4 Glass and steel

Pane glass (windows) is treated as inert building material. Float glass (uncoated) contains 10.6 ppm tin from the float bed (Frischknecht et al. 1996:A.58). Coated glass can additionally contain metals from sputtering as displayed in Tab. 4.5. As the inert material landfill model does not heed any emissions, these traces are neglected. For steel see municipal waste fractions (cf. chapter 4.4.1 'Steel' on page 37).

Tab. 4.5 Metal traces in coated pane glass.

	Metals from coating (Sprecher 1993) g/m ²	Traces from coating for 2 mm glass pane ¹ ppm
Indium	<0.1	<20
Tin	0.42	84
Silver	0.2	40
Nickel	0.15	30
Chrome	<0.1	<20

¹ Glass density 2500 kg/m³

4.6.5 Wood materials

Untreated wood and treated wood (chrome treated) see municipal waste fractions (see chapter 4.3.5 'Untreated wood' on page 35f.).

Fibre board contains 5 – 10% resin as glue. Resins types used are urea-formaldehyde resins, melamine resins or phenolic resins (Rentz et al. 1997:36). Fibre board composition is derived from 92.5 w% untreated wood and 7.5 w% glue. Glue composition is approximated with polyurethane (PU) to represent nitrogen content of urea-formaldehyde resin. PU is heeded as PU as measured in municipal waste (see chapter 4.2 'Plastic waste material' on page 32).

4.6.6 Paints

Emulsion paint

Wet emulsion paint contains 22.5% inert fillers, 22.5% titanium dioxide pigments and 34% ethylene vinylacetate vinylchloride terpolymer (von Arx 1995:86). The dry composition is inventoried. Metal traces are taken from (Bare et al. 1995:31f).

Unspecified paint

The trace composition of various waste paints, veneers etc. is given in (BZL 2000b:128, Bare et al. 1995:31f). These compositions are complemented with the theoretical composition of the matrix resin, which is taken to be phthalic acid glyceride. this composition is used for unspecified paints.

4.6.7 Bulk plastic materials

Composition of bulk building plastic materials (PE, PP, PVC) are covered with compositions from municipal waste fractions (see chapter 4.2 'Plastic waste material' on page 32).

4.6.8 Structural composite materials

This is a composite of burnable and inert material. Cement-fibre slab composition is calculated as 36% untreated wood (burnable) plus 64% cement (inert). As cement composition the values for solidifying cement are taken (chapter 4.1.6 'Solidifying cement to residual material landfill' on page 31).

4.6.9 Plaster

Wet plastic plaster contains 80 w% inert fillers and 4 w% of a ethylene vinylacetate vinylchloride terpolymer (von Arx 1995:87). The dry composition is inventoried (95.2% fillers, 4.7% terpolymer). Fillers are inventoried as inert material (lime). Theoretical composition of ethylene vinylacetate vinylchloride terpolymer is used for C, H, Cl. Traces in waste plastic plaster from (Bare et al. 1995:31) are adopted (Sn, Pb, Co, Zn, Cr, Mo, Cd).

4.6.10 Plastic sheets

PVC sealing sheet

PVC sealing sheets contain 1.46% lead from stabilisers (von Arx 1995:111&113). It is assumed that half of these stabilisers are sulfates, and the sulfur content is 0.1%. The composition is 53.9% PVC, 33.4% plasticizers, 1.7% glass fabric and 11% fillers (Weibel & Stritz 1995:13.3). PVC is heeded as PVC as measured in municipal waste (see chapter 4.2 'Plastic waste material' on page 32). Plasticizers are heeded as pure dioctyl phtalate (73.8% carbon, 9.7% hydrogen, 16.4% oxygen). Glass fabric is heeded as pure SiO₂. Fillers are heeded as natural lime (CaCO₃). Traces of Zn, Cr, Ni, Pb, As, Cd, Hg in natural lime are taken from (Graf 1994:8).

PE sealing sheet

PE sealing sheets contain 0.5% zinc from stabilisers (Weibel & Stritz 1995:12.3). The composition is 85% PE, 13.5% fillers and 1.2% glass fabric. PE is heeded as PE as measured in municipal waste (see chapter 4.2 'Plastic waste material' on page 32). Fillers are heeded as natural CaCO₃. Glass fabric is heeded as pure SiO₂.

Vapour barrier, flame-retarded

The vapour barrier is made of PE and contains 6.7% bromine from flame-retardants (hexa-bromo-cyclo-dodecan or hexa-bromo-diphenyl-ether) (Doka 2000:144). PE is heeded as PE as measured in municipal waste (see chapter 4.2 'Plastic waste material' on page 32).

Bitumen sheet

The bitumen sheet is made of 63% bitumen, 24% surface spread and filler, 6% carrier fleece and 7% elastomer (Weibel & Stritz 1995:9.3). Bitumen composition is used according to chapter 4.27.3 'Bitumen to sanitary landfill' on page 60. Surface spread and filler are asumed to be pure SiO₂. The

carrier fleece is assumed to be PE (Rentz et al. 1997:44). Elastomer is assumed to be PS. PE and PS are heeded as measured in municipal waste fractions (see chapter 4.2 'Plastic waste material' on page 32).

4.6.11 Electric wiring

Electric wiring is usually recycled for its copper content. If no recycling occurs, wires could (but should not) be incinerated. This is then a composite of burnable and inert material. Wiring composition is calculated as 61.5% wire insulation (burnable) plus 38.5% copper (inert).

Wire insulation is calculated as 66.3% PVC, 31.2% PE and contains 2.5% copper (Gewiese 1998:95). For PVC and PE the compositions encountered in municipal waste are adopted (see chapter 4.2 'Plastic waste material' on page 32). Copper wire is assumed to the pure copper.

4.7 Specific wastes by process

Wastes generated in particular parts of the ecoinvent 2000 project (e.g. wastes from aluminium production) are fully referenced in the corresponding reports. In this report, the corresponding reports are indicated, information regarding disposal options and supplemental data is given and the waste composition is repeated (in the appendix chapter 0 on page 73ff.). For full information on the waste-generating process, please consult the indicated ecoinvent 2000 project reports.

If you have difficulties in locating a module merely by its name, look also at the text in the field 'synonyms', which gives additional information. For reasons of a 80 character limitations combined with a predefined module name structure the waste names can in some cases become unclear.

4.8 Waste from furnaces and power plants

4.8.1 Wastewater from energy infrastructure production

During the production of furnaces, gas turbines, motors etc. for energy infrastructure a wastewater is produced from the metal working processes (Dones et al. 2003). No wastewater composition is available. The wastewater from pig iron production is used as a proxy (cf. chapter 4.20.1 'Wastes from pig iron production' on page 52).

4.8.2 Residue from cooling tower to sanitary landfill

In cooling towers of power plants a residue is collected which contains tensides, metal ions, surfactants from steam generation, biomass like pollen or leaves, corrosion products of the tower, lime and iron hydroxides from water preparation. A waste composition is adopted from (VGB 1990). Due to the high carbon content a disposal in a sanitary landfill is assumed here.

4.8.3 Catalytic converter for NO_x reduction to underground deposit

Converter for selective catalytic reduction (SCR) are used in stationary industrial flue gas purification. These catalysators are partially recycled for thier metal content. Used catalytic converters are also disposed in underground deposits. For disposal an inventory module 'catalytic converter NO_x reduction to underground deposit' is created (a similar module 'catalytic converter for cars to underground deposit' shall be used for mobile automotive converters). Deposition in a 200 liter steel drum without cement solidification is assumed.

4.9 Coal energy

4.9.1 Hard coal tailings to opencast backfill

Hard coal tailings (German 'Bergehalden') are mineral remains from the hard coal extraction step (Röder 2003). Part of these tailings are disposed as backfill of surface mining areas. The physical characteristics of the material are changed by the hard coal extraction process. The disposed material is much more prone to weathering than the material in its original shape and surroundings. The surface-to-volume ratio is increased and the tailings are in contact with precipitation water. Oxidation processes convert formerly stable sulfides into unstable and reactive materials. The oxidation of

sulfide forms acids, which use up the acid neutralising capacity of the material at alkaline or near-neutral pH. An acidic pH in turn increases leaching of heavy metals from the tailings. Oxidation of tailings can be delayed by very fine sized flotation tailings (German 'Flotationsberge'), and lasts in the range of several years up to decades (Wiggering & Kerth 1991).

Patrick Hofstetter devised in (Frischknecht et al. 1996:VI.42f) a coarse generic scheme to calculate emissions from hard coal tailings. After prolonged discussions, this scheme was deemed too rough by the ecoinvent administrators. A more detailed tailings model was however not conceived (see chapter 'Outlook' in part III on landfills). The result is, that hard coal tailings are currently not considered in ecoinvent 2000.

4.9.2 Hard coal ash to residual landfill

In hard coal power plants boiler and filter ash is collected (Röder 2003). Hard coal ash can be pozzolanic (binds like cement) and in some countries ashes are partially recycled. These countries are Germany, France, Netherlands, Belgium, Austria, Italy, Portugal, Spain, Poland, Czech Republic, Croatia, Slovak Republic. The remainder is landfilled. Detailed, country-specific ash compositions were calculated in Röder (2003) from country-specific coal compositions using technology-dependent transfer coefficients and enrichment factors. The remainder to 100 w-% is assumed to be oxygen. Landfilling in a residual material landfill type without cement solidification of the ash is assumed. The ashes are landfilled in the respective countries. Since no landfill models were available for those countries, the model for Swiss residual landfills was used as a proxy. The location of the disposal process module is set to the respective country.

4.9.3 Lignite ash to opencast backfill

Lignite ash has often high content of sulfur (as CaSO_4), high content of free lime, and large variations in composition. Unlike hard coal ash, lignite ash is therefore not suitable for recycling in cement production. In some countries part of the ashes from lignite incineration are disposed as backfill in opencast lignite mines (Röder 2003). These countries are Austria, Germany, France, Poland, Czech Republic, Slovak Republic, Hungary, Macedonia, Slovenia, Yugoslavia, Bosnia-Herzegovina, Spain, and Greece. Detailed, country-specific ash compositions were calculated in Röder (2003) from country-specific lignite compositions using technology-dependent transfer coefficients and enrichment factors.

The backfill is not regarded as a controlled landfill process here. **The data module only contains direct emissions.** Land use, renaturation and energy consumption for the mine backfill process are inventoried in the lignite chain. The transfer coefficients for the ash backfill emissions, however, are estimated based on emission data from residual material landfills (see part III). The ashes are not solidified with cement. As the ash is landfilled together with tailings, the carbonate buffer of the ashes is 'diluted' with tailings material. The carbonate buffer is depleted by washout from the landfill. The duration of the carbonate buffer phase determines the emission dynamics from the landfill. The average carbonate buffer is approximately 9.6 grams of carbonate per kilogram of deposited material (ash+tailings). The washout carbonate concentration is estimated to be 32 mg/l^{27} . The effective leachate volume V_{eff} is 0.0245 litres per kilogram and year²⁸. An average calcite buffer phase duration of 12'260 years is calculated²⁹. After washout of the carbonate buffer, the emissions change for most

²⁷ Based on average leaching concentrations of Ca and Mg measured in residual material landfills (cf. part III).

²⁸ Calculations see part III. Necessary parameters are mostly adopted from residual material landfills: Rain infiltration rate $I = 500 \text{ mm/m}^2\text{a}$; share of preferential flow in leachate output $w\% = 22\%$; landfill height $h = 10 \text{ m}$ residual; waste density $\delta = 1600 \text{ kg/m}^3$; water content in waste $v\% = 20 \text{ w}\%$; residence time of preferentially flown water $T_p = 0.17\text{a}$.

²⁹ $T_e = 9600 \text{ mg CO}_3/\text{kg waste} / (32 \text{ mg CO}_3/\text{l} * 0.0245/\text{kg*a}) = 12'260 \text{ years}$

pollutants (cf. residual landfill model description in part III). The resulting transfer coefficients for the lignite ash backfill are displayed in Tab. 4.6.

The backfill is assumed to be not artificially drained. All emissions occur to groundwater. The location of the process data module is set to the respective country.

Tab. 4.6 Calculated transfer coefficients for lignite ash backfill

Element	TK short-term (0-100a)		TK long-term (cumulated)		
	Mean value	Uncertainty GSD	Minimal value ¹	Mean value	Maximal value
Time frame	100 a		t _e = 12'260 a	t _g = 60'000 a	inf. time
	kg/kg		kg/kg	kg/kg	kg/kg
O	0.0003785	258.4%	0.0464	0.2271	1
H	0.0003785	258.4%	0.0464	0.2271	1
C	0.002707	223.0%	0.3319	1	1
S	0.2692	140.2%	1	1	1
N	0.1888	130.0%	1	1	1
P	0.0009363	242.1%	0.1148	0.5618	1
B	0.007835	187.3%	0.6188	0.6328	1
Cl	0.5705	122.5%	1	1	1
Br	0.9831	103.9%	1	1	1
F	0.1305	152.5%	1	1	1
I	1	100.0%	1	1	1
Ag	0.0001612	273.7%	0.01977	1	1
As	1	100.0%	1	1	1
Ba	0.0000346	301.4%	0.004242	1	1
Cd	0.00002844	305.0%	0.003486	1	1
Co	0.0007161	246.9%	0.08779	1	1
Cr	0.1508	150.6%	0.25	0.25	0.25
Cu	0.0001612	273.7%	0.01977	1	1
Hg	0.0001977	270.1%	0.02424	1	1
Mn	0.0000346	301.4%	0.004242	1	1
Mo	1	100.1%	1	1	1
Ni	0.001516	233.4%	0.1859	1	1
Pb	0.00002173	309.8%	0.002664	1	1
Sb	0.3857	118.8%	1	1	1
Se	0.3857	118.8%	1	1	1
Sn	0.00007937	286.5%	0.009731	1	1
V	0.006146	208.2%	0.5304	0.544	1
Zn	0.00005134	294.3%	0.006294	1	1
Be	0.001516	233.4%	0.1859	1	1
Sc	0.2826	135.2%	1	1	1
Sr	0.001516	233.4%	0.1859	1	1
Ti	0.001243	237.0%	0.1524	1	1
Tl	0.001516	233.4%	0.1859	1	1
W	0.3857	118.8%	1	1	1
Si	0.005643	209.8%	0.6918	0.7187	1
Fe	8.367E-06	310.4%	0.001026	0.6683	1
Ca	0.0003785	258.4%	0.0464	1	1
Al	0.001243	237.0%	0.1524	1	1
K	0.5643	122.8%	1	1	1
Mg	0.0004759	254.3%	0.05835	1	1
Na	0.6928	117.6%	1	1	1

¹ The minimal value is not used in this study (see Part III (landfills) in section 'Uncertainty of transfer coefficients and emissions' on page 36).

4.9.4 Coal ashes from small scale coal furnaces

Ashes from small scale coal furnaces have different fate than ashes from coal power plants. Such ashes are created in a household environment. Disposal with household garbage and treatment in municipal solid waste facilities (incineration or landfilling) is common. The composition of hard coal and lignite ashes is derived from coal composition (for a nominal 10% ash content) and the remainder of transfer coefficients to air ($100\% - TK_{\text{air}}$). An ash generation of 0.1 kg per kg burnt coal (hard coal and lignite) is used to derive a composition for kg element per kg ash. As unburned coal 10% carbon are assumed. The remainder to 100% is assumed to be oxygen.

Disposal in municipal solid waste incinerators *and* sanitary landfills is inventoried (In Europe approximately 82% of all municipal waste is landfilled, cf. Fig. 3.7 on page 23). In sanitary landfills a decomposition of 5% in the first 100 years is estimated (mainly wash-off).

4.10 Oil energy

4.10.1 Oil drilling waste

Drilling waste derives from drilling operations in oil and gas production. Waste composition is adopted from (solids in Tab. 6.24 of Frischknecht et al. 1996:IV.51) and complemented with 0.5% hydrogen, 10% oxygen and a water content of 71.5%. Disposal occurs either in landfills or landfarming (application on non-agricultural areas). For disposal in landfills a residual material landfill is chosen.

4.10.2 Refinery sludge

Refinery sludge is a production residue of oil refineries (Jungbluth 2003b). Waste composition is adopted from (Frischknecht et al. 1996:IV.294) with a water content of 89.5%. Refinery sludge is either incinerated in hazardous waste incinerators, or landfilled. For landfilling a sanitary landfill is assumed³⁰. Spreading on non-agricultural areas (landfarming) is not inventoried anymore (Jungbluth 2003b). A data module 'refinery sludge to landfarming' is however in the database.

4.10.3 Zeolite in inert material landfill

During the refining of oil catalysts with various metals (Co, Mo, V, Ni, W, platinum group metals) are used (Frischknecht et al. 1996:IV.291f.). Spent catalysts are recycled for their metal content (Frischknecht et al. 1996:IV.165). The catalyst carrier matrix – usually zeolite – is landfilled. No information on material composition – especially traces – is available. Zeolites are a class of tailor-made aluminosilicates. Disposal in inert material landfills is assumed.

4.10.4 Separator sludge from mineral oil storage tanks

Separator sludge is an water-oil emulsion obtained from storage tanks for mineral oil. Waste composition is adopted from (BZL 2000b:129) and from (Frischknecht et al. 1996:IV.188) based on (DGMK 1985). This composition is complemented with contents of nitrogen and fluorine from light fuel oil (Frischknecht et al. 1996:IV.16) in proportion to the carbon content. Disposal in a hazardous waste incinerator is assumed.

4.10.5 Rain water from mineral oil storage tanks

The rainwater from mineral oil storage tank sites is polluted with hydrocarbons and heavy metals (Jungbluth 2003b). In former ETH LCI data this water was assumed to be emitted directly to a natural water body (Frischknecht et al. 1996:IV.186 and IV.191). This seems unlikely. Instead a treatment of

³⁰ Landfilling of such a waste (burnable) is illegal in Switzerland.

this water in a WWTP is assumed. With the present WWTP model, the various – partly chlorinated – hydrocarbon species can only be assessed as a total CSB value.

4.10.6 Condensate from light fuel oil boiler

In condensing boilers, the water in the flue gas is condensed to recuperate the evaporation heat in steam. The condensate is collected, neutralised and discharged to the sewer (Jungbluth 2003b, BUWAL 1988). The expenditures for neutralisation of the condensate (KOH/MgOH, or ion exchange resin) are heeded in (Jungbluth 2003b). The composition of the discharged condensate applies for light fuel oil (low-Sulfur fuel oil, Heizöl extraleicht) with a sulfur content of 0.14 w%.

4.11 Natural gas energy

4.11.1 Solvents to hazardous waste incineration

During natural gas processing, various solvents are used. Waste solvents are mostly reported as an unspecified mixture with the European waste code EWC 140103 'other solvents and solvent mixes' (Faist & Heck 2003). No composition data is available. As a proxy, a solvent composition is derived from average input of the Swiss waste solvent incinerator Valorec Schweizerhalle (Jahn 2000 & 2002). This composition largely stems from chemicals production in Basel.

Waste solvent composition data for sulfur, nitrogen, oxygen and hydrogen is taken from (Jahn 2000). Data for phosphorus, carbon, the lower heating value and water content is taken from (Jahn 2002). For chlorine, bromine, fluorine, iodine, cobalt, copper, nickel, zinc, and iron the input waste composition is calculated in reverse from emissions in air and water and elemental transfer coefficients both devised in (Jahn 2002). The remainder to 100 w% is taken to be silicon. The upper heating value is estimated from (Michel 1938).

4.11.2 Soiled textiles to municipal waste incineration

During natural gas processing, used filters, absorbers, cleaning threads, and protective clothing are reported as wastes with the European waste code EWC 150201 (Faist & Heck 2003). These are textile and absorber wastes that are contaminated with hazardous substances. It is assumed that that these wastes are mainly textile in nature and contaminated with oily substances. It is assumed the waste is disposed in an incinerator equal or similar to a municipal waste incinerator. No composition of textile wastes from natural gas processing is available. Instead a waste textile composition encountered in average municipal waste complemented with an estimated 5 w% waste oil is used. Composition of natural waste textiles is taken from (Mayestre et al. 1994, Tillmann 1991, Domalski et al. 1987, Bilitewski et al. 1991, Mark et al. 1994a). Waste oil composition is documented in chapter 4.27.1 'Used mineral oil to hazardous waste incineration' on page 60.

4.11.3 Ashes to residual material landfill

During natural gas processing, scrubber residues and oil furnace ashes with the European waste code EWC 100106 and 100104 are reported (Faist & Heck 2003). Part of these ashes originate from the incineration of energy carriers. Depending on the energy carrier, these wastes are already included in the inventory chain for energy consumption and must not be inventoried again here. For the remainder no composition is known. As a proxy the landfilling of average incineration residues can be inventoried.

4.11.4 Anti-freezer liquid to hazardous waste incineration

During natural gas processing, generation of waste anti-freezer liquid with the European waste code EWC 166072 is reported (Faist & Heck 2003). No composition data or disposal option information is available. The prevalent industrial antifreeze systems are ethylene glycol (1,2-ethanediol, $C_2H_6O_2$) or propylene glycol (1,2-propanediol, $C_3H_8O_2$). Usually approximately 1:1 volumetric mixtures with

water are used. With densities of 1113 kg/m³ and 1.036 kg/m³, respectively, solutions with 52.7 w% and 50.9 w% water result. It is not known, if natural gas processing preferentially uses one or the other system. A 50:50 mixture of aqueous ethylene glycol and propylene glycol solutions (51.8 w% water content) is inventoried. From theoretical composition a content of 22.55% oxygen, 4.88% hydrogen and 20.79% carbon is calculated. No information on trace metals is available, although during use the liquid is likely to be contaminated with heavy metals. The heating values are estimated according to (Michel 1938) to be 10.91 MJ/kg (upper) and 0.55 MJ/kg (lower). It is assumed that this waste is incinerated in a hazardous waste incinerator.

4.11.5 Condensate to hazardous waste incineration

During natural gas drying a condensate liquid is reported as waste with the European waste code EWC 70104 (Faist & Heck 2003). Only a composition of pure hydrocarbons (no traces) is known. Disposal is approximated with mineral oil to hazardous waste incineration

4.11.6 Gas pipeline disposal

Gas pipelines are usually plastic-coated cast iron pipes set in underground sand beds. After use the pipeline track might be re-fitted for new pipelines. If this is not the case, the old pipelines are usually left underground (Faist & Heck 2003). The reason is usually that removal and disposal of old pipelines is expensive (Frischknecht et al. 1993:V.43). The pipeline materials themselves are deemed to be unproblematic, but possible oily condensates from the gas could pose a risk. To account for old pipeline materials left underground – mainly sand by mass – a data module 'gas pipeline to inert material landfill' was created in (Frischknecht et al. 1993:V.46). This is however not appropriate because neither a disposal energy is consumed nor an actual landfill volume is occupied by this practice. This module still exists in the database, but should not be used. Emissions similar to inert material landfills could be inventoried for such a procedure. However in this study no direct emissions from inert material landfills were calculated. Accordingly, no emissions are inventoried with on-site burial of pipelines. The disposal of condensate is detailed above.

4.12 Nuclear energy

Disposal processes of radioactive waste materials within the nuclear energy chain (re-processing, intermediate and final storage) is fully inventoried in (Dones et al. 2003).

4.12.1 Uranium mill tailings

During the refining of uranium ore and production of concentrated uranium (yellow cake) residual sludges are produced. Sludges are solidified with lime. The pH in these ponds is already high (6–2) and can be increased by oxidation of pyrite (FeS₂) to sulfuric acid. High pH enforces metal leaching. A detailed tailings model for these non-radioactive long-term emissions is documented in English in the appendix of the current ecoinvent nuclear energy report (Bauer 2009) and not here.

4.12.2 Wastes from uranium enrichment

In the uranium enrichment solid and liquid non-radioactive wastes occur. These are epoxy-polyamine ion exchange resin (49'000 kg/a), a liquid HNO₃/NaOH mixture (3'900kg/a), and a liquid nitric acid/HF mixture (16'000 kg/a).

The epoxy-polyamine ion exchange resin is bound to be a discarded anion exchange resin. It is not known with what anions this resin is charged, when it is discarded, nor any detailed composition. Anions loaded on the resin could originate from the enrichment process or from a regeneration salt, if the resin is regenerated prior to disposal. Presumably the anion exchanger is not directly involved with the produced material (uranium hexafluoride). As a proxy the incineration of spent anion exchange resin from *drinking water preparation* is inventoried.

The HNO₃/NaOH mixture is a liquid waste. The components are acid and alkaline and partly will neutralise each other. This waste will, probably after some additional neutralising treatment, likely go

to the sewer. Only NO_3 (Nitrate) is of any ecological relevance here. Unfortunately no concentrations are available and this waste flow is neglected.

The nitric acid/HF mixture is also a liquid waste, but definitely acidic. This waste will probably be neutralised with something alkaline and disposal via sewer. Ecologically relevant are nitrite NO_2 (from the nitric acid = HNO_2) and fluorine F. Again no concentrations are known and this waste flow is neglected. This liquid waste is potentially more relevant than the other due to larger volume and more pollutants.

In absence of any pollutant information it is possible to use the module 'treatment, sewage, unpolluted, to wastewater treatment, class 3'. In this module the transport and treatment of 1m^3 unpolluted water is inventoried. Due to large expenditures in infrastructure and pumping the disposal of unpolluted water alone leads to relevant burdens.

4.12.3 Uranium hexafluoride production

During the production of uranium hexafluoride (UF_6) from uranium oxide (yellow cake, U_3O_8) non-nuclear wastes are generated (Dones 2003). The wastes are incinerated on site and the resulting ashes are landfilled. The ash composition is only qualitatively given as containing "iron, calcium, magnesium, copper, and fluoride". No quantitative composition is known. As a placeholder the landfilled ash is inventoried as 'average incineration residues to residual landfill' (cf. chapter 4.1.2 'Average residual material waste mixture' on page 29). This is a composition derived from incinerated average municipal waste. In the future a specific quantitative composition shall be investigated and used, especially regarding fluoride and copper. Fluoride content is possibly underestimated with the current placeholder, while copper content is possibly overestimated.

4.13 Photovoltaic energy

See also chapter 4.21 'Silicon production' on page 53.

4.13.1 Inorganic wastes from silicon wafer production to residual material landfill

Various inorganic wastes are generated during production of silicon wafers for photovoltaic cells (Jungbluth 2003a). According to Frischknecht et al. 1996:XII.31 they contain silicon-rich sawing sludge, calcium fluoride CaF_2 from neutralisation of fluoric acid, quartz and graphite crucibles, graphite supports, and soiled silicon wastes. From the compositions given in Tab. 4.7 the elemental composition per kilogram waste is calculated.

Tab. 4.7 Inorganic waste streams from silicon wafer production from (Frischknecht et al. 1996:XII.31) based on (Hadorn & Hellriegel 1992).

Waste	Composition	g/wafer
Silicon sludge	assuming 21% water and 79% Si	5.9
CaF_2	CaF_2	1.1
SiO_2 crucible	SiO_2	5
graphite crucible	100% C	0.11
graphite support	100% C	0.3
Si waste	100% Si	0.61

4.13.2 Organic wastes from silicon wafer production to underground deposit

Various carbon-rich wastes are generated during production of silicon wafers for photovoltaic cells (Jungbluth 2003a). According to Frischknecht et al. 1996:XII.31&45 they contain oil absorbent, ethylene glycol, mineral oil, silicone oil, and paper. Disposal in an underground deposit is assumed in (Jungbluth 2003a). Deposition in a large steel container (2.25 m³) is assumed.

4.14 Solar heat energy

4.14.1 Wastewater from solar collector tube production

The production of glass tubes for solar collectors is inventoried in (Jungbluth 2003c). During manufacture (surface coating, cutting, cleaning etc.) a wastewater is discharged to a local sewer. Some limited information on the wastewater composition is available. Concentrations are 0.67 mg tin/l, 0.1 mg silver/l, 0.44 mg fluoride/l.

4.14.2 Heat carrier liquid to wastewater treatment

The heat carrier liquid in solar heat plants is a 40% aqueous solution of propylene glycole (anti-freeze agent). Common Swiss practice is to discharge used heat carrier liquids from residential solar plants to municipal sewers. Legally, anti-freeze agents have to be collected as hazardous wastes. For solar plants an exception is made due to the small amounts involved and the low toxicity of propylene glycole.

4.15 Wind energy

For wind energy generators large foundations of steel-reinforced concrete are built. After decommissioning those foundations are left on site (Burger 2003). Emissions similar to inert material landfills could be inventoried for such a procedure. However in this study no direct emissions from inert material landfills were calculated. Accordingly, no emissions are inventoried with on-site burial of concrete foundations.

4.16 Wood energy

4.16.1 Ash from natural wood incineration

Wood heating systems produce wood ash, that needs to be disposed (Bauer 2003). Ash from wood heat furnaces is partly disposed in MSWI. Usually this ash has little carbon left to burn and is essentially an inert waste. The reason wood ash is disposed as municipal waste is that there are often only relatively small amounts and logistic convenience.

Ash from larger furnaces is also landfilled directly. Landfilling in a sanitary landfill is assumed. This module is also used for the disposal of wood and bark incinerated during paper and cardboard production (Hischier 2003). A 'degradability' of 5% in the first 100 years is assumed (mainly wash-off).

Wood ash can also be used as fertiliser in agriculture or forestry on soils certified to be suitable (Noger et al. 1996). Spreading/landfarming on agricultural surface (i.e. on a food crop area) is also inventoried as a disposal option (cf. part III, chapter 11 on landfarming).

The wood ash composition for all three disposal options considered here stems from the incineration of untreated, natural wood without any chemical treatment, coating or additives (Bauer 2003). A composition representing grate (bottom) ash and cyclone ash is chosen. The unspecified remainder in the composition is assumed to be oxygen.

4.17 Paper and cardboard production

4.17.1 Ash from incineration of deinking sludge to landfill

Deinking is a process step in the recycling of printed paper (Hischier 2003). The printing inks are removed as a sludge. The sludge is incinerated and the remaining ash is landfilled. Disposal in a residual material landfill with solidification is assumed.

4.17.2 Ash from incineration of wood and bark to landfill

Ash from incineration of wood and bark during the production of paper and cardboard is partly landfilled (Hischier 2003). The waste composition from ashes of wood energy furnaces is adopted for such wastes (cf. chapter 4.16.1 'Ash from natural wood incineration' on page 49). Landfilling in a sanitary landfill is assumed.

4.17.3 Wastewater treatment sludge from paper production

During paper production, internal wastewater treatment plants (WWTP) produce a treatment sludge (Hischier 2003). In Sweden, 60 w% of these sludges is incinerated on-site. The remainder is landfilled³¹. Literature data for sludge composition is available for C, As, Cd, Cr, Cu, Hg, Ni, Pb, and Zn (Hischier 2003). The composition is complemented with proxy data from municipal wastewater treatment sludge for S, N, P, H, F, Co, Mn, Mo, Sn, Si, Fe, Ca, Al, K, Mg, and Na in amounts proportional to the carbon content (see part IV). The remainder to 100 w% is set to be oxygen.

Direct emissions from the on-site incineration of the treatment sludge are heeded within the paper production process (see Hischier 2003). Incineration ashes remain as a solid residue, which is landfilled. Landfilling in a residual material landfill is assumed. Ash composition is calculated from sludge composition and transfer coefficients of the MSWI model (see part II). A water content of 59.6% in the sludge is used. 40% of the sludge is landfilled directly. A sanitary landfill is assumed. For landfilling a water content of 25% is used.

4.17.4 Green liquor dregs from pulp production

During pulp production so called 'green liquor' is produced (Hischier 2003). Green liquor is an aqueous solution of the ashes from black liquor incineration and contains mainly sodium sulfide Na_2S , sodium carbonate Na_2CO_3 , traces of zinc, copper and chromium. The insoluble residues are removed from the pulp production process as 'green liquor dregs'. This material is landfilled. Disposal in a residual material landfill with solidification is assumed.

4.17.5 Lime residue in inert material landfill

An inorganic residue is created during the production of base paper for corrugated board (kraftliner) (Hischier 2003). The waste is characterised as 'calcium carbonate waste'. Landfilling in an inert material landfill is assumed.

4.18 Building materials production

4.18.1 Wastewater from processed wood boards production

During the production of wood boards (for building materials) several wastewater streams are generated (Werner et al. 2003). Different wastewater compositions (BOD, COD, PO_4 , Zn, Cu) for fibre board, particle board and plywood production are referenced in (Werner et al. 2003). The wastewater is generated during fibre separation (fibre board) or steaming (plywood). Therefore, all carbon is assumed to be 100% biogenic. For plywood the COD value is 0.7 kg/m^3 and the BOD is

³¹ Due to the high carbon content and burnability, this practice is prohibited in Switzerland.

4.23 kg/m³, i.e. COD < BOD, which should not occur. This is not a typo, but was checked back with the data source. The data was used accordingly.

The original data gives figures of 0.03 g copper and 380 g zinc per m³ wastewater. The latter seems an extreme figure and is probably erroneous. No other literature data could be found. Based on relative occurrence of copper and zinc in natural wood, the concentration level of zinc in wastewater was corrected to 0.13 g/m³ (0.00013 kg/m³).

4.18.2 Lime sludge from reinforced concrete board production

During the production of reinforced concrete boards, a wastewater is treated to remove CO₂ (Kellenberger et al. 2003). The resulting sludge composition is unknown. As a proxy, the lime sludge generated in decarbonising water is used (see section 'Decarbonising waste to residual material landfill' on page 57).

4.18.3 Wastewater from concrete production

In the production of concrete, during the mixing of cement, water and gravel, a wastewater is produced that must be treated. Wastewater compositions are given in (Kellenberger et al. 2003). The chromium content is Cr^{VI}. In the wastewater treatment plant Cr^{VI} is partially converted to Cr^{III} and transferred to sludge (50 w%). The remainder is emitted in the effluent as (soluble) Cr^{VI}. The wastewater is assumed to be disposed in a class 3 wastewater treatment plant.

4.18.4 Wastewater from glass production

Several glass products are inventoried in (Kellenberger et al. 2003). An average, generic wastewater composition for the whole glass industry is estimated from literature (see Kellenberger et al. 2003). Treatment in a class 2 municipal wastewater treatment plant is inventoried.

4.18.5 Wastewater from ceramic production

During the production of ceramic building materials (ceramic tiles, sanitary ceramics) a wastewater is generated (Kellenberger et al. 2003). The wastewater contains only metals in the sub-ppm range (see Kellenberger et al. 2003). Treatment in a class 3 municipal wastewater treatment plant is assumed.

4.19 Aluminium production

Several waste streams are created during the production of aluminium (Althaus et al. 2003b).

4.19.1 Redmud in residual material landfill

During alumina production from mined bauxite a so called 'redmud', an inorganic tailings slurry containing mainly oxides of iron, silicon, aluminium, sodium, and titanium is produced. The preferred disposal technology for redmud, which initially contains 7–9% solids, is to concentrate and landfill them in the nearby area. A concentration of 25–30% can be achieved by gravity settling in a tailings pond. The redmud slurry is pumped to the disposal area where it is spread in layers known as redmud stacking layers. The water drained from the redmud stacking area is collected in ponds for recycling. Water recycling is included in the alumina process (Althaus et al. 2003b). The redmud can be concentrated to 30–50% solids using a thickener reducing the redmud volume further. Redmud ponds are controlled landfills with base lining and leachate collection. The redmud disposal is therefore inventoried with the model for the residual landfill. The redmud is not solidified with cement (like e.g. landfilled incineration residues are). Still the calcium content of redmud is high enough to estimate an elevated pH and a carbonate phase of longer than 60'000 years (as in the residual landfill model). No adaptation of transfer coefficients is performed.

A generic redmud composition is derived from literature data for redmuds from Australia, Jamaica, India, Hungary, Yugoslavia, Turkey, and Suriname (Altundogan 2002, BMTPC 2001, McLaughlin 1998, NPI 1999b).

A small amount of sand is also produced during alumina production. The sand is the coarser part of the undissolved bauxite components, while redmud is the fine part of the undissolved bauxite components. It is assumed that the sand has a composition similar to redmud. Disposal of sand is accordingly approximated with the same module 'redmud to landfill'.

4.19.2 Spent pot lining to residual material landfill

During aluminium electrolysis the cathode is essentially an open carbon container (pot) supported by a ceramic material (refractory material). The carbon/ceramic composite is called the pot lining, which has an outer casing made of a steel sheet. When a pot lining has reached the end of its useful life, normally after about 6–7 years, it is replaced. The old, so called 'Spent Pot Lining' SPL contains cyanide and 7–9% fluoride from the cryolite electrolyte bath (Na_3AlF_6). Processes to produce aluminium trifluoride AlF_3 from SPL were developed. However, according to manufacturers SPL is disposed in landfills for "non-hazardous waste". Regarding the pollutant content of SPL, especially fluoride, it cannot be seen as an unproblematic waste.

During pot dismantling two fractions are generated: a first cut which is the inner carbon cathode material and a second cut, which is the supporting refractory (brick) material. For both fractions disposal in a residual landfill without cement solidification is assumed. Composition is documented in (Althaus et al. 2003b). The refractory brick composition is complemented with 369 ppm strontium derived as geometric mean from measurements in (Chemmea Bohemia 2000). Since the landfill model heeds only the fate of chemical *elements* and not compounds, such components as cyanide (CN^-) are only heeded as carbon and nitrogen which underestimates the toxic potential of this waste.

4.19.3 Filter dust from electrolysis to residual material landfill

Filter dust is collected from cyclones or electrostatic precipitators in the (primary) alumina electrolysis. Filter dust composition is given in (BBK 1999) based on (Sauer et al. 1982).

4.19.4 Dross from electrolysis to residual material landfill

Dross is a mixture of aluminium, aluminium nitride, carbide and oxide and originates in electrolysis. Disposal in a residual material landfill is assumed.

4.19.5 Tar and soot waste to residual material landfill

Smaller amounts of carbon-rich wastes (tar) are produced in alumina electrolysis. According to manufacturers they are disposed in landfills. As a proxy, disposal of bitumen in sanitary landfill is assumed.

4.20 Iron and steel production

During the production of iron and steel several waste streams are generated. The waste compositions are documented in (Althaus et al. 2003b). Production slags are partly recycled (e.g. road construction, cement production) and partly landfilled. Here only the final disposal of waste streams is described.

4.20.1 Wastes from pig iron production

A gas scrubber sludge generated in pig iron production contains heavy metals, especially zinc and lead (Althaus et al. 2003b). Disposal in a residual material landfill type with cement solidification is assumed.

The wastewater generated in pig iron production contains some heavy metals and some carbon. Discharge to a local sewer and treatment is assumed in a class 3 wastewater treatment plant.

4.20.2 Wastes from electric arc furnace steel

Electric arc furnaces (EAF) are fed with a high share of secondary metals (new and old scrap). The collected wastes from EAF are very dependent on alloy and contamination elements in the feed materials (Althaus et al. 2003b). Here a typical composition is considered.

Slag from EAF steel production is a Ca/Si/Al matrix that is especially rich in manganese and chrome. Disposal in a residual material landfill type without cement solidification is assumed.

Dusts are collected during EAF steel production typically from bag filters. Two types of dust are discerned in the disposal inventory: 1) dust from production of unalloyed (carbon) steel and 2) slag from production of (chrome) alloyed steel. The dust from alloyed steel has higher contents of Ni, Cr, V, Al, and Si. Disposal in a residual material landfill type with cement solidification is assumed.

4.20.3 Wastes from basic oxygen furnaces

In basic oxygen furnaces, unwanted traces are removed to produce high quality steels (Althaus et al. 2003b). Three different waste streams are produced during desulphurisation and steel-making. These are BOF slags, desulphurisation slag, and BOF dust. All three waste streams are partly landfilled. For the BOF dust only an incomplete composition is available, especially lacking data for chromium and nickel. BOF dust is therefore approximated with EAF dust, for which a more complete composition is known (details see Althaus et al. 2003b).

For the remaining two wastes, *one data module* is created which contains a *weighted average* of both wastes: 86 w% BOF slags and 14 w% desulphurisation slag. Disposal in a residual material landfill type with cement solidification is assumed.

4.20.4 Treatment sludge from steel rolling/drawing

In the rolling of steel and drawing of steel pipes and wires a wastewater treatment sludge is generated (Althaus et al. 2003b). The sludge originates from internal wastewater treatment of cooling and process water. Only data on pollutants in purified wastewater and total mass of removed sludge are known. The wastewater treatment operates with addition of flocculants and polyelectrolytes only. No special heavy metal precipitating agents are used. For this reason, the unknown sludge composition is derived in linear proportion to the pollutant profile in the purified wastewater. Hydrocarbons are inventoried as 90% C and 10% H. A water content of 20% is assumed in the waste. The remainder is assumed to be oxygen. Disposal in a residual material landfill type with cement solidification is assumed.

4.21 Silicon production

During the production of metallurgical grade (MG) silicon a slag is generated (Jungbluth 2003a). The slag is landfilled. Composition of major components is shown in Tab. 4.8. Disposal in inert material landfill (no landfill emissions) is assumed.

Tab. 4.8 Composition of slag from MG silicon production (Jungbluth 2003a).

Metallic silicon	Si or FeSi	20-30%
Oxidised silicon	SiO ₂	5-20%
Silicon carbide	SiC	20-40%
Calcium oxide	CaO	25-40%
Aluminium oxide	Al ₂ O ₃	3-35%

4.22 Production of various metals

During mining and smelting of metals waste streams are produced (Althaus et al. 2003b). The following procedure is applied for production of nickel, copper, zinc, platinum group metals (PGM; platinum, palladium, rhodium), lead, chromium, molybdenum, and manganese.

4.22.1 Overburden

Overburden is the material which is excavated to access an ore vein. This material does not contain ores, is therefore not processed and is usually deposited unchanged. Overburden is deposited either in the mining pit (backfill) or on the tailings stack (cf. below). The material is assumed to emit no relevant pollutants. The high pH in sulfidic tailings however is a distinctly different chemical environment than the overburden in its original location. This could lead to additional emissions from formerly inert minerals. Such changes are neglected here. The dominant mass input to tailings stacks and emissions from tailings stacks are the tailings material itself.

For overburden material deposited on tailings stacks a land use as dump site is inventoried. The disposal of overburden is inventoried in detail in (Althaus et al. 2003b) and not here.

4.22.2 Mining tailings

Tailings are the unwanted minerals mined together with the desired, metal-bearing mineral. Tailings are separated mechanically during ore beneficiation (milling) by grinding and flotation, and deposited on the surface on so called tailings stacks or impoundments. Tailings are fine ground minerals and wet with water to a degree which sometimes allows pumping as transport to stacks. Tailings stacks are several tens of meters high and usually cover a large area. Its borders are lined by dams and the excess water is led to a cascade of tailings ponds for settling of solids. Leaching water from ponds is emitted to rivers or – more frequently – to the groundwater. On the surface, the sulfide minerals in the tailings are oxidised to sulfuric acid³², which leads to a low pH and high degree of leaching of metals.

Only scarce information is available on long-term leaching behaviour of tailings. The mechanical integrity of the original mineral bulk has been compromised through grinding processes, leading to distinctly altered mechanical properties and chemical reactivity. Oxygen intrusion from the surface and microbial activity leads to progressive generation of sulfuric acid from oxidation of sulfide minerals. Due to the acidic pH of tailings in the range of 5 – 3.5, it can be expected that over long time spans (>10'000 years) a significant part of the heavy metals are emitted from the tailings stack and leached into the groundwater. A dedicated model for long-term emissions from sulfidic tailings based on field measurements, analogous to the landfill models, is documented in detail in the current ecoinvent report on metal production (Althaus et al. 2009). This waste data is applied to the metal production from sulfidic ores (copper, molybdenum, nickel, zinc, tin, lead, PGM, chromium)

4.22.3 Smelter Slags

Slags originate from smelting of purified metal ores. Slag composition depends on the ore composition and the used flux materials used to create a slag and the unwanted metals removed by smelting. Glassy, high-temperature slags might be used as construction materials or abrasives. Here only landfilling of slags is inventoried, as recycling is treated with cut-off in ecoinvent. Slags that are not recycled are assumed to be deposited in a residual material landfill without solidification.

Lead smelter slag

An average lead smelter slag composition was taken from the geometric mean of four literature sources (EPA 1995, EPA 2002, Morrison 2003, Schmidt et al. 1999). The slag composition was

³² For example oxidation of pyrite: $4 \text{FeS}_2 + 15 \text{O}_2 + 14 \text{H}_2\text{O} \rightarrow 8 \text{H}_2\text{SO}_4 + 4 \text{Fe}(\text{OH})_3$

complemented with an assumed oxygen content of 50 w-%. The remainder (27 w-%) was assumed to be silicon. The slag is assumed to be deposited in a residual material landfill without solidification.

Nickel smelter slag

From nickel metal smelting a limited slag composition (Ni, Co, Cu, Fe, Si, Ca) is available (Althaus et al. 2003b). The mass difference to 100% in the waste composition is inventoried as oxygen. This composition is only for slags from smelting of nickel and nickel-related platinum group metals. The slag is assumed to be deposited in a residual material landfill without solidification.

4.22.4 Production effluent from galvanic black chromium coating

During solar collector production, some types of collector tubes are coated (plated) with black chromium (Jungbluth 2003c). The internally pre-treated effluent of this galvanic coating contains organic pollutants and heavy metals (Cu, Cr, Ni, Zn, Pb, Ag, Cd) and is treated in a class 2 municipal wastewater treatment plant.

4.23 Inorganic chemicals production wastes

4.23.1 Solid residues from sodium dichromate production

During the production of sodium dichromate ($\text{Na}_2\text{Cr}_2\text{O}_7$) large amounts of residues are generated from chromium ore processing (Althaus et al. 2003a). Residues are neutralised with lime (CaCO_3). The residues are rich in chromite (FeCr_2O_4) and are disposed on a so called 'surface impoundment' located on-site. Surface impoundments are described as natural or man-made 'pools' to hold liquid or slurry wastes for storage, settling or aeration (EPA 2001:1.9). Over 59% of all US impoundments are without any base seal or liner (EPA 2001:2.26). Of the two US dichromate producers³³, only the smaller one has disposal facilities with HDPE liners (EPA 2000:11&26).

For dichromate production the impoundment is usually located in the abandoned chrome ore quarry (EPA 2000:22). It is assumed that these deposits are not removed after production has ceased. This disposal practice is approximated with the residual landfill model described in part III³⁴. No additional solidification with cement is assumed. A residue composition heeding P, B, Ag, As, Ba, Cd, Co, Cr, Cu, Mn, Mo, Ni, Pb, Sb, Se, Sn, V, Zn, Si, Fe, Ca, Al, K, Mg, and Na is defined in (Althaus et al. 2003a). The remainder to 100% is assumed to be oxygen.

The emission potential of chromium in the residual landfill model is limited, as 75% of chromium is assumed to be bound in very stable chromite (FeCr_2O_4). The short-term emissions (0–100 years) are 6% of the landfilled chromium, the long-term emissions (> 100 years) are another 19% of the landfilled chromium. In total only 25% of the landfilled chromium is emitted. Emission as Cr^{VI} is assumed, based on measurements in (Fruchter et al. 1990, Kersten et al. 1998). For more information see chapter 'Speciation of residual material landfill emissions' in part III (landfills) of this report.

4.23.2 Sludge from iron chloride production to underground deposit

A sludge rich in carbon, silicon, iron and sulfur originates from iron chloride production (cf. Althaus et al. 2003a). According to manufacturers, the sludge is deposited in an underground deposit. Deposition in a 200 liter steel drum with cement solidification is assumed.

³³ Elementis Chromium, Corpus Christi, TX (capacity 49'900 t/a) and Occidental Chemical Corporation "Oxychem", Castle Hayne, NC (capacity 106'200 t/a). Capacity data for 1997 from (EPA 2000).

³⁴ The residual landfill model includes a base seal and recultivation, which is not entirely appropriate here, but is of minor importance in the total landfill burdens which are dominated by leachate emissions.

4.23.3 Solid remains from titanium dioxide production

During the production of titanium dioxide (TiO_2) solid inorganic residues are produced (cf. Althaus et al. 2003a). Two different titanium dioxide production processes are regarded in ecoinvent 2000: the sulfate process and the chloride process.

The technical term for solid residues from the *sulfate process* is 'digester residues'. In essence the waste is the residue of the ilmenite ore feedstock (FeTiO_3) that will not dissolve in hot strong sulphuric acid in the digestion process. The waste is categorised as hazardous because it contains 3-5% free sulfuric acid. The waste is solidified with cement or co-produced gypsum before landfilling. Landfilling in a residual material landfill with cement solidification is assumed.

The technical term for solid residues from the *chloride process* is 'neutralised spray vessel solids'. These wastes are not considered hazardous. They are landfilled without prior treatment. Landfilling in a residual material landfill *without cement solidification* is assumed.

4.23.4 Phosphogypsum tailings stacks

During the mining of phosphor resources large volumes of phosphogypsum, a solid residue, are created (Althaus et al. 2003a). Per kg of P_2O_5 approximately 5 kg of phosphogypsum are produced. Phosphogypsum is stacked on site in up to 60 m tall piles. Phosphogypsum consists largely of gypsum (CaSO_4) but contains also fluorine, radioactive uranium and radium. The pH is very acidic and ranges at 2.6 to 1.2. No detailed model for emissions from tailings could be created for this study³⁵. An estimate of emissions from phosphogypsum stacks is described in (Althaus et al. 2003a).

4.23.5 Solid remains from phosphoric acid purification

An inorganic, solid residue originates during the purification step of phosphoric acid production (cf. Althaus et al. 2003a). It contains anhydrite (calcium sulfate) and apatite (calcium phosphate) and is rich in fluorine. According to manufacturer, the sludge is landfilled. Landfilling in a residual material landfill is chosen. Additional solidification with cement in the landfill is assumed.

4.23.6 Chlorine production

Sludge from NaCl electrolysis to residual material landfill

During the production of chlorine gas (Cl_2) from chlorine-alkali electrolysis, an inorganic brine filtration sludge is produced (Althaus et al. 2003a). The sludge mainly contains calcite (CaCO_3), magnesium hydroxide ($\text{Mg}(\text{OH})_2$) and barite (BaSO_4). For the electrolysis using mercury cells the sludge can also contain up to 150 ppm mercury. No further information was available from the manufacturers. An estimated composition of 47.5% calcite, 47.5% magnesium hydroxide, 5% barite and 0% water was assumed³⁶. For the case of mercury cells electrolysis a similar composition was assumed, but containing 75 ppm mercury. Additional solidification with cement in the landfill is assumed for both wastes.

Spent activated carbon to underground deposit

During the production of chlorine gas (Cl_2) from chlorine-alkali electrolysis, spent activated carbon is produced (Althaus et al. 2003a). The carbon absorber purifies the flue gas from the mercury cell process. The spent carbon contains 0.1 to 50 w% (sic) mercury. In a distillation step the mercury is recovered. The remaining carbon contains 20 to 200 ppm mercury. It is assumed that this remaining waste is disposed in an underground deposit. The expenditures for the mercury recycling are not contained in this module. Deposition in a BigBag (large LDPE bag) is assumed.

³⁵ It is possible to create such models from field measurements, analogous to the landfill models. Tailings models were not in the scope of this report.

³⁶ The carbon from carbonate was heeded as oxygen. Otherwise it would be interpreted as organic carbon.

4.23.7 Salt residues from potash mining to residual material landfill

During potash mining for potassium fertilisers (cf. Nemecek et al. 2003) large quantities of unusable salts are co-mined ('salt tailings'). These salt tailings consist mainly of Kieserite ($\text{MgSO}_4 \cdot \text{H}_2\text{O}$), Carnallite ($\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$), rock salt (NaCl) and Anhydrite (CaSO_4). The majority of the tailings is stacked on the surface on piles. A smaller amount is disposed internally in the potash mine as backfill. Aqueous solutions of salt tailings are also injected into deep groundwater. In this study only the surface tailings stack is inventoried. It is assumed that the stack is similar to an inorganic waste dump (landfill). For calculations of the emissions the model of residual material landfill is used. No solidification is assumed. No measurements of trace contents is available.

4.23.8 Water purification

Ion exchange resins to municipal waste incineration

During production of potable water, anion and cation exchange resins are used to treat the water. The resins can be regenerated with NaCl , but eventually have to be disposed (cf. Althaus et al. 2003a). Disposal in municipal waste incinerators is common, if the treated water is not polluted. No information on the spent resin composition was available. Exchange resins are usually copolymers of styrene. Cation exchange resins commonly have a sulfonate functional group ($\text{R}-\text{SO}_3^-$) to bind cations. For anion exchange resins a dihydroxyamine functional group is common ($\text{R}-\text{N}(\text{OH})_2^+$). For industry products, a styrene copolymer with 8% divinylbenzene (DVB) can be assumed to be typical³⁷, which results in a theoretic composition $\text{C}_{204}\text{H}_{181}(\text{SO}_3^-\text{Na}^+)_{23}$ and $\text{C}_{1234}\text{H}_{1786}(\text{N}(\text{OH})_2^+\text{Cl}^-)_{92}$. Detailed calculations see the chapter on drinking water production in Althaus et al. (2003). The spent resins are assumed to be charged with typical amounts of cations (Mg^{2+} , Ca^{2+} , K^+ , Na^+) and anions (NO_3^- , SO_4^{2-} , SiO_2^{2-} , Cl^- , HPO_4^{2-}), for cation exchange resins and anion exchange resins, respectively. This information is complemented with heavy metal trace element compositions found in polystyrene waste fraction of municipal waste and an estimated water content of 50%.

The anion exchange resin is also used as a proxy for epoxy-polyamine exchange resins of unknown composition in the uranium enrichment step. Cf. chapter 4.12.2 'Wastes from uranium enrichment' on page 47.

Decarbonising waste to residual material landfill

Decarbonising waste is carbonate-rich waste obtained during decarbonisation of water. A waste composition for slow decarbonisation from (VGB 1990) is adopted with a water content of 30%. The carbon on this waste is dominantly from inorganic carbonate. In the residual material landfill model carbon in waste leads to organic carbon emissions. This is not suitable here. The carbon in this waste is therefore registered as neutral oxygen.

This composition is also used for lime sludge from wastewater treatment in reinforced concrete board production (see chapter 4.18.2 'Lime sludge from reinforced concrete board production' on page 51).

4.24 Organic chemicals production wastes

4.24.1 Catalysts from formaldehyde manufacture

For the manufacture of formaldehyde (CH_2O), catalysts containing iron and molybdenum or vanadium oxide are used (Althaus et al. 2003a). The metals are removed from the used catalyst and recycled. The remaining catalyst matrix is landfilled. The matrix material is assumed to be SiO_2 . No measurements for traces could be obtained. As a cautious guess, a concentration of 5 ppm of

³⁷ See for example <http://www.resintech.com/index.cfm?Point=Products>.

molybdenum and vanadium is assumed for the landfilled material³⁸. Iron is assumed not to be recycled and to be present in 5 w% in the landfilled catalyst. Disposal in a residual material landfill without solidification is assumed.

4.24.2 Catalysts from ethylene oxide manufacture

For the manufacture of ethylene oxide (C_2H_4O), catalysts containing silver oxide on a porous carrier such as aluminium oxide Al_2O_3 are used (Althaus et al. 2003a). The silver is removed from the used catalyst and recycled. The remaining catalyst matrix is landfilled. No measurements for traces could be obtained. As a cautious guess, a concentration of 5 ppm of silver is assumed for the landfilled material³⁹. The remainder is assumed to be aluminium oxide Al_2O_3 . Disposal in a residual material landfill without solidification is assumed.

4.24.3 Catalysts from ethylene dichloride manufacture

For the production of ethylene dichloride⁴⁰ (EDC, CH_2Cl-CH_2Cl), catalysts with copper or iron chloride are necessary (Althaus et al. 2003a). The spent EDC catalyst contains octo-chlorodibenzofurans, which have dioxin-related toxicity. The concentration in these materials is high (approx. 1 mg TEQ TCDD per kg waste⁴¹). Copper content is in the range of 1–10% (Cavaterra & Novara 1986). An average value of 5.5% is assumed here. The remainder is assumed to be SiO_2 . Disposal in underground deposit and in hazardous waste incinerators are inventoried. Disposal in underground deposit in a 200 liter steel drum without cement solidification is assumed.

The fate of dioxin is not heeded waste-specifically in the HWI model (see part II). Instead a *constant* emission of 3 ng TEQ is attributed to *each* kg of incinerated waste, regardless of input (process-specific emission). With an input 1 mg TEQ per kg of spent catalyst, this suggests a dioxin removal of 99.9997%. This is in the vicinity of a common destruction and removal efficiency (DRE) for highly toxic compounds (99.9999%). No additional dioxin emissions are inventoried for this waste.

4.24.4 Disposal of chemical production facility infrastructure

The disposal of the infrastructure of chemical production facilities (including electronics) is inventoried in (Althaus et al. 2003a).

4.25 Agricultural products

4.25.1 Starch production effluent

During starch production from maize and potatoes a wastewater is produced, which is discharged to canalisation (Nemecek et al. 2003). Treatment in municipal wastewater treatment plant of size class 2 is assumed. For the disposal of the resulting treatment sludge 60% incineration and 40% agricultural application is assumed.

³⁸ 5% estimated concentration of silver in catalyst and an estimated removal efficiency of 99.99% in recycling.

³⁹ 5% estimated concentration of metals in catalyst and an estimated removal efficiency of 99.99% in recycling.

⁴⁰ Ethylene dichloride is the educt for the production of the vinyl chloride monomer (VCM, $CH_2=CH-Cl$), which is the building block of polyvinylchloride resins (PVC).

⁴¹ 20.3 µg I-TEQ TCDD in 20.3 g spent catalyst per ton of produced EDC (Althaus et al. 2003a).

4.26 Rail transport

4.26.1 Used rail track support material

Large masses of the gravel and sand that are used to build the supporting base (ballast) of a rail track, can be recycled after use (Spielmann et al. 2003). Usually a cleaning step is necessary to remove fines from old material. Fines are unwanted for reasons of mechanical track stability. The fines removed are assumed to be 2 w% of the total waste material, based on information from Wigro (1999). Sometimes track material is also polluted with heavy metals or organic pollutants, e.g. from herbicide use or from accidental or routine spills of chemicals or lubricants. Such material can be cleaned and reused. The pollutant fraction separated during cleaning needs to be disposed. Disposal in a residual material landfill is considered here. Since the landfill model only heeds chemical elements as input and not specific compounds, possible herbicide or mineral oil traces are not heeded here.

Tab. 4.9 Calculation of the removed parts in polluted rail track material

Pollutant	A TLV	B % at TLV	C max. conc.	D removed	E composition
	ppm	sum %	ppm	mg/kg	ppm
Arsenic	15	98% ¹	15.31	0.0361	38
Lead	50	97%	51.5	0.196	205
Cadmium	1	98%	1.02	0.00241	2.51
Cr tot.	50	80%	62.5	3.5	3655
Cr-VI	0.05	98% ¹	0.051	0.000120	0.126
Copper	40	75%	53.3	4.33	4525
Nickel	50	90%	55.6	1.056	1102
Mercury	0.5	98% ¹	0.51	0.0012	1.26
Zinc	150	95%	157.9	1.14	1195
Inert fines				947 ²	989'275
Sum				958	1'000'000

A: Threshold limit value TLV for unpolluted excavation material

B: Percentage of rail track material samples exceeding the limit value in A (160 samples in Fuhrer et al. 2000:15)

C: Estimated maximal concentration in rail track material samples, assuming a triangular probability distribution with a maximal likelihood value MLV of zero.

D: Mass of pollutant that needs to be removed in a cleaning step from 1kg *average* rail track material, assuming that on average a share of (100% minus column B) needs to be cleaned, that the treated material originally contained a concentration C of pollutants, and that the cleaned material contains a final concentration of 90% of column A. In total, a mass of 958 mg/kg needs to be removed.

E: Composition of the removed pollutants/fines in cleaned rail track material.

1: Estimate based on value for cadmium.

2: Inert fines in track material. Geometric mean of $20'000\text{ppm} \cdot (100\% \text{ minus column B})$. $20'000 \text{ ppm} = 2 \text{ w\%}$

Per kilogram of average rail track material, approximately 958 milligram of a concentrated pollutant fraction with the composition given in column E of Tab. 4.9 is separated. The calculation procedure is explained in the table footnotes. This material needs to be landfilled in a residual material landfill. Solidification with cement is assumed. Inert fines are assumed to be silicon. The concentrations in the disposed composition is rather high. The mass of pollutants per kg old rail track material (i.e. residue composition times 958 mg per kg old rail track material) should be in the right order of magnitude.

For each kilogram of all rail track material, a total of 2 w% of inert fines are removed. A small part of that (0.0947 w%) is already separated in the washing of *polluted* materials (cf. Tab. 4.9) and is landfilled. A remainder of 1.905 w% is separated from *unpolluted* materials (for mechanical reasons) and is either recycled in building material production (e.g. cement kiln) or is landfilled in an inert material landfill.

4.27 Road transport

4.27.1 Used mineral oil to hazardous waste incineration

Used mineral oil is obtained in the maintenance of transportation vehicles for road and rail. Metals traces are adopted from (VDZ 2000:18) for As, Pb, Cd, Cr, Cu, Ni, Hg, Tl and Zn. Further waste composition (C, H and P) is adopted from (BUWAL 1993, Citepa 1981).

4.27.2 Asphalt to sanitary landfill

Asphalt is a mixture of bitumen and minerals (gravel, sand stone, clay, lime etc.). Waste asphalt originates from maintenance and dismantling of transportation infrastructure (roads, ports). Waste asphalt can be reprocessed on site and only small amounts are landfilled. A bitumen content of 5.3% is adopted from (Frischknecht et al. 1996:B.18). Bitumen composition is detailed in the according to chapter 4.27.3 'Bitumen to sanitary landfill' on page 60. Mineral composition is inventoried as SiO₂.

4.27.3 Bitumen to sanitary landfill

Bitumen wastes originate e.g. from building materials (e.g. water-proof seals). Bitumen composition is adopted from (Frischknecht et al. 1996:IV.19) and complemented with sodium, aluminium, silicon, calcium, phosphor, lead, arsenic, molybdenum, cobalt, chromium, selenium, and mercury content of heavy fuel oil given in (Frischknecht et al. 1996:IV.18). This approximation is justified as bitumen and heavy fuel oil originate from the same distilling fraction during refining (Frischknecht et al. 1996:IV.130).

4.27.4 Wastewater from lorry production

A wastewater charged with sulfur, chlorine, nickel, lead, zinc in the sub-ppmw range is discharged from lorry production sites to the local sewer system (Spielmann et al. 2003). Treatment in a class 1 plant (large city sewage treatment) is assumed.

4.27.5 Catalytic converter

Catalytic converters are recycled for their platinum content. Used catalytic converters are also disposed in underground deposits. For disposal an inventory module 'catalytic converter for cars to underground deposit' is created (a similar module 'catalytic converter NO_x reduction to underground deposit' shall be used for stationary industrial converters). Deposition in a 200 liter steel drum without cement solidification is assumed.

4.27.6 Automotive shredder residue (ASR) to incineration

Automotive shredder residue (ASR, or German: RESH, 'Reststoffe aus Shredderwerken') is a mixture of fine remains from the shredding of used cars (Spielmann et al. 2003) and tractors (Nemecek et al. 2003). It mainly consists of fine plastic and metal particles and contains large quantities of heavy metals. In Switzerland approximately 50'000 to 60'000 tons of ASR are produced annually. Plans are underway to build the first European ASR recycling plant in Switzerland, which extracts metals in a recyclable form from ASR (Strässle 2002). Operation is expected to start approximately in 2005.

As an intermediary solution, ASR is incinerated in Swiss or foreign MSWIs. During 1996 to 2000, only 0.6 % were landfilled in the Swiss municipal landfill Sorval VD (IGEA 2001). Approximately 18% was disposed in the recycling plant Citron, Le Havre, that produced marketable fractions of Hg, ZnO, CdO, Mn and Fe.

To inventory the disposal of ASR the module 'plastics, mixture to municipal incineration' is used for all polymers (rubber, plastics, lacquer) along with additional modules for small metallic parts to municipal incineration (aluminium, steel, copper, zinc, lead). The small metallic parts are assumed to

be finely dispersed, since larger bulk metals are recycled. Accordingly, the metals are assumed to be only 50% inert (transferred completely to bottom ash) and 50% burnable (partly transferred to raw gas during incineration). For aluminium the aluminium composition found in MSWI is used (cf. chapter 4.4.2 'Aluminium' on page 37). For steel the iron scrap composition found in MSWI is used (cf. chapter 4.4.1 'Steel' on page 37). For copper, a composition estimated for copper particles recycled from MSWI bottom ash is used (Doka 2002:29).

4.27.7 Automobile tyres disposal

In Switzerland, used tyres are disposed via specific disposal companies. Old tyres are sorted out for export⁴² and for final disposal. The latter takes place predominantly in cement kilns and to a minor part in municipal waste incinerators. In Switzerland cement kilns burned 15'000 tons old tyres in 2000 (BUWAL 2001l).

Rubber incineration in cement kilns is treated in (Kellenberger et al. 2003). There, emissions from rubber incineration (and other alternative fuels) is *fully accounted to cement production*, i.e. waste rubber is perceived as valuable alternative fuel in the cement production process inventory and less as a burdensome waste⁴³.

This understanding conveyed by the allocation choice in the inventory for cement production is *not* reflected by involved costs: cement kilns are *paid a fee* to accept old tyres⁴⁴. This suggests that tyres are in fact regarded as a waste and not a valuable commodity (i.e. not a fuel). However, this probably reflects the circumstance that *no other facility* will accept old tyres without charge. Also zinc contents limit the use of tyres as alternative fuel in cement kilns (BUWAL 1998a), which demands attention in cement kiln operation. This latter point also establishes the value of tyres as a waste, rather than a fuel.

However, this allocation issue is managed here in order to be compatible with the inventory of cement production: tyres disposed in cement kilns (approx. 50% of all tyres) have no further burden to the road vehicle life cycle (cut-off to recycling). But tyres disposed in municipal incinerators (approx. 10% of all tyres) are inventoried with the disposal modules 'steel' and 'rubber' to municipal incineration and *the resulting burdens are fully allocated to the road vehicle life cycle*. Exported old tyres (approx. 40% of all tyres) are treated as recycling (cut-off system boundary).

4.28 Ship transport

4.28.1 Bilge oil to hazardous waste incineration

Bilge oil is a liquid pumped from the bottom of ships. It contains wasted mineral and lubricating oil residues. A small fraction (0.5 – 1 w%) of the lubricants and fuels in ship operation is transferred to bilge. Bilge composition is adopted from (Frischknecht et al. 1993:IV.112) with a water content of 90%.

⁴² Run down tyres with insufficient profile are e.g. wanted in sand desert regions.

⁴³ Also, the incineration residues of fuels in cement kilns (including residues from alternative fuels) become part of the cement product.

⁴⁴ Personal communication with Michael Spielmann, UNS-ETH Zürich, according to information from Tyre association Switzerland (Reifen Verband Schweiz) of December 11, 2002

5 Life Cycle Inventory of Municipal Waste Collection

5.1 Introduction

Prior to the disposal, waste has to be hauled to the disposal site – be it a waste incinerator or a landfill. This transport is usually *not included* in the waste disposal inventories (Frischknecht et al. 2003a)⁴⁵. The waste transport to incinerators and landfills must be inventoried with the waste-producing process. Waste of businesses and industries can be transported directly by lorry to the disposal site. In case of household or comparable waste (municipal solid waste MSW) municipal collection of the waste with special waste collection vehicles usually applies. Here the inventory of such municipal waste collection is presented to be used in the inventories of waste-producing processes.

The author of inventories of waste-producing processes has to decide if direct transport or municipal collection is more appropriate for the generated wastes. Of a total mass of 2.8 Mio. tons of MSW incinerated in Swiss MSWIs in 2000, a share of 59% (1.66 Mio. tons) were delivered by municipal collection (BUWAL 2001c).

The functional unit of the waste collection service is ton-kilometers (tkm), including an average load of 50%.

The inventory presented here is based on a lorry of the 16 t class inventoried in (Spielmann et al. 2003). Adaptations are documented below.

5.2 Fuel consumption of waste collection

Fuel consumption figures of waste collection from different studies are presented in Tab. 5.1. This fuel consumption includes 'Stop&Go' transport and hydraulic waste compression.

A collection truck has a nominal capacity of 8.2 t (Bürgin & Steinemann 1991). The lorry is gradually filled up along the collection route. When fully loaded the lorry returns to the disposal plant and returning empty to the collection route. This leads to an average load factor of approximately 50%. Per vehicle-kilometer (vkm) an average of 4.1 ton-kilometers (tkm) is accomplished.

Tab. 5.1 Waste collection data from different studies

per t MSW	Area	Distance km	Diesel consumption l/t waste	NO _x emission g/t waste
Bürgin & Steinemann 1991	Hinwil	10.4	5.1	240
Baumann & Ménard 1992	Winterthur	–	4.35	–
Gebler 1992	Germany	–	5	36
ATAL 1993	Canton Zürich	9.4	4.8	>220
AWZ ²	Zürich municipality	3.1	2.1	–
This study		10¹	4	see Tab. 5.2 below

¹ suggested standard transportation distance to MSWI in ecoinvent 2000

² communication with Peter Keller, Abfuhrwesen der Stadt Zürich, AWZ, January 23, 1996

⁴⁵ The exception is the disposal of building wastes (part V) which includes on-site demolition *and subsequent transport* of the waste material. Also in wastewater treatment (part IV) transport of wastewater in sewers is included in the inventory.

Per functional unit (tkm) 0.4 l of diesel are consumed (4 l/t / 10 km from Tab. 5.1). With a density of 0.84 kg/l the diesel consumption is 0.336 kg/tkm. A GSD value of 111% is estimated from the range in Tab. 5.1.

5.3 Direct air emissions of waste collection

5.3.1 Exhaust emissions

Exhaust emissions are based on emissions for a lorry of the 16 t class in (Spielmann et al. 2003). The 16 t class includes lorries with gross weights ranging from 3.5 – 20 t. Waste collection leads to an enforced Stop&Go driving cycle. This type of driving leads to increased emissions of volatile organic compounds VOC, particulate matter PM₁₀ and carbon monoxide CO, but reduced NO_x emissions. The diesel emission factors from normal lorry transport are adjusted here accordingly.

Tab. 5.2 Adjustment of emission factors for 16t lorry to Stop&Go driving of waste collector lorries.

Emission	Original emission factor for 16t lorry ¹ g/l diesel	Adjustment factor for Stop&Go driving ²	Adjusted emission factor for Stop&Go driving ³ g/l diesel	Adjusted emission factor for Stop&Go driving ⁴ mg/tkm
Carbon dioxide CO ₂	2646	1 ⁵	2646	1'059'000
Carbon monoxide CO	5.527	1.24	6.832	2733
Particulate Matter (PM ₁₀)	1.483	1.11	1.642	656.6
Nitrogen Oxides NO _x	25.3	0.75	18.95	7579
Nitrous Oxide N ₂ O	0.1312	1 ⁶	0.1312	52.49
Ammonia NH ₃	0.01988	1 ⁶	0.01988	7.952
Sulphur Dioxide SO ₂	0.505	1 ⁵	0.505	202
NM VOC	4.06	2.13	8.637	3455
Methane	0.0994	2.13	0.2115	84.59
Benzene	0.07952	2.13	0.1692	67.67
Toluene	0.03181	2.13	0.06767	27.07
Xylene	0.03181	2.13	0.06767	27.07
Cadmium	0.00000839	1 ⁵	0.00000839	0.00336 ⁷
Copper	0.001427	1 ⁵	0.001427	0.571
Chromium	0.00004215	1 ⁵	0.00004215	0.01686
Nickel	0.00005885	1 ⁵	0.00005885	0.02354
Selenium	0.00000839	1 ⁵	0.00000839	0.003356
Zinc	0.000839	1 ⁵	0.000839	0.3355 ⁷

1 Calculated from data given in (Spielmann et al. 2003) with a specific diesel consumption of 25.15 l/100vkm (or 211.26g/vkm) given in (Spielmann et al. 2003).

2 Average adjustment factors from stationary cycle ('ECE-13') to Stop&Go cycles ('US-Transient and TÜV-FIGE') from (BUWAL 1995a:183).

3 Adjusted emission factors from original emission factor for 16t lorry and adjustment factors for Stop&Go.

4 Derived from a fuel consumption of 0.4 l/tkm for waste collection (see chapter 5.2 'Fuel consumption of waste collection' on page 62).

5 These emission factors (per l diesel) can be assumed to remain *unchanged* in Stop&Go driving.

6 These emission factors (per l diesel) can be assumed to *change* in Stop&Go driving, but no adjustment factors could be derived from (BUWAL 1995a:183).

7 Emissions from tyre abrasion are added below.

5.3.2 Cold start emissions and petrol fumes

The emissions involved in fueling a vehicle (petrol fumes) are taken into account in the fuel module (petrol, at regional storage).

In (Spielmann et al. 2003) VOC evaporation from hot motors and cold start emissions were only inventoried for delivery vans, but not for heavy duty vehicles (>3.5 t). This data gap is not corrected here.

5.3.3 Particle emissions

Particle emissions are inventoried in three size classes in ecoinvent 2000. The complementary ranges from 0 – 2.5 µm, from 2.5 – 10 µm, and >10 µm aerodynamic diameter are discerned in the inventory. These size classes are commonly termed 'fine', 'coarse' and 'large' particles, respectively.

Exhaust emissions

A PM₁₀ value for exhaust emissions is derived in Tab. 5.2. The TSP distribution profile is adopted from TSP emissions for a 16 t lorry from (Spielmann et al. 2003): 88.7% fine, 7.5% coarse, and 3.9% large. Other sources of PM are added below.

Tab. 5.3 PM emission factors for waste collection.

PM category	PM profile	Emission factors g PM/tkm
fine <2.5µm	88.7%	0.6057 ¹
coarse 10µm-2.5µm	7.5%	0.0509 ¹
large >10 µm	3.9%	0.0263
TSP	100%	0.683

1 Please note that the sum of fine and coarse particles gives the emission factor of 0.6566 g PM₁₀/tkm as given in Tab. 5.2.

Tyre abrasion

Values of 0.2 g PM₁₀ and 0.768 g TSP per vkm are adopted from (Hüglin 2000) for Swiss heavy duty vehicles in urban Stop&Go traffic. A fraction of 5% of PM₁₀ is assumed to be PM_{2.5}. A factor of 4.1 tkm per vkm is used to calculate emission factors per tkm.

Heavy metals are present in the tyre, which are inventoried additionally. Concentrations of heavy metals in tyre material is 1.6% zinc, 26 ppm lead and 6 ppm cadmium. Based on total TSP matter from tyres, emissions to air of 3 mg zinc, 0.0049 mg lead, and 0.0011 mg cadmium per tkm waste collection are calculated.

Brake abrasion

Values from (Lükewille 2001) are adopted for European heavy duty vehicles (7.1 mg/vkm PM_{2.5}, 22.8 mg/vkm PM₁₀, 27.5 mg/vkm TSP). A factor of 4.1 tkm per vkm is used to calculate emission factors per tkm.

Though not considered a major source of particles, the materials involved in brake abrasion are different, than for tyre abrasion. The involved toxic materials have to be inventoried additionally to the particle emission (Frischknecht et al. 2003a). Trimbacher & Neinavaie (2002:30) found materials like brass (Cu-Zn) and copper, as well as iron, graphite and silicates to be typical for deposited particles originating from brake abrasion in road traffic. Österle et al. (2000) give the composition of one case study sample of a braking lining material to be composed of 48% barite (BaSO₄), 14% vermiculite (SiO₂), 19% phenolic resin, 4.6% antimony sulfide Sb₂S₃, 6.4% aramide, 7.5% natural rubber, 0.3% Sulfur (S), 0.1% Zinc oxide ZnO. The braking layer is however just a part of the emitted braking particles. Due to lack of further quantitative information the *composition* of emitted brake lining particles is not inventoried.

Road abrasion and re-suspended road dust

The values given in (Spielmann et al. 2003) are adopted without change. A factor of 4.1 tkm per vkm is used to calculate emission factors per tkm.

Tab. 5.4 Synopsis of particle emissions from all sources.

g/tkm	PM _{2.5}	PM _{2.5} -PM ₁₀	PM _{>10}	Sum TSP
From fuel	<u>0.606</u>	<u>0.0509</u>	0.0263	<u>0.683</u>
Tyre abrasion	0.00244	0.0463	<u>0.139</u>	0.187
Brake abrasion	0.00173	0.00383	0.00115	0.00671
Road abrasion	0.0051	0.00417	0.00927	0.0185
Total	0.615	0.105	0.175	0.896

Underlined figures indicate dominant contributions

Exhaust emissions are generally the largest source of particles (indicated by underlining in Tab. 5.4), except for PM_{>10} where tyre abrasion is dominant.

5.3.4 Emission category

Seeing that waste collection vehicles operate mainly in populated areas to collect waste of municipalities, all air emissions are inventoried as being emitted in areas with high population density.

5.4 Infrastructure for waste collection vehicle

In (Spielmann et al. 2003) the expenditures for construction, maintenance and disposal of vehicle infrastructure is inventoried per one single vehicle unit. For the 16 t lorry class an average load of 2.93 tons is assumed and a vehicle lifetime mileage 540'000 vkm. This amounts to a lifetime service performance of 1'582'200 tkm per vehicle unit. Per tkm with a 16 t lorry a fraction of $6.27 \cdot 10^{-7}$ vehicle units is inventoried.

These figures change for the waste collection vehicle. It is estimated here that the total mass of the collection vehicle is 30% higher than that of a generic 16 t lorry. The total mass of the waste collection vehicle is 21t. Expenditures for lorry construction and disposal, but not maintenance, are increased accordingly. The average load is 4.1 tons and a vehicle lifetime mileage 540'000 vkm is assumed⁴⁶, resulting in a lifetime service performance of 2'214'000 tkm per vehicle unit. Per tkm a fraction of $4.52 \cdot 10^{-7}$ 16 t lorry maintenance units is inventoried. For construction and disposal the figure is a factor 1.3 larger.

⁴⁶ This same figure is used for all lorries, 16 t, 28 t, and 40 t class, in (Spielmann et al. 2003).

Tab. 5.5 Expenditures for construction, maintenance and disposal of waste collection vehicle infrastructure

	Mass increase in waste collection vehicle compared to 16 t lorry	Units of 16 t lorry infrastructure per tkm with waste collection vehicle
construction	30%	0.000000587
maintenance	0%	0.000000452
disposal	30%	0.000000587

5.5 Road infrastructure

In (Spielmann et al. 2007:92) the expenditures for construction, maintenance and disposal of roads are given in meter-years, i.e. one meter of average road mix for the duration of one year, denoted as 'ma'. Construction and disposal of the road infrastructure is allocated according to the yearly gross-tonne kilometric performance on the roads. For different classes of vehicles, different infrastructure expenditures result. The calculations of (Spielmann et al. 2007:92) are adapted here to the waste collection lorry.

Total road length in Switzerland is 71'600 km. Annual gross-tonne kilometric performance on the roads is 134'000 Million Gtkm/a. Thus per Gtkm an infrastructure of 0.000537 ma is necessary. The waste collection lorry has a gross weight of 21 tons and an average load of 4.1 tons, the Gtkm/tkm ratio is therefore 5.12. Per tkm a road infrastructure of 0.002735 ma is necessary ($=0.000537 \text{ ma/Gtkm} * 5.12 \text{ Gtkm/tkm}$). This figure relates to the construction and disposal of the road infrastructure.

The annual *maintenance* of the road infrastructure is allocated according to the yearly vehicle-kilometric performance on the roads. Annual vehicle-kilometric performance on Swiss roads is 62'761 million vkm. Thus per vkm an infrastructure of 0.001141 ma is necessary ($=71'600'000 \text{ ma}/62'761 \text{ Mio vkm}$). The waste collection lorry has an average load of 4.1 tons, therefore the vkm/tkm-ratio is 0.244 ($=1/4.1 \text{ t}$). Per tkm a road infrastructure maintenance of 0.000278 ma is necessary ($=0.001141 \text{ ma/vkm} * 0.244 \text{ vkm/tkm}$). This figure relates only to the maintenance of the road infrastructure.

5.6 Uncertainties

For uncertainty ranges of all exchanges the same GSD values as derived in (Spielmann et al. 2003) for 16t lorries are used without modification. The original data covers special vehicles like collection vehicles.

5.7 Inventoried unit process data

Tab. 5.6 Inventoried exchanges for municipal waste collection

Name	Location	Infrastructure Process	Unit	transport, municipal waste collection, lorry 21t	StandardDeviation95% GSD ²
Location				CH	
InfrastructureProcess				0	
Unit				tkm	
diesel, at regional storage	CH	0	kg	0.336	123.2%
Carbon dioxide, fossil			kg	1.05857654	107.1%
Carbon monoxide, fossil			kg	0.00273271	500.4%
Nitrogen oxides			kg	0.00757928	150.4%
Dinitrogen monoxide			kg	5.2485E-05	150.4%
Ammonia			kg	7.9523E-06	150.4%
Sulfur dioxide			kg	0.00020199	107.1%
NM VOC, non-methane volatile organic compounds, unspecified origin			kg	0.00345477	150.4%
Methane, fossil			kg	8.4593E-05	150.4%
Benzene			kg	6.7674E-05	150.4%
Toluene			kg	2.707E-05	150.4%
Xylene			kg	2.707E-05	150.4%
Cadmium			kg	4.4798E-09	510.8%
Copper			kg	5.7097E-07	510.8%
Chromium			kg	1.6859E-08	510.8%
Nickel			kg	2.3539E-08	510.8%
Selenium			kg	3.3559E-09	510.8%
Zinc			kg	3.3327E-06	510.8%
Lead			kg	4.8702E-09	510.8%
Particulates, < 2.5 um			kg	0.00061497	200.4%
Particulates, > 2.5 um, and < 10um			kg	0.00010526	201.7%
Particulates, > 10 um			kg	0.00017529	206.1%
lorry 21t, municipal waste collection	CH	1	unit	4.517E-07	110.0%
operation, maintenance, road	CH	1	ma	0.000278	110.0%
road	CH	1	ma	0.002735	110.0%
disposal, road	RER	1	ma	0.002735	110.0%

Tab. 5.7 Inventoried exchanges for infrastructure of municipal waste collection lorry

Name	Location	Infrastructure Process	Unit	lorry 21t, municipal waste collection	StandardDeviation95% GSD ²
Location				CH	
InfrastructureProcess				1	
Unit				unit	
lorry 16t	RER	1	unit	1.3	110.0%
maintenance, lorry 16t	RER	1	unit	1	201.3%
disposal, lorry 16t	RER	1	unit	1.3	110.0%

6 Data Quality Considerations

6.1 The Pedigree approach

Within the ecoinvent 2000 project each LCI figure in the database is given a measure of uncertainty. This measure is given either as a standard deviation (for normal distribution), as a geometric standard deviation (GSD^2 , for lognormal distribution), as a range with minimal and maximal values (for triangular and rectangular distributions). The lognormal distribution is the default distribution type.

Often in LCI, the collected data is too sparse to conduct proper uncertainty calculations. To obtain a measure of uncertainty a generic method to judge a particular data item was devised for the ecoinvent 2000 project. This so called 'Pedigree approach' is based on (Weidema & Wesnaes 1996). The Pedigree approach in ecoinvent 2000 is described in (Frischknecht et al. 2003a).

6.2 Mathematical Operations

In this project, calculation tools for inventories are created. These tools shall not only calculate mean values for inventories, but also the respective uncertainty data. This necessitates that mathematical operations for uncertainty data are known. At various places, uncertain values are involved in mathematical operations like addition or multiplication. For example, air emissions are derived from waste composition multiplied by the transfer coefficients. Or addition is used, e.g. for the combined transport distances of consumed auxiliary materials. It is necessary to know, how the geometric standard deviation GSD is calculated in such operations.

Multiplication

If lognormally distributed values are multiplied, the resulting distribution is strictly lognormal again and the GSD can be calculated exactly. If GSD_i are the geometric standard deviations of log-normally distributed factors of a multiplication, the GSD of the resulting product GSD_{prod} can be calculated from the following formula:

$$\text{Eq. 6.1} \quad \ln^2(GSD_{prod}) = \sum_i \ln^2(GSD_i)$$

and thus

$$\text{Eq. 6.2} \quad GSD_{prod} = e^{\sqrt{\sum_i \ln^2(GSD_i)}}$$

The mean value of the product m_{prod} is the usual product of all mean values of the factors m_i .

$$\text{Eq. 6.3} \quad m_{prod} = \prod_i m_i$$

Addition

If lognormally distributed values are added, the resulting distribution is not formally defined and the GSD cannot be calculated with a general and exact formula. However, approximations are known that calculate a GSD based on the assumption, that the resulting sum is lognormally distributed again. If m_i are the (geometric) mean values of the lognormally distributed summands and GSD_i are the corresponding geometric standard deviations, the GSD of the sum GSD_{sum} can be calculated according to the Wilkinson-Fenton approximation (Fenton 1960):

$$\text{Eq. 6.4} \quad GSD_{sum} = e^{\sqrt{\frac{av_{sum}}{am_{sum}} + 1}} \quad (\text{GSD of the sum})$$

where

$$\text{Eq. 6.5} \quad am_{sum} = \sum_i m_i \cdot e^{\ln^2(GSD_i)/2} \quad (\text{Arithmetic mean of the sum})$$

$$\text{Eq. 6.6} \quad av_{sum} = \sum_i am_i \cdot (e^{\ln^2(GSD_i)} - 1) \quad (\text{Arithmetic variance of the sum})$$

$$\text{Eq. 6.7} \quad am_i = m_i \cdot e^{\ln^2(GSD_i)/2} \quad (\text{Arithmetic mean of summand } i)$$

After Wilkinson's approximation, the proper mean value of the sum should be derived from the arithmetic mean of the sum am_{sum} and GSD_{sum} using the formula:

$$\text{Eq. 6.8} \quad m_{sum} = am_{sum} / e^{\ln^2(GSD_{sum})/2}$$

An 'improper' way – improper within the Wilkinson-Fenton approximation – to calculate the mean value of the sum would be to use the simple sum of the mean values:

$$\text{Eq. 6.9} \quad m_{sum} = \sum_i m_i$$

The 'proper' mean in Eq. 6.8 is not equal to the 'improper' simple sum in Eq. 6.9. The discrepancy is usually small, especially if small values are added to large values. The difference can become significant, if the summands all have the same order of magnitude and have larger uncertainties associated with them. For an LCI this is quite a disturbing fact. It means that a *sum* of inventoried figures can increase *merely by the decision that the summands have a lognormal and not a normal distribution* (in the case of normal distribution the simple sum as in Eq. 6.9 would be the correct value for the mean of the sum). In other words the 'proper' Wilkinson-Fenton mean in Eq. 6.8 violates mass balance considerations.

For example in the MSWI three outputs are joined and landfilled as residual material (boiler ash, precipitator ash and scrubber sludge). To calculate the GSD of the resulting sum of elements the Wilkinson-Fenton approximation is used (Eq. 6.4 – Eq. 6.7). If, for example, the 'proper' formula according to Wilkinson-Fenton (Eq. 6.8) were used for the mean value of the sum, the masses of mercury and cadmium in the combined residual waste would increase by 25% compared to the simple sum (Eq. 6.9).

Accordingly, using the 'proper' formula Eq. 6.8 would suggest that mass balances of the MSWI violate the principle of mass conservation. This seems nonsensical. Hence, in order to inventory mass-conserving mean values also for sums I choose to deviate from the Wilkinson-Fenton approximation in this study and use the 'improper' simple sum (Eq. 6.9) for the mean value of the sum.

Division

No exact or approximate formula for the calculation of GSD values of divisions of lognormally distributed values is available. A very coarse method to derive a uncertainty estimate for divisions is developed here.

Let's consider a division of two lognormally distributed values A and B with mean values m_A and m_B . The upper confidence value of A is $u_A = m_A \cdot GSD_A^2$. The lower confidence value of B is $l_B = m_B / GSD_B^2$. An upper confidence value of the result $C = (A/B)$ could be estimated to be $u_C = u_A / l_B$. Assuming that the result C is still lognormally distributed⁴⁷, the upper confidence value of C can be used to calculate the GSD of C, since $u_C = m_C \cdot GSD_C^2$. Thus,

⁴⁷ Generally, this is not likely, since dividing by a lognormally distributed and hence left-slanted distribution B will result in a *right-slanted* distribution for C, which will no longer be lognormally distributed.

$$\text{Eq. 6.10 } GSD_C = \sqrt{u_C / m_C}$$

And with $u_C = u_A / l_B = (m_A * GSD_A^2 * GSD_B^2 / m_B)$ and $m_C = (m_A / m_B)$

$$\text{Eq. 6.11 } GSD_C = \sqrt{\frac{(m_A \cdot GSD_A^2 \cdot GSD_B^2)}{m_B} \cdot \frac{m_B}{m_A}} = GSD_A \cdot GSD_B$$

The formula suggests that the GSD of the result is the simple product of the GSDs of the two constituting terms. This makes sense. Assuming that GSD_B is 100%, the GSD_C is exactly equal to GSD_A . The division with a *certain* figure ($GSD_B = 100\%$) is merely a shrinking of the scale for A in the probability distribution graph. The GSD_A , which is a *relative* measure for the uncertainty of A, is unaffected by this mere 'change of scale', i.e. $GSD_C = GSD_A$. Also for a certain A ($GSD_A = 100\%$) and uncertain B, the formula makes sense⁴⁸.

The formula increases quickly for large GSD values. This is an effect of combining the upper confidence value of A with the lower confidence value of B⁴⁹.

Subtraction

No exact or approximate formula for the calculation of GSD values of subtractions of lognormally distributed values is available. A very coarse method to derive an uncertainty estimate for differences is developed here.

Let's consider a difference of two lognormally distributed values A and B with mean values m_A and m_B . The upper confidence value of A is $u_A = m_A * GSD_A^2$. An upper confidence value of the result $C = (A-B)$ could be estimated to be $u_C = u_A - m_B$, if B has no uncertainty. Assuming that the result C is still lognormally distributed⁵⁰, the upper confidence value of C can be used to calculate the GSD of C, since $u_C = m_C * GSD_C^2$. Thus,

$$\text{Eq. 6.12 } GSD_C = \sqrt{u_C / m_C}$$

And with $u_C = u_A - m_B = m_A * GSD_A^2 - m_B$ and $m_C = (m_A - m_B)$

$$\text{Eq. 6.13 } GSD_C = \sqrt{(m_A \cdot GSD_A^2 - m_B) / (m_A - m_B)}$$

This formula features some characteristics which are correct for uncertainties of differences. The formula results in *greatly enlarged* GSD values for the result C, if A is close to B, i.e. if C is the *relatively small* difference of two large values. On the other hand if $B=0$ the GSD_C for the result C is accurately equal to GSD_A (see Fig. 6.1).

⁴⁸ However the same two corollary points also hold for an unused ad hoc formula $GSD_C = GSD_A + (GSD_B - 1)$.

⁴⁹ Combining these two values results in a uncertainty range which is larger than the usual 95%-probability range.

⁵⁰ Generally, this is not likely, since subtracting a lognormally distributed and hence left-slanted distribution B with large uncertainty will result in a right-slanted distribution for C, which will no longer be lognormally distributed.

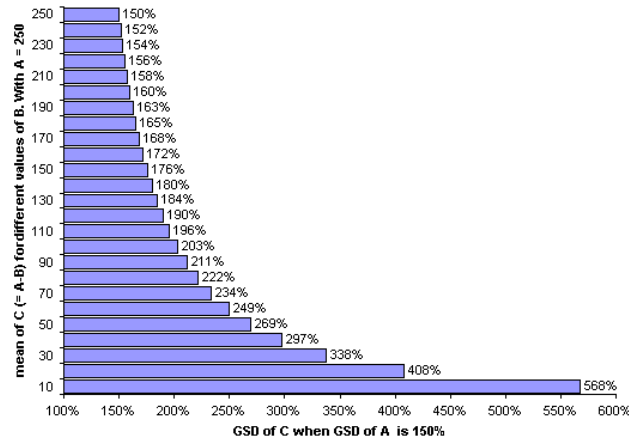


Fig. 6.1 GSD_C for the difference $C = (A-B)$ for $A=250$ and different values of B (with $GSD_A = GSD_B = 1.50$). Note how the GSD_C increases if C is small, i.e. when C is the difference of two large numbers

There would be other ways of estimating a GSD_C value. For example, the u_C could be estimated from the difference of the mean value of A minus the lower value for B (i.e. $u_C = m_A - m_B/GSD_B^2$). Several other possibilities exist, which seem a priori equally reasonable. Numerical testing for different values of m_A , m_B , GSD_A , and GSD_B however showed that the approach in Eq. 6.13 gives reasonable results, especially regarding following general prerequisites:

1. GSD_C is always larger than GSD_A , which is reasonable, as the GSD is a measure of *relative* uncertainty (i.e. not an *absolute* measure. It relates to the mean value). The relative uncertainty GSD_C must be larger than GSD_A , since the mean of C is smaller than the mean of A .
2. Granted that $m_C = (m_A - m_B)$ is positive, the term in the square root is never negative even for extreme values of m_A , m_B , and GSD_A . Otherwise, this would result in nonsensical imaginary numbers.

Other approaches do not fulfil these two prerequisites and are therefore not suitable as a general formula. As a disadvantage, the approach in Eq. 6.13 does not include the uncertainty of B , i.e. GSD_B is not used at all. This is equal to assuming that B has no uncertainty, which will usually not be the case. The uncertainty of C will be even larger if A and B are uncertain. So the Eq. 6.13 calculates a lax estimate for GSD_C . Attempts to include GSD_B in the formula for GSD_C failed to fulfil the two general prerequisites noted above.

Within this study, subtraction is only used in heat balance and waste heat calculations. So this approximation is not applied in essential parts of the LCI, like for pollutants.

Abbreviations

APME	Association of Plastics Manufacturers in Europe
BOD	Biological oxygen demand (German 'Biologischer Sauerstoffbedarf, BSB'). A measure for the content of readily degradable hydrocarbons. Equates to the amount of oxygen taken up in microbial decomposition of hydrocarbons – usually in dark incubation at 20°C and over 5 days (BOD ₅). Given as mass of absorbed oxygen (O ₂). Usually smaller than COD.
COD	Chemical oxygen demand (German 'Chemischer Sauerstoffbedarf, CSB'). A measure for the total content of degradable hydrocarbons. Equates to the amount of oxygen taken up in oxidation of hydrocarbons using a oxidation agent like potassium permanganate KMnO ₄ or potassium dichromate K ₂ Cr ₂ O ₇ . Given as mass of absorbed oxygen (O ₂). Usually larger than BOD.
DOC	Dissolved organic carbon. Given as mass of carbon (C _{org}). Smaller or equal than TOC (excludes organic carbon in undissolved solids and particulates).
EDC	1,2-ethylene dichloride, CH ₂ Cl-CH ₂ Cl.
GSD	(also SDg) Geometric Standard Deviation. Uncertainty measure for lognormally distributed values. The GSD cannot be smaller than 1 (i.e. 100%). The squared GSD value (GSD ²) can be used to calculate 95% confidence intervals around the mean. More information on the uncertainty calculations in ecoinvent 2000 can be found in chapter 1.2.3 'Uncertainty values GSD and GSD ² ' on page 12 and the methodology report (Frischknecht et al. 2003a).
MSWI	Municipal Solid Waste Incinerator (Swiss expression: KVA Kehrichtverbrennungsanlage, German expression: MVA Müllverbrennungsanlage)
PGM	platinum group metals (platinum, palladium, rhodium)
PM	particulate matter (air emissions), usually given as PM _{2.5} and PM ₁₀ , aerodynamic cross-section <2.5 micrometers or <10 micrometers respectively, Total suspended particles TSP is also particulate matter. In ecoinvent 2000 the complementary fractions from 0 – 2.5 µm (fine), from 2.5 – 10 µm (coarse), and >10 µm (large) aerodynamic diameter are discerned in the inventory.
SKN	SKN signifies the soluble Kjeldahl nitrogen. Kjeldahl nitrogen is a sum parameter for organic bound nitrogen (N _{org}) and ammonium (NH ₄ ⁺).
TEQ	Toxicity equivalents for polychlorinated dioxins and furans. Many TEQ factors are given by different scientific/government groups (USEPA TEQ, Nordic TEQ, Canadian TEQ, and i-TEQ. I-TEQ stands for international.). All of the factors are somewhat different and the way to handle not detects is different in each of these. All of them are trying to equate the non-TCDD isomers/congeners to 2,3,7,8-TCDD toxicity.
TKN	TKN signifies the total (soluble + particulate) Kjeldahl nitrogen. Kjeldahl nitrogen is a sum parameter for organic bound nitrogen (N _{org}) and ammonium (NH ₄ ⁺).
TLV	(legal) threshold limit value
TOC	Total organic carbon. Given as mass of carbon (C _{org}). Larger or equal than DOC, since it includes also carbon in undissolved solids or particulates.
TSP	Total suspended particles (air emissions). See also particulate matter PM.
VCM	Vinyl chloride monomer, CH ₂ =CH-Cl. Building block of polyvinylchloride resins
vkm	vehicle-kilometer

7 Appendix

Remark on displayed vs. significant digits

Figures in the tables of this report often feature several digits. This is not to imply that all the digits are significant or that the data displayed is very precise (it is mostly not). Showing several digits helps to minimise the avoidable accumulation of rounding mistakes along the chain of calculations within this study, and in possible future studies referring to the data published in this study. See also chapter 1 in part I.

7.1 Synopsis of solid waste compositions

The following tables give the waste compositions and characteristics of solid waste materials, as described in chapter 4 'Compositions of waste fractions' on page 28ff. The first column in the first table gives a description of the entry; the second column gives an abbreviation. In the tables on the subsequent pages only the abbreviation is repeated for space reasons. The entries 'Burnability', 'Share of metallic iron', 'Share of biogenic carbon', 'Degradability', and 'Solidification' are additional informations used in the calculation tool.

The uncertainty of the waste composition is determined in generic fashion described in chapter 'Uncertainty of waste composition data' in part II (incineration).

Tab. A. 1 Solid waste compositions

Burnability (1=burnable, 0=inert)	Burn	0	1	1	1	1	1
Name for waste:	Name	average residual materials from MSWI	PE	PP	PS	PVC	PET
German name for waste:	Ger.	Reststoffe aus Siedlungsabfall verbrennung	PE	PP	PS	PVC	PET-Flaschen
Comment	Com.						
upper heating value in MJ/kg	Ho		42.82	36.16	38.88	21.67	23.13
lower heating value in MJ/kg	Hu		42.47	32.78	38.67	21.51	22.95
Water content in kg/kg	H2O	0	0.004	0.159	0.002	0.002	0.002
Oxygen in kg/kg	O	0.344284119	0.038435229	0.032453843	0.03885217	0.016911274	0.368754661
Hydrogen in kg/kg	H		0.122492832	0.103430192	0.078046398	0.054921039	0.063379707
Carbon in kg/kg	C	0.083667394	0.822052914	0.694122992	0.868489598	0.440610989	0.553639029
Sulfur in kg/kg	S	0.028664206	0.000426897	0.000360462	0.000678747	0.001056955	0.000147866
Nitrogen in kg/kg	N		0.001297189	0.001095317	0.001942609	0.001056955	0.006456816
Phosphor in kg/kg	P	0.007811402					
Boron in kg/kg	B	0.000182671					
Chlorine in kg/kg	Cl	0.008536717	0.001461533	0.001234085	0.001132119	0.475285734	0.000788619
Bromium in kg/kg	Br	0.00088481	9.58959E-06	8.09723E-06	0.000882119	7.38389E-05	6.90041E-05
Fluorine in kg/kg	F	0.001359681	1.43844E-05	1.21459E-05	1.42496E-05	1.55062E-05	1.18293E-05
Iodine in kg/kg	I	1.72727E-08					
Silver in kg/kg	Ag	2.01588E-05					
Arsenic in kg/kg	As	2.0658E-05	1.82202E-06	1.53847E-06	1.84179E-06	2.0042E-06	1.87297E-06
Barium in kg/kg	Ba	0.000296431	0.00023974	0.000202431	0.000180786	0.000196728	4.92887E-05
Cadmium in kg/kg	Cd	0.000846912	3.39456E-05	2.86629E-05	2.33365E-05	2.75462E-06	3.25305E-06
Cobalt in kg/kg	Co	1.48199E-05	1.67818E-06	1.41702E-06	3.06221E-05	3.33224E-05	3.10519E-05
Chromium in kg/kg	Cr	0.002208858	1.24665E-05	1.05264E-05	3.9613E-05	4.31061E-05	5.42175E-06
Copper in kg/kg	Cu	0.009684757	4.13672E-05	3.49296E-05	0.000107563	2.82234E-05	1.18293E-05
Mercury in kg/kg	Hg	8.50685E-05	4.79479E-08	4.04862E-08	8.23109E-07	8.95692E-07	8.87196E-08
Manganese in kg/kg	Mn	0.002666712	3.00474E-05	2.53713E-05	7.74694E-05	8.43008E-05	1.77439E-05
Molybdenum in kg/kg	Mo	1.88094E-05					
Nickel in kg/kg	Ni	0.000395447	9.58959E-07	8.09723E-07	1.99679E-05	2.17287E-05	3.94309E-06
Lead in kg/kg	Pb	0.020800087	2.25959E-05	1.90795E-05	1.49353E-05	1.60024E-05	5.52033E-06
Antimony in kg/kg	Sb	0.001017446	1.00691E-05	8.5021E-06	4.16502E-05	4.5323E-05	0.000159202
Selenium in kg/kg	Se	1.5513E-05	2.01381E-06	1.70042E-06	2.03566E-06	2.21517E-06	2.07012E-06
Tin in kg/kg	Sn	0.002075626	3.81404E-06	3.22049E-06	2.54026E-05	0.000391413	9.85773E-07
Vanadium in kg/kg	V	7.44419E-05	0.002205606	0.001862364	0.000279322	0.000303952	0.001281505
Zinc in kg/kg	Zn	0.071908472	0.000342006	0.000288782	0.000834608	5.97746E-05	6.86755E-05
Beryllium in kg/kg	Be		4.79479E-07	4.04862E-07	4.84681E-07	5.2742E-07	4.92887E-07
Scandium in kg/kg	Sc						
Strontium in kg/kg	Sr		8.48679E-05	7.16605E-05	8.57885E-05	9.33534E-05	8.72409E-05
Titanium in kg/kg	Ti		0.000958959	0.000809723	0.000969362	0.001054841	0.000985773
Thallium in kg/kg	Tl		3.83584E-07	3.23889E-07	3.87745E-07	4.21936E-07	3.94309E-07
Tungsten in kg/kg	W						
Silicon in kg/kg	Si	0.090527405					
Iron in kg/kg	Fe	0.061648786	0.001534334	0.001295557	0.003611518	0.003929985	9.85773E-05
Calcium in kg/kg	Ca	0.08836641	0.002589189	0.002186253			0.000295732
Aluminium in kg/kg	Al	0.081225396	0.000191792	0.000161945	0.000193872	0.000210968	0.000197155
Potassium in kg/kg	K	0.049565635					
Magnesium in kg/kg	Mg	0.014926581	9.58959E-05	8.09723E-05			
Sodium in kg/kg	Na	0.02619855	0.001405354	0.00118665	0.001420599	0.001545869	0.001444651
sum wet mass	sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Share of iron in waste that is metallic/recyclable	Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Share of carbon in waste that is biogenic	C bio	61.1%	0.0%	0.0%	0.0%	0.0%	0.0%
Degradability of waste in a municipal landfill within 100 years	Degr.		1.0%	1.0%	1.0%	1.0%	1.0%

Solidification of waste to residual landfill with cement? (1=Yes,0=No)	Sol.	1					
used in following waste module materials	Mod.	average incineration residue.	polyethylene; bitumen sheet.	polypropylene.	polystyrene; bitumen sheet.	polyvinylchloride.	polyethylene terephthalate.

Tab. A. 2 Solid waste compositions (Cont.)

Burn	1	1	1	1	1	1	0	1
Name	PU	Mixed various plastics	rubber	Plastics from electronic consumer goods	Plastics from electronic industrial goods	PVF	tin sheet inert	tin volatile
Ger.	PU	Plastik Mischung	Gummi	Plastik aus Elektronikprodukten Haushalt	Plastik aus Elektronikprodukten Industrie	PVF	Weissblechdose inert	Zinn volatil
Com.			C, H theoretic. traces from BZL 2000b				inert Sn	volatile Sn
Ho	30.90	34.05	29.30	36.29	36.24	28.09	0.00	0.00
Hu	30.67	30.79	27.19	34.78	34.78	27.89	0.00	0.00
H2O	0.002	0.153		0.011128901	0.007213701	0.002	0	0
O	0.178704839	0.073717146		0.041	0.062	0.016911274		
H	0.084275578	0.105910077	0.106691333	0.075	0.073	0.071065595		
C	0.640697464	0.633594205	0.853530666	0.821	0.808	0.570132741		
S	0.000203074	0.001406743	0.0037	0.0003	0.0004	0.001056955		
N	0.060922104	0.006111991		0.0131	0.0135	0.001056955		
P				0.0006	0.0004			
B								
Cl	0.024368842	0.018130514	0.02	0.0063	0.0064		0.000751465	
Br	7.09336E-05	6.5319E-05		0.0068	0.0042	7.38389E-05		
F	1.48961E-05	1.3717E-05		0.00013	0.00037	0.329619425		
I								
Ag								
As	1.92534E-06	1.77294E-06		0.00001	0.000005	2.0042E-06		
Ba	0.000188987	0.000174028		0.000137	0.000217	0.000196728		
Cd	8.23008E-05	7.57864E-05	0.000008	0.000115	0.000186	2.75462E-06	3.00586E-05	
Co	3.20113E-05	2.94775E-05		2.94775E-05	2.94775E-05	3.33224E-05		
Cr	4.141E-05	3.81323E-05		0.000034	0.000071	4.31061E-05		
Cu	0.000321889	0.00029641		0.003194114	0.003194114	2.82234E-05	0.000199389	
Hg	8.6045E-07	7.92343E-07	0.000000001	0.0000014	0.0000003	8.95692E-07		
Mn	8.09839E-05	7.45738E-05		0.000004	0.0000098	8.43008E-05		
Mo				0.000002	0.000004			
Ni	2.08738E-05	1.92216E-05		0.000703	0.000299	2.17287E-05		
Pb	0.000473273	0.000435812	0.00007	0.000127	0.000165	1.60024E-05	0.000380742	
Sb	4.35397E-05	4.00934E-05		0.004795227	0.004795227	4.5323E-05		
Se	2.12801E-06	1.95957E-06		1.95957E-06	1.95957E-06	2.21517E-06		
Sn	2.6555E-05	2.44531E-05		0.000267	0.000139	0.000391413	0.00195381	1
V	0.000291993	0.000268881		0.0000005	0.000001	0.000303952		
Zn	0.000566535	0.000521692	0.016	0.000361	0.00052	5.97746E-05	0.000202395	
Be	5.06669E-07	4.66564E-07		4.66564E-07	4.66564E-07	5.2742E-07		
Sc								
Sr	8.96804E-05	8.25819E-05		8.25819E-05	8.25819E-05	9.33534E-05		
Ti	0.001013338	0.000933128		0.004187	0.004767	0.001054841		
Tl	4.05335E-07	3.73251E-07		3.73251E-07	3.73251E-07	4.21936E-07		
W								
Si				0.003547	0.002383			
Fe	0.003775358	0.003476525		0.001483	0.001673	0.003929985	0.996509193	
Ca				0.003177	0.003047			
Al	0.000202668	0.000186626		0.00202	0.002773	0.000210968		
K				0.000233	0.000001			
Mg								
Na	0.001485046	0.0013675		0.000128	0.000151	0.001545869		
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%
C bio	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%		
Degr.	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	50.0%	50.0%
Sol.			0.01					
Mod.	polyurethane.	plastics, mixture.	rubber, unspecified.	plastic, consumer electronics.	plastic, industr. electronics.	polyvinylfluoride	tin sheet.	tin sheet.

Tab. A.3 Solid waste compositions (Cont.)

Burn	0	0	0	0	1	1	1	1
Name	MSWI iron scrap	Alu in MSW	Glass	inert material (as cement)	newspaper	packaging paper	average paper	cardboard
Ger.	Eisenschrott in Siedlungsabfall	Alu in Siedlungsabfall	Glasverpackungen	Inertmaterial (Zement)	Zeitungspapier	Verpackungspapier	Papiermischung	Karton
Com.	inert	inert		to MSWLF	burn			
Ho	0.00	0.00	0.14	0.00	15.19	16.77	16.61	17.91
Hu	0.00	0.00	0.05	0.00	14.11	14.12	14.12	15.92
H2O	0	0	0.02	0.005969195	0.109441594	0.11098	0.110826159	0.104366667
O	0.121255		0.483070643	0.672764757	0.386622916	0.377787455	0.378671002	0.392677523
H				0.066666667	0.05385766	0.053851149	0.0538518	0.057790832
C					0.40830533	0.403516816	0.403995667	0.433272346
S	0.00076			0.004938398	0.001631106	0.001381767	0.001406701	0.001776803
N					0.000678411	0.004098584	0.003756567	0.002583521
P	0.0008			0.000231955	1.21671E-05	0.000124153	0.000112955	
B						1.91005E-05	1.71905E-05	
Cl	0.00075	0.004825	0.00019589		0.0013433	0.001892363	0.001837456	0.007173307
Br								
F						2.16313E-05	1.94682E-05	
I								
Ag					4.77325E-07		4.77325E-08	
As				3.39566E-06		2.27615E-06	2.04853E-06	1.16799E-06
Ba					1.3571E-05	0.000125427	0.000114241	5.71929E-05
Cd	0.000003	0.000003		6.0687E-07	1.38908E-07	1.83229E-06	1.66295E-06	9.33128E-07
Co				1.7555E-06	9.12533E-07	7.1627E-07	7.35896E-07	9.18062E-07
Cr	0.0012		0.00030363	2.13859E-05	1.03102E-05	1.49223E-05	1.44611E-05	9.24026E-06
Cu	0.01357	0.0000085	7.8356E-06	3.86139E-06	2.99894E-05	6.4327E-05	6.08932E-05	3.51128E-05
Hg				1.71153E-08	8.18458E-08	1.45374E-07	1.39021E-07	3.01313E-07
Mn	0.0024			0.000189757	4.67966E-05	3.94903E-05	4.0221E-05	9.5196E-05
Mo						4.77513E-06	4.29762E-06	
Ni	0.00105			1.64636E-05	2.31903E-06	1.02347E-05	9.44313E-06	1.06401E-05
Pb	0.00038	0.0000255	3.86883E-05	2.93709E-05	9.02836E-05	7.92728E-05	8.03739E-05	2.8963E-05
Sb	0.00021			8.81004E-08		7.87897E-07	7.09107E-07	4.60443E-09
Se						2.86508E-06	2.57857E-06	3.10729E-06
Sn	0.00067			1.32927E-06				6.75208E-06
V	0.00025			2.31322E-05				
Zn	0.000202	0.000029	3.9178E-06	8.05319E-05	5.55631E-05	0.000132494	0.000124801	6.33327E-05
Be						1.08236E-06	9.74127E-07	1.12992E-06
Sc								
Sr						6.47508E-05	5.82757E-05	4.50086E-05
Ti						0.000191005	0.000171905	
Tl				1.92563E-07		1.91005E-06	1.71905E-06	
W								
Si	0.03095		0.359089998	0.038426691	0.020871273	0.02129709	0.021254509	
Fe	0.81533			0.010045634	0.000139033	0.001305044	0.001188443	
Ca			0.07281778	0.18306136		0.003820106	0.003438095	
Al	0.01022	0.995109		0.013408124	0.01216711	0.012415344	0.012390521	
K					0.004679658	0.000955026	0.00132749	
Mg						0.004775132	0.004297619	
Na			0.064471618	0.004115331		0.001020923	0.000918831	
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	100.0%		0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
C bio			0.0%		100.0%	100.0%	100.0%	100.0%
Degr.	50.0%	50.0%	0.0%	0.0%	16.2%	27.0%	27.0%	32.4%
Sol.								
Mod.	steel.	aluminium.	glass; PVC sealing sheet; PE sealing sheet.	inert material.	newspaper.	packaging paper.	paper.	packaging cardboard.

Tab. A. 4 Solid waste compositions (Cont.)

Burn	1	1	1	1	0	0	1	1
Name	soiled textiles	chrome-preserved wood electricity pole	chrome-preserved building wood	natural wood	cement hydrated	Gypsum natural	organics in plastic plaster	Emulsion paint (remains)
Ger.	Textilien mit Öl	chromimprägnierter Holz-Strommasten	chromimprägniertes Bauholz	Holz natürlich	abgebundener Zement	Gips	Kunstharz in Kunststoffputz	Dispersionsfarbe
Com.						Naturgips	ethylene vinylacetate vinylchloride terpolymer plus traces (plastic plaster = 95.2% fillers (lime), 4.7% terpolymer)	Dispersionsfarbe trocken
Ho	19.78	15.36	15.36	15.36	0.00	0.00	25.62	11.14
Hu	14.45	13.99	13.99	13.99	0.00	0.00	25.46	10.51
H2O	0.229675	0.173451289	0.173800125	0.174078611	0.005969195	0.193990188	0	0
O	0.287940353	0.370838309	0.371584118	0.372179521	0.672764757	0.375047697	0.179563174	0.214758043
H	0.049839463	0.049982499	0.050083021	0.050163271	0.066666667		0.072947539	0.031708028
C	0.345153252	0.400020456	0.400824955	0.401467211			0.538689521	0.268328469
S	0.014523047	0.000288795	0.000197999	0.000125514	0.004938398	0.187523848		
N	0.070863926	0.000983427	0.000985405	0.000986984				
P	0.0000375	0.000108574	0.000108792	0.000108966	0.000231955			
B		5.89191E-05	2.73296E-05	2.11072E-06				
Cl	0.00179129	0.000329758	0.000330421	0.00033095		0.000404146	0.198922328	0.086465353
Br								
F		2.10304E-05	2.10727E-05	2.11065E-05		0.000505183		
I								
Ag								
As	0.00000006	4.20609E-07	4.21454E-07	4.2213E-07	3.39566E-06	1.97021E-06		4.41222E-07
Ba								
Cd	0.00000004	1.9979E-07	2.00192E-07	2.00512E-07	6.0687E-07	9.59847E-08	0.000266382	1.68494E-08
Co		8.64592E-08	8.66331E-08	8.67719E-08	1.7555E-06		0.001973889	
Cr	0.00000056	0.2281%	0.1013%	6.57822E-07	2.13859E-05	7.57774E-06	0.000815128	1.24796E-06
Cu	4.46394E-05	0.00078637	0.000351399	4.14815E-06	3.86139E-06	7.1736E-06		
Hg	6E-11	3.18611E-07	3.19252E-07	3.19763E-07	1.71153E-08	1.1114E-07		8.54422E-09
Mn		5.29271E-05	5.30335E-05	5.31185E-05	0.000189757			
Mo		8.27209E-07	8.28873E-07	8.30201E-07			0.000426211	
Ni	0.00000016	5.53803E-07	5.54917E-07	5.55806E-07	1.64636E-05	7.94147E-06		9.55273E-07
Pb	2.56025E-05	2.77672E-05	2.7823E-05	2.78676E-05	2.93709E-05	2.23796E-06	0.00234416	6.49461E-07
Sb					8.81004E-08			
Se								
Sn					1.32927E-06		0.002826311	
V					2.31322E-05			
Zn	0.000105078	1.77427E-05	1.77784E-05	1.78069E-05	8.05319E-05	1.40946E-05	0.001225356	6.54437E-06
Be								
Sc								
Sr								
Ti								0.135
Tl	0.00000003				1.92563E-07			
W								
Si					0.038426691			0.149807317
Fe		1.54229E-05	1.54539E-05	1.54786E-05	0.010045634			
Ca		0.000130416	0.000130679	0.000130888	0.18306136	0.242487735		0.113922927
Al		6.30935E-06	6.32204E-06	6.33217E-06	0.013408124			
K		6.51764E-05	6.53074E-05	6.54121E-05				
Mg		0.000196846	0.000197242	0.000197558				
Na		0.000334082	0.000156133	1.40715E-05	0.004115331			
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
C bio	100.0%	100.0%	100.0%	100.0%		0.0%	0.0%	0.0%
Degr.	12.0%	1.5%	1.5%	1.5%		100.0%	1.0%	1.0%

Sol.					0			
Mod.	textiles, soiled.	wood pole, chrome preserved.	building wood, chrome preserved.	wood untreated; cement-fibre slab.	cement-fibre slab; cement, hydrated.	gypsum.	plastic plaster.	emulsion paint; emulsion paint remains.

Tab. A.5 Solid waste compositions (Cont.)

Burn	1	1	1	1	1	0	1	0
Name	paint (remains)	PVC sealing sheet	EPS insulation	PE sealing sheet	Vapour barrier, flame-retarded	Filler (lime)	plastiziser	inert material
Ger.	Anstrichstoffe	PVC für Dichtungsbahn	EPS-Isolation	PE für Dichtungsbahn	PE für Dampfbremse, flammgeschützt	Naturkalk	Weichmacher	Inertstoff (SiO ₂)
Com.	unspec. paints	The composition is 53.9% PVC, 33.4% plasticizers, 1.7% glass fabric SiO ₂ and 11% fillers CaCO ₃ .	1	85% PE, 13.5% fillers and 1.2% glass fabric.	1	for sheets and plastic plasted (traces from Graf 1994)	for sheets	gravel/glass for sheets
Ho	10.86	21.08	33.20	42.59	40.38		35.61	0.00
Hu	10.14	20.93	32.20	42.24	40.05		33.68	0.00
H ₂ O	0	0.001944201	0.001988354	0.003977831	0.003732036			
O	0.290303031	0.016439459	0.038625933	0.038222211	0.035860412	0.48	0.164102564	0.533333333
H	0.036287879	0.053388771	0.077591933	0.121813947	0.114286908		0.097435897	
C	0.651511819	0.428318179	0.863432368	0.817496897	0.766982723	0.12	0.738461538	
S		0.001855288	0.000674795	0.000424531	0.000398299			
N		0.001027466	0.001931297	0.00129	0.001210289			
P								
B								
Cl	0.0016	0.462025517	0.001125526	0.001453432	0.001363623			
Br		7.17788E-05	0.0067	9.53644E-06	0.067			
F		1.50735E-05	1.41666E-05	1.43047E-05	1.34208E-05			
I								
Ag								
As	0.0000025	1.94828E-06	1.83106E-06	1.81192E-06	1.69996E-06	1.54919E-06		
Ba		0.000191239	0.000179733	0.000238411	0.000223679			
Cd	5.06216E-06	2.67776E-06	2.32006E-05	3.37575E-05	3.16716E-05	5.91608E-08		
Co	0.000388235	3.23927E-05	3.04438E-05	1.66888E-06	1.56576E-06			
Cr	0.001040736	4.19034E-05	3.93823E-05	1.23974E-05	1.16313E-05	4.38178E-06		
Cu	0.000059	2.7436E-05	0.000106937	4.1138E-05	3.8596E-05			
Hg	0.0000006	8.70702E-07	8.18316E-07	4.76822E-08	4.47359E-08	0.00000003		
Mn		8.19488E-05	7.70183E-05	2.98809E-05	2.80345E-05			
Mo	3.86194E-05							
Ni	0.000015	2.11225E-05	1.98517E-05	9.53644E-07	8.94717E-07	3.3541E-06		
Pb	1.525%	0.027087199	1.48484E-05	2.24707E-05	2.10822E-05	2.28035E-06		
Sb		4.40585E-05	4.14077E-05	1.00133E-05	9.39453E-06			
Se		2.15336E-06	2.02381E-06	2.00265E-06	1.87891E-06			
Sn	0.000256782	0.000380493	2.52547E-05	3.7929E-06	3.55853E-06			
V		0.000295472	0.000277695	0.002193382	0.00205785			
Zn	0.3251%	5.8107E-05	0.000829748	0.005882353	0.000319095	2.29783E-05		
Be		5.12706E-07	4.81859E-07	4.76822E-07	4.47359E-07			
Sc								
Sr		9.07489E-05	8.5289E-05	8.43975E-05	7.91825E-05			
Ti		0.001025411	0.000963717	0.000953644	0.000894717			
Tl		4.10164E-07	3.85487E-07	3.81458E-07	3.57887E-07			
W								
Si								0.466666667
Fe		0.003820341	0.003590488	0.001525831	0.001431548			
Ca				0.002574839	0.002415737	0.4		
Al		0.000205082	0.000192743	0.000190729	0.000178943			
K								
Mg				9.53644E-05	8.94717E-05			
Na		0.00150274	0.001412327	0.001397566	0.001311208			
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%			0.0%
C bio	0.0%	0.0%	0.0%	0.0%	0.0%		0.0%	
Degr.	1.0%							0.0%
Sol.								

Mod.	paint; paint remains.	PVC sealing sheet.	expanded polystyrene.	PE sealing sheet.	vapour barrier, flame-retarded.	PVC sealing sheet; PE sealing sheet; plastic plaster.	PVC sealing sheet.	bitumen sheet; asphalt.
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Tab. A. 6 Solid waste compositions (Cont.)

Burn	1	0	1	1	1	1	1	1
Name	bitumen	wiring copper	wiring plastic	bilge oil	separator sludge	refinery sludge	hazardous waste avg.	waste oil
Ger.	Bitumen	Kupfer aus Leitungen	Kabelisolation	Bilgen/Bunkeröl	Leichtstoffabscheiderschlamm	Raffinerieschlämme	sondermüll avg	Altöl
Com.	elements from bunkeroil				from oil tank storage			
Ho	41.39		26.81	4.10	0.90	4.73		41.80
Hu	39.38		25.95	1.92	0.00	2.45	17.00	34.70
H2O	0.01356		0.0355	0.9	0.9	0.888	0.25	0.1
O	0.01		0.022437981	0.0013	0.000197	0.001017085	0.04	
H	0.1		0.07216679	0.01	0.0026595	0.013730647	0.061	0.12
C	0.83085974		0.530495592	0.084	0.016745	0.086452224	0.416	0.778268999
S	0.04		0.000806423	0.00365	0.0000985	0.000508542	0.032	
N	0.005		0.001068991	0.001	2.71961E-06	0.001017085	0.0074	
P	0.000004			4E-10	0.000149	0.00047803	0.0022	0.00075
B					0.0000021		0.000007	
Cl			0.305153153		0.0004		0.104	
Br			5.02323E-05					
F			1.4281E-05		7.7703E-09		0.0037	
I								
Ag								
As	0.0000007		1.83462E-06		0.0000005			0.0000012
Ba			0.000198455		0.000027			
Cd			1.20074E-05		0.00000066	1.01708E-05	0.00000037	0.0000008
Co	0.00000043		2.18698E-05		0.0000104	2.03417E-05	0.000074	
Cr	0.00000035		3.1397E-05		0.0000713		0.000123948	0.0000112
Cu	0.000005	1	0.024205301		0.0005481	0.000101708	0.000267472	0.0001
Hg	0.00000002		5.88706E-07		0.00000022	2.03417E-06	0.00000074	1.2E-09
Mn			6.31117E-05		0.0000405			
Mo	0.00000056				0.0000005			
Ni	0.0001		1.42199E-05	0.000000079	0.0000376	0.000152563	0.000126808	0.0000032
Pb	0.000001		1.70766E-05	0.000000001	0.0000276	0.000457688	0.00029664	0.000184
Sb			3.2095E-05		0.00000049			
Se	0.0000002		2.02774E-06					
Sn	0.000089		0.000252091		0.0000017			
V	0.00025		0.000860301	0.0000331	0.00000085			
Zn			0.000141506		0.0002706	0.001017085	0.00237827	0.00068
Be			4.82796E-07					
Sc								
Sr			8.54548E-05		0.00003			
Ti			0.000965591					
Tl			3.86237E-07					0.0000006
W								
Si	0.000006			0.0000006			0.080424752	
Fe	0.000075		0.002982477	0.0000028	0.0034			
Ca	0.000006		0.00078116	0.0000006	0.0182	0.007		
Al	0.000007		0.000193118		0.0001335			
K					0.000148			
Mg			2.89318E-05		0.00167			
Na	0.000035		0.001415074	0.0000035	0.000071			
sum	100.00%	100.00%	100.00%	100.00%	94.49%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%			
C bio	0.0%		0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Degr.	1.0%					60.0%		
Sol.								
Mod.	bitumen sheet; asphalt; bitumen.	copper.	wire plastic.	bilge oil.	separator sludge.	refinery sludge.	hazardous waste.	used mineral oil.

Tab. A. 7 Solid waste compositions (Cont.)

Burn	1	1	0	0	0	0	0	0
Name	Anti-Freeze liquid	waste solvents mixture	cooling tower residue	hard coal tailings	AT hard coal ash	BE hard coal ash	CZ hard coal ash	DE hard coal ash
Ger.	Frostschutzflüssigkeit	Lösemittel	Kühlturm Rückstände	Steinkohle berge	AT Steinkohleasche	BE Steinkohleasche	CZ Steinkohleasche	DE Steinkohleasche
Com.				separate model PH	AT	BE	CZ	DE
Ho	10.92	23.68						
Hu	0.56	21.70						
H ₂ O	0.518	0.165	0.3		0	0	0	0
O	0.2255	0.2	0.274410042		0.490564121	0.607806759	0.777384065	0.575262847
H	0.0488	0.08166						
C	0.2079	0.477	0.038096293					
S		0.00642	0.051429996	0.0084	0.003246168	0.001987448	0.001768173	0.003806907
N		0.01036						
P		0.001		0.0006	0.000810945	0.001200755	0.000353453	0.00142698
B					0.000283612	0.000167784	0.000123836	0.000149073
Cl		0.02444837	2.28578E-05	0.0003	2.26633E-05	2.10034E-05	9.65422E-06	4.12973E-05
Br		0.002505247			6.48954E-05	2.28485E-05	2.82908E-05	4.39197E-05
F		4.49209E-05			2.93721E-06	9.0701E-05	7.07269E-07	9.12291E-05
I		0.000100239			4.03884E-06	1.85114E-06	1.76817E-06	2.22871E-06
Ag								
As			2.66674E-05	0.00002	1.26265E-05	1.91086E-05	5.26573E-06	3.8874E-05
Ba				0.0008	0.001620428	0.001420697	0.000707269	0.001027291
Cd			2.28578E-06	0.0000005	4.02533E-06	1.23303E-06	1.76273E-06	1.46889E-06
Co		4.1029E-06		0.00002	5.72516E-05	3.58997E-05	2.47417E-05	5.85508E-05
Cr			0.000224768	0.0001	7.39125E-05	0.000143331	3.18271E-05	0.000132544
Cu		5.1445E-06	0.000476204	0.00005	0.000122523	7.48709E-05	5.30098E-05	0.000128776
Hg			1.18099E-06	1.53832E-10	8.50052E-08	1.12909E-07	3.53816E-08	2.12869E-07
Mn				0.0002	0.001613018	0.000435621	0.000707269	0.000439269
Mo				0.000025	2.47579E-05	1.18285E-05	1.06036E-05	3.25358E-05
Ni		4.25536E-06	0.004000111	0.00005	0.000162662	9.25543E-05	7.06434E-05	0.00013523
Pb			0.00010286	0.00005	0.000323848	6.8448E-05	0.000141091	0.000164674
Sb					1.21735E-05	3.66065E-06	5.29635E-06	7.75164E-06
Se				0.00005	8.27146E-06	5.65359E-06	3.59259E-06	9.01084E-06
Sn					8.8976E-05	5.89969E-05	3.88998E-05	5.83965E-05
V			0.008000222	0.00005	0.000324958	0.000163278	0.000141381	0.000234782
Zn		4.1844E-05	0.000838118	0.0001	0.000403689	0.000138445	0.000176273	0.000209545
Be				2.3E-09	1.62231E-05	1.07472E-05	7.07269E-06	1.21134E-05
Sc					1.07613E-05	3.26588E-05	4.59772E-06	3.46431E-05
Sr				0.0001	0.001613683	0.001454391	0.000707088	0.000855578
Ti				0.006	0.012110219	0.006299839	0.005304519	0.00593318
Tl					8.06859E-06	4.35413E-06	3.53271E-06	3.91611E-06
W					8.17414E-06	1.90615E-05	3.53635E-06	1.53831E-05
Si		0.031321225	0.08369455		0.202885482	0.156972454	0.088408644	0.176704477
Fe		8.46517E-05	0.111911726	0.03	0.056856865	0.044324752	0.02475442	0.054015208
Ca			0.073513347	0.004	0.016419569	0.013217871	0.007072692	0.021975403
Al			0.032259393	0.1	0.137521756	0.107781241	0.060117878	0.107168515
K			0.006483301	0.03	0.016199383	0.012180153	0.007072692	0.012286441
Mg			0.005743929	0.008	0.048394042	0.037141655	0.021218075	0.030735721
Na			0.008762147	0.005	0.008103187	0.006587933	0.003536346	0.006756027
sum	100.02%	100.00%	100.00%	19.39%	100.00%	100.00%	100.00%	100.00%
Fe met.		0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
C bio	0.0%	0.0%	0.0%	0.0%				
Degr.			1.5%					
Sol.				0	0	0	0	0
Mod.	antifreezer liquid.	solvents mixture.	residue from cooling tower.		hard coal ash.	hard coal ash.	hard coal ash.	hard coal ash.

Tab. A. 8 Solid waste compositions (Cont.)

Burn	0	0	0	0	0	0	0	0
Name	ES hard coal ash	FR hard coal ash	HR hard coal ash	IT hard coal ash	NL hard coal ash	PL hard coal ash	PT hard coal ash	SK hard coal ash
Ger.	ES Steinkohleasche	FR Steinkohleasche	HR Steinkohleasche	IT Steinkohleasche	NL Steinkohleasche	PL Steinkohleasche	PT Steinkohleasche	SK Steinkohleasche
Com.	ES	FR	HR	IT	NL	PL	PT	SK
Ho								
Hu								
H ₂ O	0	0	0	0	0	0	0	0
O	0.746173418	0.54066223	0.536273523	0.625531081	0.503854765	0.729244954	0.545993815	0.704100628
H								
C								
S	0.003500452	0.002629417	0.002916667	0.001666667	0.003442189	0.002120441	0.003030303	0.002598753
N								
P	0.000462888	0.001026802	0.000747759	0.000820838	0.001167271	0.000380837	0.001086182	0.000461389
B	0.000158237	0.000219459	0.000263284	0.000222274	0.000255284	5.40482E-05	0.000216666	0.000251987
Cl	1.36523E-05	3.59349E-05	0.000075	1.7478E-05	2.42994E-05	3.80178E-05	1.91922E-05	3.60681E-05
Br	2.05232E-05	3.98251E-05	0.00006	3.00051E-05	4.07667E-05	4.56868E-05	3.90252E-05	2.85022E-05
F	4.65102E-05	8.23177E-05	0.0000375	7.3405E-05	9.24024E-05	3.23453E-05	7.67385E-05	3.16101E-05
I	1.57181E-06	2.79185E-06	0.00000375	2.32474E-06	3.07022E-06	3.80505E-06	2.65021E-06	2.58255E-06
Ag								
As	2.09172E-05	5.00803E-05	7.36155E-05	1.49726E-05	2.59054E-05	4.48238E-05	1.7351E-05	2.02048E-05
Ba	0.000895155	0.001494417	0.0015	0.001120607	0.001446703	0.000763757	0.001181015	0.001125762
Cd	1.08197E-06	8.1967E-07	3.68277E-06	8.57904E-07	1.28024E-06	1.50368E-06	1.00542E-06	1.40401E-06
Co	1.96765E-05	3.93323E-05	3.7388E-05	2.45843E-05	3.48012E-05	2.47477E-05	2.77002E-05	2.53563E-05
Cr	8.33449E-05	0.000148126	0.00015	0.000113513	0.000152462	7.61891E-05	0.000139982	0.000118111
Cu	4.59423E-05	7.41106E-05	0.000112063	5.6225E-05	7.95038E-05	9.48632E-05	6.07636E-05	6.03624E-05
Hg	5.95222E-08	1.35332E-07	7.52241E-08	8.37217E-08	1.07361E-07	4.59205E-08	9.08966E-08	5.3913E-08
Mn	0.000242257	0.000429536	0.000525	0.000321169	0.000418721	0.000497075	0.000319616	0.000656879
Mo	7.60851E-06	1.25796E-05	2.24328E-05	8.2233E-06	1.31973E-05	1.14132E-05	1.26458E-05	9.82135E-06
Ni	6.23132E-05	0.000112379	0.000148969	7.31322E-05	0.000101594	7.58661E-05	8.79085E-05	9.06172E-05
Pb	7.49313E-05	0.000132493	0.000295518	4.86738E-05	6.09159E-05	6.81375E-05	4.35255E-05	9.09247E-05
Sb	2.39301E-06	1.40472E-05	7.43277E-06	2.63031E-06	3.90525E-06	5.67605E-06	3.22893E-06	5.16562E-06
Se	3.40927E-06	1.0231E-05	6.22598E-06	8.24911E-06	1.31867E-05	6.34198E-06	1.72827E-05	4.14504E-06
Sn	2.9927E-05	3.44651E-05	0.000015	5.81749E-05	7.59059E-05	0.000190033	3.72524E-05	0.000176437
V	0.000114229	0.000211818	0.000299104	0.000134466	0.000179039	0.000140858	0.000158973	0.000164352
Zn	0.000103488	0.000631724	0.000368277	8.51381E-05	0.000127979	0.000187699	9.32746E-05	0.00010995
Be	7.64432E-06	1.32544E-05	0.000015	1.01428E-05	1.29631E-05	6.87245E-06	1.07478E-05	6.269E-06
Sc	1.95036E-05	4.19493E-05	3.00179E-05	3.39656E-05	4.91919E-05	2.05549E-05	5.33683E-05	2.36322E-05
Sr	0.000876864	0.001515477	0.001497759	0.001173366	0.001345171	0.000762691	0.000982635	0.000929164
Ti	0.00340097	0.006333147	0.0045	0.005797181	0.007701485	0.00382583	0.007280586	0.006772961
Tl	3.72879E-06	6.52163E-06	7.45518E-06	5.0956E-06	6.76812E-06	3.79592E-06	6.26587E-06	3.22847E-06
W	9.80342E-06	1.73673E-05	0.0000075	1.50105E-05	2.00174E-05	5.72493E-06	1.65927E-05	3.29104E-06
Si	0.101865742	0.185266268	0.1875	0.15	0.199633759	0.095419847	0.185484325	0.116943867
Fe	0.028246841	0.052186583	0.0525	0.042069045	0.057244689	0.038100395	0.051451849	0.028771699
Ca	0.008147349	0.014818739	0.015	0.012359033	0.016175502	0.011429004	0.014258358	0.009842852
Al	0.069292599	0.126298568	0.1275	0.103782509	0.137791212	0.083861406	0.124702555	0.081958545
K	0.008114072	0.014736618	0.015	0.011678136	0.015538323	0.00421783	0.0150443	0.008916901
Mg	0.02380751	0.043325312	0.045	0.036248101	0.044925609	0.022901254	0.041058121	0.029425181
Na	0.004123386	0.007345126	0.0075	0.006393642	0.007940054	0.00533563	0.006984109	0.006231345
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
C bio								
Degr.								
Sol.	0	0	0	0	0	0	0	0
Mod.	hard coal ash.	hard coal ash.	hard coal ash.	hard coal ash.	hard coal ash.	hard coal ash.	hard coal ash.	hard coal ash.

Tab. A.9 Solid waste compositions (Cont.)

Burn	0	0	0	0	0	1	0	0
Name	hard coal ash small scale	lignite ash small scale	drilling waste	inorganic Waste Si wafer production	wood ash pure	sludge from pulp and paper production	composition of paper sludge ash	Green liquor dregs from pulp production, to landfill
Ger.	Steinkohle ashe Kleinf Feuerung	Braunkohleasche Kleinf Feuerung	Bohrabfall	Abfall, Si- Waferherst., anorg.	Holzasche gemischt, rein	Klärschlamm (Papierprod.) in Deponie	Asche aus Papierprod.- Klärschlamm	Asche aus Deinkingschlam mverbrennung
Com.	based on coal composition, combustion transfer coefficients and an ash content of 10%	based on coal composition, combustion transfer coefficients and an ash content of 10%			MSWI, MWLF and landfarming	wet		
Ho	0.47	2.58	0.77		0.00	5.03		
Hu	0.00	0.00	0.00		0.00	2.65		
H2O			71.4696%	0.093801843		0.25	0	0.25
O	0.29803325	0.08379703	10.0000%	0.204813108	0.4450055	0.427889007	0.030082957	0.24
H			0.5000%			0.029256976		
C	0.1	0.1	3.0000%	0.031490015	0.012	0.157500053	0.00916471	0.293
S			0.6000%		0.0092	0.00336699	0.041259464	0.048
N						0.026558832	0.006222479	
P	0.00098	0.000392	0.0500%		0.0098	0.012074076	0.146232473	
B	0.00035	0.0021	0.0100%					
Cl			3.5000%		0.0032	2.58375E-08	1.24085E-08	
Br	0.000097	0.0000388						
F			0.0500%	0.041159557		1.92608E-06	9.61409E-06	
I	0.0000164	0.0000164						
Ag								
As	0.000092	0.000046	0.0004%		0.0000067	1.4625E-06	1.02148E-05	
Ba	0.00196	0.00098	0.5000%					
Cd	0.0000045	0.00000549			0.0000142	0.000000255	1.54033E-06	
Co	0.000049	0.000147			0.000018	4.77864E-06	3.21451E-05	
Cr	0.000192	0.0001344	0.0050%		0.0001949	0.00001275	9.82709E-05	0.00073
Cu	0.00014625	0.0000585			0.0001632	0.000084585	0.000560169	0.0003
Hg	0.0000001	0.0000001			0.0000001	0.00000018	1.01747E-06	
Mn	0.000693	0.000198	0.0400%		0.019956	0.000156556	0.001069465	
Mo	0.0000285	0.0000665			0.0000037	2.82817E-06	2.24032E-05	
Ni	0.000194	0.0001455			0.0000552	0.000008925	6.00849E-05	
Pb	0.000376	0.000047			0.000065	0.00004695	0.000267552	
Sb	0.0000095	0.000038						
Se	0.000007	0.000014						
Sn	0.0000198	0.00001584				1.18618E-05	7.95676E-05	
V	0.000388	0.0003395			0.0000395			
Zn	0.000475	0.000266	0.0150%		0.001662	0.000131498	0.000865797	0.00215
Be	0.0000196	0.0000098						
Sc	0.0000396	0.00000495						
Sr	0.00194	0.000679	0.0100%					
Ti	0.00588	0.00098			0.001383			
Tl	0.0000095	0.00000019						
W								
Si	0.245	0.4704	0.1000%	0.585409626	0.0826	0.017626147	0.199030391	
Fe	0.0686	0.05292	2.0000%		0.022833	0.020951245	0.158415468	
Ca	0.0196	0.196	4.0000%	0.04332585	0.2844	0.029909453	0.221031929	
Al	0.1666	0.04606	1.0000%		0.0208	0.005851395	0.050413329	
K	0.0196	0.00588	0.3500%		0.0545	0.002344639	0.014896898	
Mg	0.0588	0.0294	0.8000%		0.0321	0.003357918	0.02941549	
Na	0.0098	0.00882	2.0000%			0.012848688	0.090756556	0.16582
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
C bio	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	100.0%
Degr.	5.0%	5.0%			5.0%	60.0%		
Sol.	0	0		1				1

7. Appendix

Mod.	hard coal ash from stove.	lignite ash from stove.	drilling waste.	waste, Si waferprod., inorg.	wood ash mixture, pure.	sludge from pulp and paper production.	ash from paper production WWTP sludge.	green liquor dregs.
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Tab. A. 10 Solid waste compositions (Cont.)

Burn	0	0	0	0	0	1	0	0
Name	Ash, from Incineration of Deinking Sludge, to landfill	Nickel smelting slag	dust from unalloyed electric steel production to landfill	slag from electric steel production to landfill	BOF waste mix	sludge from pig iron production to landfill	dust from electric chromium steel production to landfill	Sludge from Steelrolling
Ger.	Bodensatz Grünlauge Zellstoffprod.	Schlacke aus Nickel-Verhüttung	Staub, v. Elektrostahl unlegiert	Schlacke, v. Elektrostahl unlegiert	Blasstahl Produktionsabfall allmix	Schlamm, Roheisenproduktion	Staub, v. Elektrostahl legiert	Abwasser-Schlamm vom Stahlwalzen
Com.			slag in english name should read dust		2nd corrected composition 7.8.03		slag in english name should read dust	
Ho								
Hu								
H2O			0.105410884	0	0	0.08585	0.105410884	0.2
O	0.423285048	0.432122222	0.075250262	0.273901768	0.346844341	0.118420431	0.14320574	0.728319171
H								0.001101712
C		0.00345	0.01197851	0.003386352		0.31	0.00695566	0.009915409
S	0.004		0.007187106	0.000205233	0.004168564	0.0245	0.001854843	
N								
P			0.001672885	0.001343547	0.033251674	0.001178358	0.000222614	
B								
Cl			0.026352721				0.026352721	0.007822674
Br								
F			0.004408092				0.000278226	
I								
Ag								
As			0.000397687				0.000397687	
Ba								
Cd	0.00000005		0.000574968				0.00041734	0.000874601
Co		0.001116667	1.43742E-05				1.43742E-05	
Cr	0.000001		0.003933977	0.010531669	0.003994883		0.095182288	0.003148565
Cu	0.0000044	0.0008	0.002635272	0.000245932			0.001437503	0.001692303
Hg	2.5E-09		5.27054E-06				5.27054E-06	0.00017492
Mn			0.029686076	0.063578192	0.010423199	0.0013	0.032321362	0.003567688
Mo								
Ni	0.0000007	0.0019	0.000287484	0.001612825			0.027822642	0.004947493
Pb	0.000002		0.032581546			0.014	0.011592767	0.001749203
Sb								
Se								
Sn								
V			0.000335398	0.00103469			0.001854843	
Zn	0.0000068		0.215613173	0.000164885		0.0505	0.055645284	0.001308984
Be								
Sc								
Sr								
Ti				0.003078502				
Tl								
W								
Si	0.1983	0.164111111	0.014558183	0.067154829	0.083851754	0.028046735	0.036849117	
Fe		0.385	0.359355288	0.215495126	0.105499302	0.21	0.324597487	0.027098536
Ca	0.25	0.0115	0.065063416	0.271356584	0.376641839	0.076829666	0.072910534	
Al	0.1244		0.002535904	0.029871078	0.013266562	0.014290052	0.012271204	0.00827874
K			0.010739479	0.00093706		0.00182633	0.010739479	
Mg			0.017336503	0.052599856	0.022057883	0.061811793	0.019574588	
Na			0.012085541	0.003501873		0.001446635	0.012085541	
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
C bio		0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Degr.			0.0%	0.0%	0.0%	0.0%	0.0%	
Sol.	1	0	0	0	1	1	1	1
Mod.	ash from deinking sludge.	nickel smelter slag.	dust, unalloyed EAF steel.	dust, unalloyed electr. steel.	basic oxygen furnace wastes.	sludge, pig iron production.	slag, alloyed EAF steel.	sludge from steel rolling.

Tab. A. 11 Solid waste compositions (Cont.)

Burn	0	1	0	0	0	0	0	0
Name	redmud from bauxite digestion	spent pot liner, carbon fraction	spent pot liner, refractory fraction	filter dust, aluminium electrolysis	dross, aluminium electrolysis	reduced residues dichromate prod	residue from TiO2 production (sulfate process)	residue from TiO2 production (chloride process)
Ger.	Rotschlamm Alumina produktion	Kohlekathode Alu-Elektrolyse	Wanne Keramik Alu-Elektrolyse	Filterstaub Alu-Elektrolyse	Schlacke Alu-Elektrolyse	Rückstände Dichromatprod.	Rückstände aus TiO2-Produktion (Sulfat-Verfahren)	Rückstände aus TiO2-Produktion (Chlor-Verfahren)
Com.		1st cut	2nd cut					
Ho		22.20	0.00	0.18	0.00			
Hu		22.20	0.00	0.18	0.00			
H2O		0	0	0	0		0.296	0.56
O	0.379730241	0.049911771	0.392510388	0.38224	0.287058824	0.350633019	0.3196	0.2
H								
C	0.007606696	0.664635875		0.11586	0.0075			0.09
S	0.0008						0.0286	
N		0.013294237	0.000330121		0.020487805			
P	0.001475014	0.000140138	8.02178E-05	0.00058		0.000893793		0.000893793
B	0.00001					7.19327E-06	7.19327E-06	7.19327E-06
Cl		0.000592543	0.000399402					
Br								
F	0.00113	0.07900575	0.085871329	0.09051				
I								
Ag						0.000000714	0.000000714	0.000000714
As	0.000029					6.2521E-07	6.2521E-07	6.2521E-07
Ba						0.000149038	0.000149038	0.000149038
Cd	0.0000045					1.17476E-05	1.17476E-05	1.17476E-05
Co	0.000022					1.34527E-06	1.34527E-06	1.34527E-06
Cr	0.000750973			0.00005		0.0454		
Cu	3.67423E-05			0.00057		0.00121276		
Hg	0.000000025							
Mn	0.000363318	7.64707E-05	7.73172E-05			0.00025936		0.00025936
Mo						1.95506E-06		1.95506E-06
Ni	0.0000054			0.00051		0.0011		
Pb	1.41421E-05			0.00039		0.00050243		0.00050243
Sb	0.0000003					2.25636E-05		2.25636E-05
Se	0.000005					3.19689E-07		3.19689E-07
Sn						7.34399E-05		7.34399E-05
V	0.000535082			0.00048		0.0002		0.0002
Zn	3.66606E-05			0.00003		0.00077		0.00077
Be	0.0000007							
Sc								
Sr			0.000369222					
Ti	0.040202659		0.000359161	0.00039			0.14	0.08
Tl								
W								
Si	0.104474281	0.013852298	0.172736178	0.00046		0.0312	0.1281	0.02
Fe	0.296416769		0.012567635	0.00545		0.214	0.0184	0.025
Ca	0.022445764	0.024706901	0.039254905			0.0916	0.0214	0.022
Al	0.084217078	0.050830906	0.195589279	0.3382	0.684953372	0.0992	0.0177	
K		0.004099157	0.00248672			0.002059697		0.002059697
Mg				0.0009		0.0477	0.03	
Na	0.059687655	0.098757188	0.097853375	0.06308		0.113		
sum	100.00%	99.99%	100.05%	99.97%	100.00%	100.00%	100.00%	100.20%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
C bio	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Degr.							0.0%	0.0%
Sol.	0	1	1	1	0	1	1	0
Mod.	redmud from bauxite digestion.	carbon SPL, Al elec.lysis.	refractory SPL, Al elec.lysis.	filter dust Al electrolysis.	dross from Al electrolysis.	residues Na-dichromate prod..	residue from TiO2 prod. SO4.	residue from TiO2 prod. Cl.

Tab. A. 12 Solid waste compositions (Cont.)

Burn	0	0	0	0	0	1	1	0
Name	salt tailings potash mining	Brine filtration sludge without mercury cells	Brine filtration sludge with mercury cells	residue from H3PO4 purification	decarbonising waste	cation exchange resin f. water	anion exchange resin f. water	EDC Oxychlor catalyst
Ger.	Rückstände aus Kalibergbau	Schlamm aus NaCl- Elektrolyse o. Hg	Schlamm aus NaCl- Elektrolyse m. Hg	Feststoff aus H3PO4- Raffinierung	Rückstand Entkarbonisieru ng	Kationentausch erharz aus Wasseraufberei tung	Anionentausch erharz aus Wasseraufberei tung	Katalysator für EDC Prod
Com.	as proxy for tailings stack. Kieserite (MgSO4·H2O), Carnallite (KCl·MgCl2·6H 2O), rock salt (NaCl) and Anhydrite (CaSO4)			anhydrite (calcium sulfate) and apatite (calcium phosphate)				to HWI and UTD
Ho						11.35	16.06	0.00
Hu						1.47	6.08	0.00
H2O	0				0.3	0.496917753	0.499712096	0
O	0.098835418	0.559207076	0.559165135	0.41906	0.469002599	0.11025	0.07032	0.504
H		0.016295026	0.016293804	0.000165		0.01906	0.04016	
C						0.25464	0.32713	
S	0.04952149	0.006857241	0.006856727	0.192714	0.001866667	0.07364	0.02398	
N					0.00077	0.00056957	0.02864	
P				0.015345	0.000840493		0.00001	
B								
Cl	0.49857313				0.0001925	0.000331936	0.00788	
Br						0.000258636		
F				0.04819		4.17797E-06		
I								
Ag								
As				0.0000489	0.00000385	5.4001E-07		
Ba		0.029428278	0.02942607			5.30062E-05		
Cd				0.000071	1.0605E-06	6.84223E-06	1.59196E-06	
Co						8.97837E-06		
Cr				0.001537	0.00001925	1.16145E-05		
Cu				0.000189	0.00002975	3.15374E-05	1.6311E-05	0.055
Hg			0.000075	0.000000474	1.785E-07	2.41335E-07		
Mn				0.000063		2.27139E-05		
Mo								
Ni				0.000238	0.0000182	5.85457E-06		
Pb				0.000014	0.00002975	4.37902E-06	9.24817E-06	
Sb						1.22118E-05		
Se						5.96853E-07		
Sn						7.44802E-06	0.000226207	
V				0.000908		8.18967E-05		
Zn				0.001328	0.000077	0.000244706	3.45452E-05	
Be						1.42108E-07		
Sc								
Sr						2.51531E-05		
Ti						0.000284216		
Tl						1.13686E-07		
W								
Si				0.028646	0.010634033		0.00188	0.441
Fe				0.005589	0.013953607	0.001058892		
Ca	0.008241325	0.190227818	0.190213551	0.27402	0.18760699	0.03344		
Al				0.002795	0.004816093	5.68431E-05		
K	0.006363883				0.00063922	0.00094		
Mg	0.047830255	0.197984563	0.197969714	0.009083	0.00949876	0.00265		
Na	0.290634498					0.00538		
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
C bio	0.0%			0.0%	0.0%	0.0%	0.0%	0.0%

Degr.								
Sol.	0	1	1	1				
Mod.	salt tailings potash mining.	sludge, NaCl electrolysis.	sludge, NaCl electrolysis Hg.	H3PO4 purification residue.	decarbonising waste.	cation exchange resin f. water.	anion exchange resin f. water.	catalyst for EDC production.

Tab. A. 13 Solid waste compositions (Cont.)

Burn	0	0	0	1	0	1	0	1
Name	Formox catalyst carrier	Ethylene oxide catalyst carrier	polluted rail ballast residue	Al in ASR burnable	Al in ASR inert	Fe in ASR burnable	Fe in ASR inert	Zn in ASR burnable
Ger.	Katalysatorträger Formaldehydprod.	Katalysatorträger Ethylenoxidprod.	Schadstofffraktion aus Bahnschotter	Al in RESH brennbar	Al in RESH inert	Fe in RESH brennbar	Fe in RESH inert	Zn in RESH brennbar
Com.	to residual	to residual						
Ho	0.00	0.00						
Hu	0.00	0.00						
H2O	0	0	0	0	0	0	0	
O	0.506661333	0.470585882				0.121255	0.121255	
H								
C								
S						0.00076	0.00076	
N								
P						0.0008	0.0008	
B								
Cl				0.004825	0.004825	0.00075	0.00075	
Br								
F								
I								
Ag		0.000005						
As			3.77224E-05					
Ba								
Cd			2.51483E-06	0.000003	0.000003	0.000003	0.000003	
Co								
Cr			0.003655027			0.0012	0.0012	
Cu			0.004525272	0.0000085	0.0000085	0.01357	0.01357	
Hg			1.25741E-06					
Mn						0.0024	0.0024	
Mo	0.000005							
Ni			0.00110231			0.00105	0.00105	
Pb			0.000205091	0.0000255	0.0000255	0.00038	0.00038	
Sb						0.00021	0.00021	
Se								
Sn						0.00067	0.00067	
V	0.000005					0.00025	0.00025	
Zn			0.001195441	0.000029	0.000029	0.000202	0.000202	1
Be								
Sc								
Sr								
Ti								
Tl								
W								
Si	0.443328667		0.989275365			0.03095	0.03095	
Fe	0.05					0.81533	0.81533	
Ca								
Al		0.529409118		0.995109	0.995109	0.01022	0.01022	
K								
Mg								
Na								
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	100.0%	0.0%	0.0%			0.0%	100.0%	
C bio	0.0%	0.0%	0.0%					
Degr.								
Sol.	0	0	1					
Mod.	catalyst base CH2O production.	catalyst base Eth.oxide prod..	pollutants from rail ballast.	aluminium in car shredder residue.	aluminium in car shredder residue.	steel in car shredder residue.	steel in car shredder residue.	zinc in car shredder residue.

Tab. A. 14 Solid waste compositions (Cont.)

Burn	0	1	0	1	0	0	1	1
Name	Zn in ASR inert	Cu in ASR burnable	Cu in ASR inert	Pb in ASR burnable	Pb in ASR inert	sulfidic tailings nickel mine	avg wastewater treatment sludge	paper
Ger.	Zn in RESH inert	Cu in RESH brennbar	Cu in RESH inert	Pb in RESH brennbar	Pb in RESH inert	Sulfideische Bergehalden Nickelmine	CH Klärschlamm (Faulschlamm)	fraction in MSW from Hellweg 2000
Com.							from WWT calculation (including dummy values for missing elements, for calculation of WWT sludge incineration)	3 Proben Altpapier nach BUWAL
Ho								15.30
Hu								14.20
H2O							0.63	0.0575
O						0.428189925	0.001240914	0.461217112
H							0.016310097	0.049
C							0.130480778	0.386
S		2.83724E-12	2.83724E-12			0.166225	0.011612743	0.00066
N							0.017574819	0.000943
P						0.000375	0.016366634	0.000123
B							0.000000123	0.0000189
Cl							0.000000123	0.000189
Br							0.000000123	
F							0.000000123	
I							0.000000123	
Ag		0.000489052	0.000489052			0.00000375	0.000000123	
As						0.00005765	1.46559E-06	
Ba						0.00008	0.000000123	0.0000283
Cd						0.000004525	1.04293E-06	0.00000943
Co						0.00013365	6.01216E-06	
Cr		2.73457E-06	2.73457E-06			0.0000875	4.54593E-05	0.000032
Cu		0.924873057	0.924873057			0.00109125	0.00020926	0.0000283
Hg							1.04292E-06	2.83E-08
Mn		0.001483172	0.001483172			0.0011125	0.000196968	0.0000471
Mo							3.55822E-06	0.000005
Ni		0.004062329	0.004062329			0.00261575	1.95702E-05	0.00000943
Pb		4.48233E-09	4.48233E-09	1	1	0.000089	5.79744E-05	0.0000471
Sb		0.003438133	0.003438133				0.000000123	0.000005
Se						0.0000645	0.000000123	
Sn		0.008733941	0.008733941				1.49237E-05	
V		2.17804E-11	2.17804E-11				0.000000123	
Zn	1	0.001075481	0.001075481			0.0006695	0.000570112	0.0000943
Be							0.000000123	
Sc							0.000000123	
Sr						0.0000355	0.000000123	
Ti						0.0024	0.000000123	
Tl							0.000000123	
W							0.000000123	
Si		0.006428249	0.006428249				0.022176027	0.021
Fe		0.009181089	0.009181089			0.33695	0.100139479	0.00066
Ca						0.0143175	0.037630052	0.00377
Al		0.040232759	0.040232759			0.0255475	0.011116362	0.0123
K						0.002575	0.000000123	0.000943
Mg						0.014125	0.004224705	0.00471
Na						0.00325	0.000000123	0.00066
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.						0.0%	0.0%	0.0%
C bio						0.0%	100.0%	100.0%
Degr.						0.0%	60.0%	27.0%

Sol.						0	0	
Mod.	zinc in car shredder residue.	copper in car shredder residue.	copper in car shredder residue.	lead in car shredder residue.	lead in car shredder residue.			municipal solid waste.

Tab. A. 15 Solid waste compositions (Cont.)

Burn	1	1	1	1	1	0	1	0
Name	Mixed cardboard	plastics	laminated materials	laminated packaging, e.g. tetra bricks	combined goods e.g. dippers	glass	textiles	minerals
Ger.	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000
Com.	Mixed cardboard (Finnveden 2001)	Kunststoffe im CH Gemeinde-Abfall	Verbundstoffe plastic coated paper; 1% Verbundwaren, 2% Verbundverpackung	Verbundverpackung: Tetrapack 5)	Verbundwaren: Windeln 2) plastic coated paper	Glas	Textilien	Mineralien 1) teilw. Belevi/Eigene Ber.
Ho	19.00	32.87	18.43	18.62	3.85	0.00	16.04	0.00
Hu	16.60	30.00	17.10	17.10	1.84	0.00	14.50	0.00
H2O	0.21	0	0.0471	0.08857	0.8015	0	0.25	0
O	0.259806328	0.01997955	0.394041959	0.301920581	0.078770938	0.490473	0.2909	0.57
H	0.05451	0.145	0.0617	0.06674	0.01234		0.0502	
C	0.4503	0.79822795	0.453	0.4819	0.0906		0.3723	0.05
S	0.000948	0.0016	0.0008	0.001414	0.00016	0.0024	0.0028	
N	0.002054	0.003	0.0018	0.002185	0.00036		0.0311	
P	0.0003713		0.00009225		0.00001845			
B			0.000014175		0.000002835			
Cl	0.001343	0.027	0.00689175	0.005831	0.009358	0.0002	0.0027	0.011
Br		0.000075	0.00001875	0.000001831	0.00000375			
F		0.000014	0.0000035	0.000002747	0.0000007			
I								
Ag								
As				0.000000876				
Ba	0.0000487		0.000021225	0.00008868	0.000004245	0.002		
Cd	3.002E-08	0.00005	1.95725E-05	0.000007289	8.869E-07			
Co				0.000001009				
Cr	0.00001106	0.00005	0.0000365	0.000009311	0.0000073			
Cu	0.00002133	0.000075	0.000039975	0.00003454	0.000006116	0.00001		
Hg	1.422E-08	0.0000005	1.46225E-07	2.351E-07	2.9245E-08			
Mn			0.000035325	0.00007714	0.000007065			
Mo	0.000005		0.00000375		0.00000075			
Ni	0.000006478	0.000018	1.15725E-05	0.000008163	2.3145E-06			
Pb	0.00000316	0.00009	0.000057825	0.00002694	0.00001284	0.00001		
Sb		0.00002	0.00000875	0.000001926	0.00000175			
Se				0.00000233		0.000003		
Sn				0.000005792				
V				0.000421				
Zn	0.0000316	0.0007	0.000245725	0.0001138	0.00003058	0.000004		
Be								
Sc								
Sr								
Ti								
Tl								
W								
Si			0.01575		0.00315	0.3379		0.285
Fe		0.0039	0.00147	0.000293	0.000294	0.001		
Ca	0.01106		0.0028275	0.0004945	0.0005655	0.042		0.05
Al		0.0002	0.009275	0.04983	0.001855	0.0079		0.024
K	0.00948		0.00070725		0.00014145	0.0002		
Mg			0.0035325	0.00001831	0.0007065	0.0027		
Na			0.000495		0.000099	0.1132		0.01
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
C bio	100.0%	0.0%	80.0%	80.0%	50.0%	0.0%	80.0%	0.0%
Degr.	32.4%	1.0%	18.0%	18.0%	18.0%	0.0%	12.0%	0.0%
Sol.								

Mod.	municipal solid waste.	municipal solid waste.	municipal solid waste.	municipal solid waste.	municipal solid waste.	municipal solid waste.	municipal solid waste.	municipal solid waste.
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Tab. A. 16 Solid waste compositions (Cont.)

Burn	1	1	0	1	1	1	1	0
Name	natural products	compostable material	inert metals	electronic goods	volatile metals	batteries	electronic goods	lead smelter slag
Ger.	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	fraction in MSW from Hellweg 2000	Bleihütten-Schlacke
Com.	Naturprodukte 6) (8% + 1.0035% Verbundwaren wie Schuhe, Möbel etc.)	Organisches Nahrungsmittel im Abfall (nach K.Schleiss)	Metalle 3)	Problem Müll 4) Leiterplatten, elektronische Geräte	Metalle 3)	Verbundwaren Problem Müll Batterien 4)	Verbundwaren. Problem Müll Leiterplatten (Zimmermann et al.1996), Hg aus Elektr. Material	geometric mean from 4 literature sources
Ho	10.47	5.72	0.00	13.12	0.00	12.18	13.12	
Hu	9.00	4.00	0.00	12.00	0.00	12.00	12.00	
H2O	0.390600063	0.6		0.16303	0		0.16303	0.00
O	0.239075296	0.126427543		0.207984781	0.002006524	0.079477095	0.207984781	0.595057335
H	0.0306375	0.02		0.0384038		0.009044514	0.0384038	
C	0.25776	0.1624		0.317022		0.1	0.317022	
S	0.001637133	0.001499266		0.0006668			0.0006668	0.027
N	0.00773125	0.004		0.001445			0.001445	
P	0.0065025	0.00113		0.000260279			0.000260279	0.00158
B	0.00000512	0.00001024		5.67E-08			5.67E-08	
Cl	0.00224375	0.004		0.00862726		0.209190548	0.00862726	
Br	0.000003	0.000006		0.000000075			0.000000075	
F	0.00011	0.0002		0.000050014		0.00005	0.000050014	
I		0.000000055						
Ag				0.00021			0.00021	0.000052
As	1.8625E-06	0.000002						0.000998711
Ba	0.000039375		0.002	3.41749E-05	0.002		3.41749E-05	0.00181
Cd	8.3625E-07	0.000000138	0.0000065	5.51059E-05	0.000001025	0.015842778	5.51059E-05	2.99333E-05
Co	0.0000025	0.000005						0.000123
Cr	2.79688E-05	0.000008	0.0098	0.000467888	0.00323	0.011333333	0.000467888	0.00015
Cu	2.70125E-05	0.000018	0.020784951	0.125315053	0.020784951	0.005733	0.125315053	0.00130384
Hg	3.5125E-07	0.00000007	0.00001	1.01054E-06	0.0001	0.000536667	1.01054E-06	
Mn	7.22688E-05	0.0000043		0.000490141	0.02325	0.077464789	0.000490141	0.0543
Mo	0.000003	0.0000004		0.000003515			0.000003515	0.000057
Ni	6.02875E-06	0.00000542	0.002	0.013104581	0.000175	0.008666667	0.013104581	0.000087
Pb	0.000175986	0.00001856	0.0075	0.029807469	0.01558	0.00006667	0.029807469	0.0124972
Sb			0.00032	0.000200035	0.0009		0.000200035	0.000123
Se	8.12378E-07	4.99755E-07						
Sn	3.99804E-06	7.99609E-06	0.00065	0.015575	0.0001025		0.015575	
V	1.49927E-06	2.99853E-06						0.000069
Zn	0.00012062	0.00005824	0.0124	0.017711732	0.075	0.1652	0.017711732	0.021754999
Be								
Sc								0.000007
Sr								0.001283
Ti								0.00169
Tl								
W								
Si	0.019990215	0.03998043		0.000063			0.000063	0.280026981
Fe	0.0006	0.0006	0.81453	0.03900588	0.72687	0.178	0.03900588	
Ca	0.0243375	0.0218		0.00785331			0.00785331	
Al	0.004997554	0.009995107	0.1	0.0059371	0.1		0.0059371	
K	0.003	0.0035		0.006658829		0.139393939	0.006658829	
Mg	0.00516	0.00282	0.03	0.00001413	0.03		0.00001413	
Na	0.005125	0.0015		0.00000198			0.00000198	
sum	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Fe met.	0.0%	0.0%	83.0%	100.0%	0.0%	100.0%	60.0%	0%
C bio	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0%
Degr.	27.0%	27.0%	10.0%	5.0%	0.0%	0.0%	0.0%	0%
Sol.								
Mod.	municipal solid waste.	municipal solid waste.	municipal solid waste.	municipal solid waste.	municipal solid waste.	municipal solid waste.	municipal solid waste.	lead smelter slag

7.2 Synopsis of wastewater compositions

The following tables give the compositions of wastewaters, as described in chapter 4 'Compositions of waste fractions' on page 28ff. The first column in the first table gives a description of the entry; the second column gives an abbreviation. In the tables on the subsequent pages only the abbreviation is repeated for space reasons. The entries 'Capacity class', 'Share of sludge to incineration', 'Share of sludge to agriculture', 'residential sewage collection', 'Share of biogenic carbon' are additional informations used in the calculation tool.

Tab. A. 17 Wastewater compositions

Name		Name	average sewage		fibre board production effluent		particle board production effluent	
German name		Ger.	Durchschnittlich es Abwasser		Abwasser Faserplattenpro- duktion		Abwasser Spanplattenpro- duktion	
			mean amount	GSD	mean amount	GSD	mean amount	GSD
Chemical Oxygen Demand COD as O ₂	kg/m ³	COD	0.1556	122%	100	123%		
Biological Oxygen Demand BOD ₅ as O ₂	kg/m ³	BOD ₅	0.1036	122%			0.196	140%
Dissolved organic carbon DOC as C	kg/m ³	DOC	0.04575	122%				
Total organic carbon TOC as C	kg/m ³	TOC	0.0673	122%				
Sulfate SO ₄ as S	kg/m ³	SO ₄	0.044	122%				
Particulate S part as S	kg/m ³	S part.	0.002	122%				
Total S tot. as S	kg/m ³	S tot.		122%				
Ammonia NH ₄ as N	kg/m ³	NH ₄	0.01495	122%				
Nitrate NO ₃ as N	kg/m ³	NO ₃	0.00105	122%				
Nitrite NO ₂ as N	kg/m ³	NO ₂	0.0004	122%				
Particulate N part. as N	kg/m ³	N part.	0.003278689	122%				
Organic soluble N org. solv. as N	kg/m ³	N org.	0.008391734	122%				
Soluble Kjeldahl SKN as N	kg/m ³	SKN		122%				
Total Kjeldahl TKN as N	kg/m ³	TKN		122%				
Total Nitrogen N-tot. as N	kg/m ³	N tot.		122%				
Phosphate PO ₄ as P	kg/m ³	PO ₄	0.002458964	122%	0.00116	108%		
Particulate P-part. as P	kg/m ³	P part.	0.000614741	122%				
Total P-tot. as P	kg/m ³	P tot.		122%				
Boron B	kg/m ³	B						
Chlorine Cl	kg/m ³	Cl	0.030031224	224%				
Bromium Br	kg/m ³	Br						
Fluorine F	kg/m ³	F	3.27689E-05	224%				
Iodine I	kg/m ³	I						
Silver Ag	kg/m ³	Ag						
Arsenic As	kg/m ³	As	0.0000009	224%				
Barium Ba	kg/m ³	Ba						
Cadmium Cd	kg/m ³	Cd	2.80629E-07	224%				
Cobalt Co	kg/m ³	Co	1.61775E-06	224%				
Chromium Cr	kg/m ³	Cr	1.22321E-05	224%				
Copper Cu	kg/m ³	Cu	3.74433E-05	224%	0.00003	226%		
Mercury Hg	kg/m ³	Hg	2.00043E-07	224%				
Manganese Mn	kg/m ³	Mn	0.000053	224%				
Molybdenum Mo	kg/m ³	Mo	9.57442E-07	224%				
Nickel Ni	kg/m ³	Ni	6.5891E-06	224%				
Lead Pb	kg/m ³	Pb	8.63145E-06	224%				
Antimony Sb	kg/m ³	Sb						
Selenium Se	kg/m ³	Se						
Tin Sn	kg/m ³	Sn	0.0000034	224%				
Vanadium V	kg/m ³	V						
Zinc Zn	kg/m ³	Zn	0.000109353	224%	0.00013 ^a	226%		
Beryllium Be	kg/m ³	Be						
Scandium Sc	kg/m ³	Sc						
Strontium Sr	kg/m ³	Sr						
Titanium Ti	kg/m ³	Ti						
Thallium Tl	kg/m ³	Tl						
Tungsten W	kg/m ³	W						
Silicon Si	kg/m ³	Si	0.003126298	224%				
Iron Fe	kg/m ³	Fe	0.007092772	224%				
Calcium Ca	kg/m ³	Ca	0.050833712	224%				
Aluminium Al	kg/m ³	Al	0.001037845	224%				
Potassium K	kg/m ³	K	0.000398898	224%				

Magnesium Mg	kg/m3	Mg	0.005707072 224%		
Sodium Na	kg/m3	Na	0.002185974 224%		
Capacity class of WWTP	–	class	1–5	3	3
Share of sludge to incineration	w% of dry substance	incin.			
Share of sludge to agriculture	w% of dry substance	agri.			
Inventory including residential sewage collection		resid.			
Share of biogenic carbon in sewage		C bio.	100%	100%	100%

a Corrected from an original value of 0.38 kg/m³, which was deemed to high, based on the Zn/Cu ratio in natural wood.

Tab. A. 18 Wastewater compositions (Cont.)

Name	plywood production effluent		glass production effluent		black chrome coating effluent		condensate from light oil boiler		heat carrier liquid, 40% C3H8O2	
Ger.	Abwasser Sperrholzplatte nproduktion		Abwasser Glasproduktion		Abwasser Schwarzchrom beschichtung		Kondensat aus Leichtöl- Heizung		Wärmeträgerflü ssigkeit, 40% C3H8O2	
	mean amount	GSD	mean amount	GSD	mean amount	GSD	mean amount	GSD	mean amount	GSD
COD	0.7	177%	0.115	124%	0.002	124%		128%	650	125%
BOD5	4.23	177%		124%	0.001	124%		128%		123%
DOC				124%		124%		128%		123%
TOC				124%		124%	0.025	128%		123%
SO4			0.333	124%		124%	0.084	128%		127%
S part.				124%		124%		128%		127%
S tot.				124%		124%		128%		123%
NH4			0.009	124%		124%		128%		127%
NO3				124%		124%	0.002	128%		127%
NO2				124%		124%		128%		127%
N part.				124%		124%		128%		123%
N org.				124%		124%		128%		127%
SKN				124%		124%		128%		127%
TKN				124%		124%		128%		123%
N tot.				124%		124%		128%		127%
PO4	0.00116	108%		124%		124%		128%		127%
P part.				124%		124%		128%		127%
P tot.				124%		124%		128%		123%
B			0.000524272	224%		224%		227%		225%
Cl				224%		224%	0.01	227%		225%
Br				224%		224%		227%		224%
F			0.03	224%		224%	0.00045	227%		224%
I				224%		224%		227%		224%
Ag				224%	0.000005	224%		227%		224%
As			0.0003	224%		224%		227%		224%
Ba			0.003	224%		224%		227%		226%
Cd			0.00005	224%	0.000005	224%		227%		226%
Co				224%		224%		227%		226%
Cr			0.0005	224%	0.00035	224%	0.0004	227%		226%
Cu	0.00003	226%	0.0005	224%	0.00049	224%	0.00003	227%		226%
Hg				224%		224%		227%		226%
Mn				224%		224%		227%		226%
Mo				224%		224%		227%		226%
Ni			0.0005	224%	0.0003	224%	0.002	227%		226%
Pb			0.0005	224%	0.00005	224%		227%		224%
Sb			0.0003	224%		224%		227%		226%
Se				224%		224%		227%		226%
Sn			0.0005	224%		224%		227%		224%
V				224%		224%		227%		224%
Zn	0.00013 ^a	226%	0.0005	224%	0.00021	224%	0.00002	227%		226%
Be				224%		224%		227%		226%
Sc				224%		224%		227%		226%
Sr				224%		224%		227%		226%
Ti				224%		224%		227%		226%
Tl				224%		224%		227%		226%
W				224%		224%		227%		226%
Si				224%		224%		227%		226%
Fe				224%		224%	0.002	227%		226%
Ca				224%		224%		227%		226%
Al				224%		224%	0.02	227%		226%
K				224%		224%		227%		226%
Mg				224%		224%		227%		226%
Na				224%		224%		227%		226%
class	3		2		2		2		2	
incin.										
agri.										

7. Appendix

resid.					
C bio.	100%	0%	0%	0%	0%

a Corrected from an original value of 0.38 kg/m³, which was deemed to high, based on the Zn/Cu ratio in natural wood.

Tab. A. 19 Wastewater compositions (Cont.)

Name	tube collector production effluent		rainwater mineral oil storage		maize starch production effluent		potato starch production effluent		pig iron production effluent	
Ger.	Abwasser Kollektorrohrpr oduktion		Regenwasser Mineralöl- Tanklager		Abwasser Maisstärkeprod uktion		Abwasser Kartoffelstärkep roduktion		Abwasser Roheisenprodu ktion	
	mean amount	GSD	mean amount	GSD	mean amount	GSD	mean amount	GSD	mean amount	GSD
COD		125%	0.07	136%	7.34	217%	2	217%	0.0000203	150%
BOD5		125%		136%	6.13	154%	1.35	154%		
DOC		125%		136%						
TOC		125%		136%					0.0000041	150%
SO4		125%	0.011	136%						
S part.		125%		136%						
S tot.		125%		136%						
NH4		125%	0.00028	136%						
NO3		125%	0.001	136%						
NO2		125%		136%						
N part.		125%		136%						
N org.		125%		136%						
SKN		125%		136%						
TKN		125%		136%						
N tot.		125%		136%	0.569	113%	0.006	113%		
PO4		125%		136%						
P part.		125%		136%						
P tot.		125%	0.00066	136%	0.105	132%				
B		225%	0.000016	231%						
Cl		225%	0.34	231%						
Br		225%		231%						
F	0.000444	225%		231%						
I		225%		231%						
Ag	0.0001	225%		231%						
As		225%		231%						
Ba		225%	0.00003	231%						
Cd		225%	0.0000005	231%						
Co		225%		231%						
Cr		225%	0.000008	231%					0.000002	150%
Cu		225%	0.000015	231%					0.0000042	150%
Hg		225%		231%						
Mn		225%	0.00012	231%						
Mo		225%		231%						
Ni		225%	0.000005	231%					0.000003	150%
Pb		225%	0.0000025	231%					0.000002	150%
Sb		225%		231%						
Se		225%		231%						
Sn	0.000667	225%		231%						
V		225%		231%						
Zn		225%	0.00028	231%					0.00002	150%
Be		225%		231%						
Sc		225%		231%						
Sr		225%	0.00021	231%						
Ti		225%		231%						
Tl		225%		231%						
W		225%		231%						
Si		225%		231%						
Fe		225%	0.0014	231%						
Ca		225%	0.036	231%						
Al		225%	0.00035	231%						
K		225%	0.004	231%						
Mg		225%	0.0022	231%						
Na		225%	0.212	231%						
class	2		2		2		2		3	
incin.					60%		60%			
agri.					40%		40%			

7. Appendix

resid.					
C bio.	0%	0%	100%	100%	0%

Tab. A. 20 Wastewater compositions (Cont.)

Name	concrete production effluent		lorry production effluent		ceramic production effluent	
Ger.	Abwasser Betonproduktio n		Abwasser Lastkraftwagen -Produktion		Abwasser Keramikprodukt ion	
	mean amount	GSD	mean amount	GSD	mean amount	GSD
COD						
BOD5						
DOC	0.01216	125%				
TOC	0.01216	125%				
SO4	0.00100233	125%				
S part.						
S tot.			0.000114	224%		
NH4						
NO3	0.0032108	125%				
NO2						
N part.						
N org.						
SKN						
TKN						
N tot.						
PO4	0.0000672	125%				
P part.						
P tot.						
B						
Cl	0.02969	125%	0.000114	224%		
Br						
F						
I						
Ag						
As	0.000001	125%				
Ba					0.00048	300%
Cd					0.000008	300%
Co					0.000025	300%
Cr	0.00002134	225%			0.000025	300%
Cu					0.000005	300%
Hg						
Mn						
Mo						
Ni			0.00027	224%	0.000025	300%
Pb	0.000001	225%	0.000044	224%	0.000025	300%
Sb						
Se						
Sn						
V						
Zn	0.00003462	225%	0.000484	224%	0.000012	300%
Be						
Sc						
Sr						
Ti						
Tl						
W						
Si						
Fe						
Ca						
Al						
K	0.3182	125%				
Mg						
Na	0.08275	125%				
class	3		1		3	
incin.						
agri.						

resid.			
C bio.	0%	0%	

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